

Cross-Domain State Preservation Functor

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Abstract

We mechanize cross-domain state preservation in Isabelle/HOL. Preservation morphisms have identity and composition laws, and their composition is associative. An authenticated layer lifts merge and blinding from the `ADS_Functor` Merkle interface; its extraction laws apply on extractable endpoints and are instantiated on blindable positions and a recursive Canton transaction-tree model. A degree-indexed tower supplies composable natural transformations and separates a load-bearing over-provisioning reconciliation theorem from degree-free validity preservation. A guarded composition proves the existence of a bounded run from any finite-domain unlocked state to a valid state under deterministic fair discharge; recovery is selected non-constructively and is not an executable protocol. The priority layer proves unique maximal selection for finite non-empty candidate sets and aggregate pending-count decrease under an in-roster fair-leader assumption. Static Byzantine-count assumptions provide an honest-tagged roster witness, but no VRF, BFT execution, network, or request-identity semantics are formalized. Concrete instances cover a regulatory state model and a TCP-inspired toy endpoint/tracker example. The synchronization model is atomic and excludes concurrent lock contention.

Contents

1	Overview	5
1.1	Theory Dependency Structure	7
1.2	Related Work and Scope	7
1.3	Design Decisions	8
1.4	Proof Automation	8
2	Generic State Machine Locale	9
3	Cross-Domain State Preservation Locale	10
4	Symmetric State Preservation	13

5	Multi-Domain State Preservation	14
6	Regulatory State and Action Definitions	17
7	Fundamental Properties of the Transition Function	19
7.1	Invariant I1: CONFISCATED is terminal	19
7.2	Invariant I2: CONFISCATE reaches CONFISCATED from any non-terminal state	19
7.3	Invariant I3: Determinism (inherent from function definition)	19
7.4	SEIZED to FROZEN exclusion	19
7.5	FROZEN to RESTRICTED exclusion	19
7.6	Transition completeness: every non-terminal state has at least one valid action	20
7.7	Reachability from ACTIVE	20
7.8	Return paths to ACTIVE	21
8	State Machine Instance	21
9	Asset and Global State Modeling	22
10	State Accessors and Predicates	22
11	Global State Validity	23
12	Synchronization Protocol	23
12.1	Lock acquisition	23
12.2	State update across chains	24
12.3	The synchronization function	24
13	Synchronization Properties	25
13.1	No self-loops	25
13.2	Properties of update_all_chains	25
13.3	Lock mutual exclusion	26
14	Main Theorem: Cross-Chain Regulatory Consistency	26
15	Valid State Preservation	28
15.1	Part 1: consistent_state preservation	28
15.2	Part 2: no_locked_without_reason preservation	31
15.3	The valid_state preservation theorem	32
16	Heterogeneous-Action State Preservation Instance	32
17	Layer-Crossing Symmetric State Preservation Instance	37
18	Multi-Domain Locale Instantiation	44

19 Priority-Based Deterministic Selection	46
20 Fair-Leader Aggregate Progress	49
21 Authority Level and Action Severity	53
22 Priority Key	53
23 Extended Message Type	54
24 Priority Construction	54
25 Node and System Model	54
26 D-quencer System Locale	55
27 Priority System Instantiation	56
28 Fair-Leader Aggregate-Progress Instantiation	60
29 Conditional Safety alongside the Liveness Results	61
30 Guarded Invariants	63
31 Eventual Dischargers	63
32 Compositional Safety and Liveness	64
33 Guarded Bounded Convergence	68
34 Discharge Methods for Instance Obligations	71
35 Functor Laws on State-Preservation Morphisms	73
35.1 Identity morphism	73
35.2 Composition of morphisms	73
35.3 Associativity of morphism composition	74
36 Composition Exercised: A Heterogeneous Three-Domain Chain	75
37 Proof Automation Exercised on the Regulatory Instances	78
38 State Glue: Join and Refinement	81
39 Authenticated Cross-Domain State	82
40 Authenticated Cross-Domain State: the Oraclizer Instance	85

41 Inconsistency Measure on Global States	89
42 Structural Effect of Synchronization	91
43 The Critical Path: Synchronization Reduces Inconsistency	95
44 Existence of a Reducing Synchronization	97
45 Terminal-Faithful Safe Recovery	98
46 The Oraclizer as a Converging Composition	104
47 Guarded Bounded Convergence for the Oraclizer	106
48 Global Model Witnesses for the Liveness Hosts	111
49 Sync Degree Stratification	114
50 Degree Functors and Their Natural Transformations	117
51 Functoriality of the Degree Transition Functor on Action Words	121
52 Degree Classes and Hierarchy Monotonicity	126
53 A TCP-Inspired Endpoint State Machine	133
54 The Connection-Tracker Representation	134
55 The Tracked Event Subset and the Tracker Transition	135
56 Discharging the Instance with the Generic Methods	136
57 Auxiliary list predicate for synchronization sequences	139
58 Composite functor laws: cross-domain preservation after the ADS Merkle functor	139
59 Authenticity along blinding paths and merge folds	142
59.1 Need-to-know blinding along a sequence	142
59.2 Re-revealing merge along a sequence	145
60 Instantiation on the ADS blindable-position functor	147
61 Instantiation on the top-level Canton transaction commitment	151

62 Recursive instantiation on the Canton transaction tree	154
62.1 Extraction commutes with blinding and merge	155
62.2 The recursive transaction-tree instance	158
63 Path-level Merkle inclusion for the recursive view tree	164
64 Conclusion	166

1 Overview

Methodological positioning and contributions. Regarding state machines as objects and the structure-preserving synchronization maps of `State_Preservation.thy` as morphisms, `Functor_Laws.thy` proves the three category axioms on these morphisms: identity is a preservation morphism (`preservation_id`), preservation morphisms compose (`preservation_compose`), and their composition is associative (`preservation_assoc`). The functorial reading is carried at two further places: each transition system is a functor from its one-object action category into partial maps on states, its two functor laws carried by the identity word and word concatenation (`apply_actions_append`); and `Hierarchy.thy` exhibits a tower of degree-indexed functors with natural transformations between them. This is the cross-domain counterpart of the “closed under composition” structure of Lochbihler and Marić’s `ADS_Functor` entry [2], whose authenticated-data-structure functor is single-domain; the present entry carries the functorial reading across heterogeneous domains and couples it with that authenticated layer through two glue predicates, `state_join` and `state_refines`, that lift the ADS merge and blinding (partial-order) operations to the global-state level.

We claim no new proof technique. The contribution is a mechanization in three layers: (i) the cross-domain state-preservation *functor* and its category axioms, above; (ii) *guarded bounded convergence* under a deterministic in-roster fair-leader assumption in a static roster/count context — from an *arbitrary* finite-domain, unlocked initial global state, with no assumption that the cross-chain consistency invariant holds initially, there exists a run reaching a valid state within a bounded number of evolution steps; the safe recovery is selected by Hilbert choice rather than by an executable queue or BFT algorithm (`oraclizer_guarded_bounded_convergence`, `Functor_Laws.thy`); and (iii) a *tower* of synchronization-degree functors connected by natural transformations (`degree_natural_transformation`) that are closed under composition (`nt_compose`, via `nt_vertical_compose`), as developed in `Hierarchy.thy`.

The entry consists of ten theory files; the theory text carries the detailed exposition alongside the formal material.

State_Preservation.thy domain-independent locales for state machines, pairwise state preservation (with naturality in both the enabled and the disabled direction), symmetric bidirectional preservation, and multi-domain preservation — instantiable by any domain with finite states and actions, deterministic transitions, and optional terminal states.

Regulatory_Instance.thy the regulatory safety instance: five states, seven actions, transition exclusions with legal justification, an atomic synchronization function with a Boolean lock guard, and the main safety theorems — *regulatory homomorphism*, *valid state preservation*, and *multi-domain parametric instantiation* — together with heterogeneous-action and layer-crossing bridge instances.

Priority_Resolution.thy domain-independent priority and aggregate-progress locales: deterministic selection from a finite message set with injective priorities (**priority_system**), and bounded aggregate pending-count decrease under periodic honest-tagged scheduling (**fair_leader_system**).

DQuencer_Instance.thy the regulatory scheduling instance: a four-component priority key with proven injectivity under well-formedness bounds, a Byzantine fault model with the BFT threshold $n \geq 3f + 1$, unique maximal finite-set selection, bounded aggregate pending-count decrease, closed-count completion, and the safety-side corollary *conditional safety preservation*.

Proof_Automation.thy reusable Eisbach discharge methods with entry points **discharge_state_machine** and **discharge_preservation**, driven by three named theorem collections; see the *Proof Automation* subsection below.

Composition.thy domain-independent locales composing a guarded safety invariant with an eventual-progress scheduler, extended by a natural-number progress measure to *guarded bounded convergence* from an arbitrary carrier state (**bounded_convergence_from_arbitrary**).

Functor_Laws.thy the category axioms, the ADS glue predicates (**state_join**, **state_refines**), the **authenticated_state** locale over the **merkle_interface** with soundness theorems whose proofs invoke the Merkle interface explicitly, extraction laws on extractable endpoints, and the existential fair-discharge convergence instantiation (**oraclizer_guarded_bounded_convergence**).

Hierarchy.thy the synchronization-degree tower: a forgetful natural transformation closed under composition (**degree_forget**,

`degree_natural_transformation`, `nt_compose`), reduction containment, a degree-2 boundary grounded only in a strict integer-timestamp order, and a genuinely one-directional degree monotonicity (`over_provisioning_guarantees`, `no_downward_safety`). Its degree is a coupling-breadth index, not a formalization of product S0–S3 operational meanings such as directionality, causal execution, atomic binding, or mutual rollback.

Canton_Bridge.thy the categorical laws of the authenticated layer on its extractable sub-preorder (`cdsp_ads_compose`, `cdsp_ads_merge_assoc`), authenticity along blinding paths and merge folds, and instantiations on the blindable-position functor of `ADS_Functor.ADS_Construction` and on a recursive model of the Canton transaction tree, with a declared consensus-scope limit.

External_Instance.thy an instance outside the regulatory domain — a TCP-inspired toy endpoint lifecycle and abstract tracker, using state and event names drawn from RFC 793 [3] but neither trace-conformant to that specification nor a model of a concrete contrack implementation — importing only the generic theories and discharged by the generic methods: evidence that the generic layer carries no hidden regulatory assumptions.

1.1 Theory Dependency Structure

The four base theories form two generic–instance pairs that meet at `DQuencer_Instance.thy`: `State_Preservation.thy` is imported by `Regulatory_Instance.thy`, and `Priority_Resolution.thy`, which depends only on `Main`, is imported by `DQuencer_Instance.thy` together with the regulatory instance. The compositional theories sit on top: `Composition.thy` and `Proof_Automation.thy` import `State_Preservation.thy`; `Functor_Laws.thy` imports both instances, `ADS_Functor.Merkle_Interface`, `Composition.thy`, and `Proof_Automation.thy`; `Hierarchy.thy` and `Canton_Bridge.thy` import `Functor_Laws.thy`; and `External_Instance.thy` imports only `Proof_Automation.thy`, deliberately independent of the regulatory instance. The dependency on `ADS_Functor` is the entry’s only external session dependency beyond the Isabelle distribution (`HOL-Library`, `HOL-Eisbach`); `Canton_Bridge.thy` additionally imports `ADS_Functor.ADS_Construction` and `ADS_Functor.Canton_Transaction_Tree`.

1.2 Related Work and Scope

The authenticated layer directly reuses the Isabelle/HOL interfaces and constructions of `ADS_Functor` [2]; this dependency is declared in the ses-

sion graph and exercised by the merge, blinding, and inclusion proofs. The threshold $n \geq 3f + 1$ is standard background for Byzantine-fault-tolerant replication [1], but the present entry uses only static roster cardinalities and node tags. It does not formalize the PBFT protocol, adversarial executions, or probabilistic leader election. The external instance only borrows names from the historical TCP specification [3]; its collapsed teardown is neither an RFC subset nor a verification of TCP or a concrete connection tracker.

1.3 Design Decisions

The regulatory state model incorporates several deliberate design decisions based on legal precedence and operational considerations:

- The state machine models seven regulatory `reg_actions`: the four escalation actions `FREEZE`, `SEIZE`, `CONFISCATE`, `RESTRICT` and their de-escalation counterparts `UNFREEZE`, `UNRESTRICT`, `RELEASE`. `RECOVER` and `LIQUIDATE` are excluded as they involve force transfers or external DEX interactions rather than regulatory state transitions.
- The direct transition `SEIZED` $\rightarrow$ `FROZEN` is excluded because seizure is a strictly stronger legal constraint than freezing; relaxation requires explicit release first.
- The direct transition `FROZEN` $\rightarrow$ `RESTRICTED` is excluded to enforce passage through `ACTIVE`.
- Locking is modelled as an atomic Boolean guard on the synchronization function: `sync` acquires the lock, updates, and releases in a single atomic step. Under this atomic abstraction the guard expresses the *intended* exclusion of competing regulatory actions on the same asset rather than serialising them dynamically (there is no interleaving in the model to serialise); the dynamic correctness of contention-time locking is outside this entry. The guard does not address double-spending, which is handled at a different layer.
- Deadlock (a concurrency phenomenon) does not arise within the model’s scope: the atomic `sync` model has no concurrent lock contention. Forced lock release under contention is out of scope; the entry states no proved deadlock-freedom theorem.

1.4 Proof Automation

The entry ships a reusable proof-automation layer, `Proof_Automation.thy`: two Eisbach methods discharge the obligations of `state_machine` and `state_preservation` instances on finite, enumerated domains — `discharge_state_machine` and `discharge_preservation`. An instance

theory declares the defining equations of its state/action sets and transition functions into three named theorem collections (`discharge_simps` for equational and conditional-rewrite content, `discharge_intros` for closure rules, `discharge_dels` for default rewrites that must be suppressed to keep structured state maps opaque), after which whole instances are discharged by a single method invocation.

The methods are exercised against the entry's own instances, with the statements kept identical to their manual counterparts. Four instances are discharged by pure search, with no registered interpretation to fall back on — the escalation-scoped DAML state machine (6 obligations), the escalation-scoped layer-crossing bridge (11 obligations), the composed three-domain morphism re-derived from atomic obligations without invoking `preservation_compose` (11 obligations), and a forward escalation-scoped bridge with no manual counterpart, which the method constructs outright (11 obligations) — closing 39 of 39 atomic obligations, with no obligation weakened or left to a manual fallback. Three further one-line restatements of the registered interpretations are discharged from the registrations. The methods and their collections are part of the entry's reusable surface: an importer instantiating the locales on its own domain populates the collections and reuses the same two entry points, as `External_Instance.thy` does.

The synchronization model is atomic; lifting it to a partially synchronous network with message delays, redelivery, and out-of-order arrival is left to subsequent entries.

```
theory State-Preservation
imports Main
begin
```

2 Generic State Machine Locale

A state machine with finite state set, deterministic partial transition function, and optional terminal states. Terminal states accept no transitions.

```
locale state-machine =
  fixes states :: 's set
    and actions :: 'a set
    and transition :: 's  $\Rightarrow$  'a  $\Rightarrow$  's option
    and terminal :: 's set
  assumes finite-states: finite states
    and finite-actions: finite actions
    and terminal-subset: terminal  $\subseteq$  states
    and terminal-absorbing:  $\llbracket s \in \text{terminal}; a \in \text{actions} \rrbracket \Longrightarrow \text{transition } s \ a = \text{None}$ 
    and transition-closed:  $\llbracket s \in \text{states}; a \in \text{actions}; \text{transition } s \ a = \text{Some } s' \rrbracket \Longrightarrow s' \in \text{states}$ 
```

and *transition-domain*: $\llbracket s \notin \text{states} \rrbracket \implies \text{transition } s \ a = \text{None}$
begin

Determinism is inherent: transition is a function, not a relation.

lemma *transition-deterministic*:

transition $s \ a = \text{Some } s1 \implies \text{transition } s \ a = \text{Some } s2 \implies s1 = s2$
by *simp*

Sequential composition of actions (partial, returns None on first failure).

fun *apply-actions* :: $'s \Rightarrow 'a \ \text{list} \Rightarrow 's \ \text{option}$ **where**

apply-actions $s \ [] = \text{Some } s$
| *apply-actions* $s \ (a \ \# \ as) = (\text{case } \text{transition } s \ a \ \text{of}$
 $\text{None} \Rightarrow \text{None}$
 $\text{Some } s' \Rightarrow \text{apply-actions } s' \ as)$

lemma *apply-actions-closed*:

$\llbracket s \in \text{states}; \forall a \in \text{set } as. a \in \text{actions}; \text{apply-actions } s \ as = \text{Some } s' \rrbracket$
 $\implies s' \in \text{states}$

by (*induction as arbitrary: s*) (*auto split: option.splits dest: transition-closed*)

Note: *apply-actions* $?s \ [] = \text{Some } ?s$ already provides the simp rule *apply-actions* $s \ [] = \text{Some } s$, so no separate [simp] lemma is needed.

lemma *apply-actions-terminal*:

$\llbracket s \in \text{terminal}; a \in \text{actions} \rrbracket \implies \text{apply-actions } s \ (a \ \# \ as) = \text{None}$
by (*simp add: terminal-absorbing*)

Concatenation of action words is the Kleisli composition of the two segment maps: together with *apply-actions* $?s \ [] = \text{Some } ?s$ (the identity word), this is the functor reading of a transition system on the free monoid of action words.

lemma *apply-actions-append*:

apply-actions $s \ (as \ @ \ bs) =$
 $(\text{case } \text{apply-actions } s \ as \ \text{of } \text{None} \Rightarrow \text{None} \mid \text{Some } s' \Rightarrow \text{apply-actions } s' \ bs)$
by (*induction as arbitrary: s*) (*auto split: option.splits*)

Outside the state set no action sequence makes progress: the domain discipline **transition_domain** lifts from single transitions to words.

lemma *apply-actions-outside-domain*:

$s \notin \text{states} \implies \text{apply-actions } s \ (a \ \# \ as) = \text{None}$
by (*simp add: transition-domain*)

end

3 Cross-Domain State Preservation Locale

Two state machines connected by a synchronization map. The key property (naturality / homomorphism) states that synchronizing after a local transition yields the same result as transitioning after synchronization.

More precisely: for state machines (S_src, T_src) and (S_tgt, T_tgt) connected by sync (mapping source states to target states), the following diagram commutes:

$$\begin{array}{ccc} s_src & \xrightarrow{T_src(a)} & s_src' \\ \downarrow \text{sync} & & \downarrow \text{sync} \\ s_tgt & \xrightarrow{T_tgt(a)} & s_tgt' \end{array}$$

That is: $\text{sync } (T_src \ s \ a) = T_tgt \ (\text{sync } s) \ a$

This is the structural preservation guarantee: a transition performed at the source domain, when synchronized, produces the same result as performing the corresponding transition at the target domain.

```

locale state-preservation =
  source: state-machine statess actionss transitions terminals +
  target: state-machine statest actionst transitiont terminalt
  for statess :: 's set and actionss :: 'a set
    and transitions :: 's  $\Rightarrow$  'a  $\Rightarrow$  's option
    and terminals :: 's set
    and statest :: 't set and actionst :: 'b set
    and transitiont :: 't  $\Rightarrow$  'b  $\Rightarrow$  't option
    and terminalt :: 't set +
  fixes state-map :: 's  $\Rightarrow$  't
    and action-map :: 'a  $\Rightarrow$  'b
  assumes state-map-well-defined: s  $\in$  statess  $\implies$  state-map s  $\in$  statest
    and action-map-well-defined: a  $\in$  actionss  $\implies$  action-map a  $\in$  actionst
    and terminal-preservation: s  $\in$  terminals  $\implies$  state-map s  $\in$  terminalt
  — The naturality / homomorphism condition:
    and naturality:
       $\llbracket s \in \text{states}_s; a \in \text{actions}_s; \text{transition}_s \ s \ a = \text{Some } s' \rrbracket$ 
       $\implies \text{transition}_t \ (\text{state-map } s) \ (\text{action-map } a) = \text{Some } (\text{state-map } s')$ 
    and naturality-none:
       $\llbracket s \in \text{states}_s; a \in \text{actions}_s; \text{transition}_s \ s \ a = \text{None} \rrbracket$ 
       $\implies \text{transition}_t \ (\text{state-map } s) \ (\text{action-map } a) = \text{None}$ 
begin

```

The fundamental theorem: state preservation extends to sequences of actions. If a sequence of transitions is valid at the source, the mapped sequence is valid at the target with the mapped final state.

theorem sequential-preservation:

```

  assumes s  $\in$  statess
    and  $\forall a \in \text{set } as. a \in \text{actions}_s$ 
    and source.apply-actions s as = Some s'
  shows target.apply-actions (state-map s) (map action-map as) = Some
    (state-map s')
  using assms
proof (induction as arbitrary: s)
  case Nil
  then show ?case by simp
next
  case (Cons a as)
  then obtain s-mid where

```

```

    step: transitions s a = Some s-mid and
    rest: source.apply-actions s-mid as = Some s'
  by (auto split: option.splits)
from Cons.premis(1) Cons.premis(2) step have s-mid ∈ statess
  using source.transition-closed by auto
from Cons.premis(1) Cons.premis(2) step
  have mapped-step: transitiont (state-map s) (action-map a) = Some (state-map
s-mid)
  using naturality by auto
from ⟨s-mid ∈ statess⟩ Cons.premis(2) rest
  have target.apply-actions (state-map s-mid) (map action-map as) = Some
(state-map s')
  using Cons.IH by auto
with mapped-step show ?case by simp
qed

```

theorem sequential-preservation-none:

```

  assumes s ∈ statess
  and ∀ a ∈ set as. a ∈ actionss
  and source.apply-actions s as = None
  shows target.apply-actions (state-map s) (map action-map as) = None
  using assms
proof (induction as arbitrary: s)
  case Nil
  then show ?case by simp
next
  case (Cons a as)
  show ?case
  proof (cases transitions s a)
    case None
    then have transitiont (state-map s) (action-map a) = None
      using naturality-none Cons.premis(1) Cons.premis(2) by auto
    then show ?thesis by simp
  next
    case (Some s-mid)
    then have mid-in: s-mid ∈ statess
      using source.transition-closed Cons.premis(1) Cons.premis(2) by auto
    from Some Cons.premis(3) have source.apply-actions s-mid as = None
      by simp
    then have ih: target.apply-actions (state-map s-mid) (map action-map as) =
None
      using Cons.IH mid-in Cons.premis(2) by auto
    from Some have transitiont (state-map s) (action-map a) = Some (state-map
s-mid)
      using naturality Cons.premis(1) Cons.premis(2) by auto
    with ih show ?thesis by simp
  qed
qed
qed

```

Terminal states are preserved: if a source state is terminal, its image is

terminal.

corollary *terminal-image-absorbing*:

assumes $s \in \text{terminal}_s$ and $a \in \text{actions}_s$
 shows $\text{transition}_t (\text{state-map } s) (\text{action-map } a) = \text{None}$

proof –

from *assms*(1) have $s \in \text{states}_s$ using *source.terminal-subset* by *auto*

from *assms* have $\text{transition}_s s a = \text{None}$

using *source.terminal-absorbing* by *auto*

then show *?thesis*

using *naturality-none* $\langle s \in \text{states}_s \rangle$ *assms*(2) by *auto*

qed

end

4 Symmetric State Preservation

For bidirectional synchronization, both directions must preserve structure. This locale composes two **state_preservation** instances with inverse maps, establishing that synchronization in either direction is structure-preserving.

locale *symmetric-state-preservation* =

forward: *state-preservation*

states_s actions_s transition_s terminal_s

states_t actions_t transition_t terminal_t

state-map action-map +

backward: *state-preservation*

states_t actions_t transition_t terminal_t

states_s actions_s transition_s terminal_s

state-map-inv action-map-inv

for $\text{states}_s :: 's$ set and $\text{actions}_s :: 'a$ set

and $\text{transition}_s :: 's \Rightarrow 'a \Rightarrow 's$ option

and $\text{terminal}_s :: 's$ set

and $\text{states}_t :: 't$ set and $\text{actions}_t :: 'b$ set

and $\text{transition}_t :: 't \Rightarrow 'b \Rightarrow 't$ option

and $\text{terminal}_t :: 't$ set

and $\text{state-map} :: 's \Rightarrow 't$ and $\text{action-map} :: 'a \Rightarrow 'b$

and $\text{state-map-inv} :: 't \Rightarrow 's$ and $\text{action-map-inv} :: 'b \Rightarrow 'a$ +

assumes *roundtrip-state-src*: $s \in \text{states}_s \implies \text{state-map-inv} (\text{state-map } s) = s$

and *roundtrip-state-tgt*: $t \in \text{states}_t \implies \text{state-map} (\text{state-map-inv } t) = t$

and *roundtrip-action-src*: $a \in \text{actions}_s \implies \text{action-map-inv} (\text{action-map } a) = a$

and *roundtrip-action-tgt*: $b \in \text{actions}_t \implies \text{action-map} (\text{action-map-inv } b) = b$

begin

Under symmetric preservation with roundtrip maps, **state_map** is injective on the state set – distinct source states map to distinct target states. This guarantees no information loss during synchronization.

lemma *state-map-injective*:

$\llbracket s1 \in \text{states}_s; s2 \in \text{states}_s; \text{state-map } s1 = \text{state-map } s2 \rrbracket \implies s1 = s2$

using *roundtrip-state-src* **by** *metis*

The roundtrip pair upgrades injectivity to a bijection between the two state sets and between the two action sets — the formal content of “no information loss” for bidirectional synchronization.

lemma *state-map-bij: bij-betw state-map states_s states_t*
proof (*rule bij-betw-byWitness*[**where** $f' = \text{state-map-inv}$])
 show $\forall s \in \text{states}_s. \text{state-map-inv} (\text{state-map } s) = s$
 using *roundtrip-state-src* **by** *blast*
 show $\forall t \in \text{states}_t. \text{state-map} (\text{state-map-inv } t) = t$
 using *roundtrip-state-tgt* **by** *blast*
 show $\text{state-map} ' \text{states}_s \subseteq \text{states}_t$
 using *forward.state-map-well-defined* **by** *auto*
 show $\text{state-map-inv} ' \text{states}_t \subseteq \text{states}_s$
 using *backward.state-map-well-defined* **by** *auto*
qed

lemma *action-map-bij: bij-betw action-map actions_s actions_t*
proof (*rule bij-betw-byWitness*[**where** $f' = \text{action-map-inv}$])
 show $\forall a \in \text{actions}_s. \text{action-map-inv} (\text{action-map } a) = a$
 using *roundtrip-action-src* **by** *blast*
 show $\forall b \in \text{actions}_t. \text{action-map} (\text{action-map-inv } b) = b$
 using *roundtrip-action-tgt* **by** *blast*
 show $\text{action-map} ' \text{actions}_s \subseteq \text{actions}_t$
 using *forward.action-map-well-defined* **by** *auto*
 show $\text{action-map-inv} ' \text{actions}_t \subseteq \text{actions}_s$
 using *backward.action-map-well-defined* **by** *auto*
qed

end

5 Multi-Domain State Preservation

Generalization to N domains sharing the same abstract state machine. This models scenarios where a single asset exists on multiple domains (e.g., blockchain networks), all of which must reflect the same state.

The key property: if one domain transitions, the synchronization brings ALL other domains to the same resulting state.

We model this as: all domains are instances of the same abstract state machine, connected by identity-like state/action maps. The "same abstract state machine" assumption holds when the protocol enforces a uniform state model across all domains.

locale *multi-domain-preservation* =
 fixes *domains* :: 'd set
 and *states* :: 's set
 and *actions* :: 'a set
 and *transition* :: 's $\Rightarrow$ 'a $\Rightarrow$'s option

```

and terminal :: 's set
and domain-state :: 'd  $\Rightarrow$  'id  $\Rightarrow$  's option
    — Current state of asset 'id in domain 'd
assumes fin-domains: finite domains
and sm: state-machine states actions transition terminal
and consistent-init:
    — Precondition: all domains holding an asset agree on its state
     $\llbracket d1 \in \text{domains}; d2 \in \text{domains};$ 
       $\text{domain-state } d1 \text{ aid} = \text{Some } s1; \text{domain-state } d2 \text{ aid} = \text{Some } s2 \rrbracket$ 
     $\Rightarrow s1 = s2$ 

```

begin

Which domains hold a given asset.

```

definition connected-domains :: 'id  $\Rightarrow$  'd set where
  connected-domains aid = {d  $\in$  domains. domain-state d aid  $\neq$  None}

```

The global consensus state of an asset (well-defined by **consistent_init**).

```

definition consensus-state :: 'id  $\Rightarrow$  's option where
  consensus-state aid =
    (if connected-domains aid = {} then None
     else domain-state (SOME d. d  $\in$  connected-domains aid) aid)

```

The finiteness of the domain roster bounds every connected set.

lemma *connected-domains-finite*: *finite (connected-domains aid)*

proof —

```

  have connected-domains aid  $\subseteq$  domains
  unfolding connected-domains-def by auto
  then show ?thesis using fin-domains by (rule finite-subset)

```

qed

The consensus state is well-defined: by **consistent_init** the choice of witnessing domain is immaterial, so the **SOME** in the definition is sound — every connected domain reports the consensus value.

lemma *consensus-state-well-defined*:

```

assumes d  $\in$  connected-domains aid
shows consensus-state aid = domain-state d aid

```

proof —

```

  have some-in: (SOME d'. d'  $\in$  connected-domains aid)  $\in$  connected-domains aid
    using assms by (rule someI)
  from assms have ne: connected-domains aid  $\neq$  {} by auto
  obtain s where s: domain-state d aid = Some s
    using assms unfolding connected-domains-def by auto
  obtain s' where s': domain-state (SOME d'. d'  $\in$  connected-domains aid) aid
    = Some s'
    using some-in unfolding connected-domains-def by auto
  have d  $\in$  domains (SOME d'. d'  $\in$  connected-domains aid)  $\in$  domains
    using assms some-in unfolding connected-domains-def by auto
  then have s' = s using consistent-init s s' by blast
  then show ?thesis

```

unfolding *consensus-state-def* **using** *ne s s'* **by** *simp*
qed

After synchronizing an action on an asset, all connected domains reflect the new state. This is the cross-domain consistency theorem. Connectivity (**connected_domains**) is read from the locale-fixed **domain_state**; the theorems below apply **sync_all** to that same map.

definition *sync-all* ::

$'d \Rightarrow 'a \Rightarrow 'id \Rightarrow ('d \Rightarrow 'id \Rightarrow 's \text{ option}) \Rightarrow ('d \Rightarrow 'id \Rightarrow 's \text{ option}) \text{ option}$

where

sync-all source action aid ds =
 (case *ds source aid* of
 None $\Rightarrow$ None
 | Some *s* $\Rightarrow$
 (case *transition s action* of
 None $\Rightarrow$ None
 | Some *s'* $\Rightarrow$
 Some ($\lambda d \text{ aid'}$.
 if *aid' = aid* $\wedge d \in \text{connected-domains aid}$
 then Some *s'*
 else *ds d aid'*)))

Terminal states admit no synchronization: the state-machine assumption **sm** transports terminal absorption to **sync_all**.

lemma *sync-all-terminal-none*:

assumes *domain-state source aid = Some s*

and *s* $\in$ *terminal* **and** *action* $\in$ *actions*

shows *sync-all source action aid domain-state = None*

proof –

have *transition s action = None*

using *state-machine.terminal-absorbing[OF sm assms(2) assms(3)]* .

then show *?thesis*

unfolding *sync-all-def* **using** *assms(1)* **by** *simp*

qed

The cross-domain consistency theorem: after **sync_all**, every connected domain agrees on the new state.

theorem *cross-domain-consistency*:

assumes *source* $\in$ *domains*

and *domain-state source aid = Some s*

and *s* $\in$ *states*

and *action* $\in$ *actions*

and *transition s action = Some s'*

and *sync-all source action aid domain-state = Some ds'*

and *d* $\in$ *connected-domains aid*

shows *ds' d aid = Some s'*

proof –

from *assms(2,5,6)* **have**

ds' = ($\lambda d \text{ aid'}$.


```

    if  $aid' = aid \wedge d \in \text{connected-domains } aid$ 
    then Some s'
    else domain-state d aid')
  unfolding sync-all-def by auto
  with assms(7) show ?thesis by simp
qed

sync_all does not affect other assets.

theorem sync-isolation:
  assumes sync-all source action aid domain-state = Some ds'
    and  $aid' \neq aid$ 
  shows  $ds' d aid' = \text{domain-state } d aid'$ 
proof -
  from assms(1) obtain s s' where
    domain-state source aid = Some s
    transition s action = Some s'
  unfolding sync-all-def by (auto split: option.splits)
  then have  $ds' = (\lambda d aid'. \text{domain-state } d aid')$ .
  if  $aid' = aid \wedge d \in \text{connected-domains } aid$ 
  then Some s'
  else domain-state d aid')
  using assms(1) unfolding sync-all-def by auto
  with assms(2) show ?thesis by auto
qed

end

end

```

```

theory Regulatory-Instance
  imports State-Preservation
begin

```

6 Regulatory State and Action Definitions

```

datatype reg-state = ACTIVE | FROZEN | SEIZED | CONFISCATED | RE-
STRICTED

```

```

datatype reg-action = FREEZE | SEIZE | CONFISCATE | RESTRICT
  | UNFREEZE | UNRESTRICT | RELEASE

```

The seven constructors are the four escalation actions (*FREEZE*, *SEIZE*, *CONFISCATE*, *RESTRICT*) and their de-escalation counterparts (*UNFREEZE*, *UNRESTRICT*, *RELEASE*). *RECOVER* and *LIQUIDATE*, two of the product's six enforcement actions, are excluded from *reg-action*: they involve force transfers or external DEX interactions rather than regulatory state transitions, and are modelled at a different layer.

The regulatory transition function. Partial: returns None for invalid (state, action) combinations. The transition table encodes legal precedence: seizure is stronger than freezing, confiscation is terminal and universally reachable, and intermediate states must return to Active before lateral transitions.

```
fun reg-transition :: reg-state  $\Rightarrow$  reg-action  $\Rightarrow$  reg-state option where
  — From ACTIVE: all forward actions valid
  | reg-transition ACTIVE FREEZE      = Some FROZEN
  | reg-transition ACTIVE SEIZE       = Some SEIZED
  | reg-transition ACTIVE CONFISCATE  = Some CONFISCATED
  | reg-transition ACTIVE RESTRICT    = Some RESTRICTED
  | reg-transition ACTIVE UNFREEZE    = None
  | reg-transition ACTIVE UNRESTRICT = None
  | reg-transition ACTIVE RELEASE     = None
  — From FROZEN: unfreeze, escalate to SEIZED or CONFISCATED
  | reg-transition FROZEN UNFREEZE    = Some ACTIVE
  | reg-transition FROZEN SEIZE       = Some SEIZED
  | reg-transition FROZEN CONFISCATE  = Some CONFISCATED
  | reg-transition FROZEN FREEZE     = None
  | reg-transition FROZEN RESTRICT    = None
  | reg-transition FROZEN UNRESTRICT = None
  | reg-transition FROZEN RELEASE     = None
  — From SEIZED: release or final confiscation only
  | reg-transition SEIZED RELEASE     = Some ACTIVE
  | reg-transition SEIZED CONFISCATE  = Some CONFISCATED
  | reg-transition SEIZED FREEZE     = None
  | reg-transition SEIZED SEIZE       = None
  | reg-transition SEIZED RESTRICT    = None
  | reg-transition SEIZED UNFREEZE    = None
  | reg-transition SEIZED UNRESTRICT = None
  — From CONFISCATED: terminal no transitions out
  | reg-transition CONFISCATED FREEZE = None
  | reg-transition CONFISCATED SEIZE  = None
  | reg-transition CONFISCATED CONFISCATE = None
  | reg-transition CONFISCATED RESTRICT = None
  | reg-transition CONFISCATED UNFREEZE = None
  | reg-transition CONFISCATED UNRESTRICT = None
  | reg-transition CONFISCATED RELEASE = None
  — From RESTRICTED: unrestrict, escalate to FROZEN or CONFISCATED
  | reg-transition RESTRICTED UNRESTRICT = Some ACTIVE
  | reg-transition RESTRICTED FREEZE     = Some FROZEN
  | reg-transition RESTRICTED CONFISCATE = Some CONFISCATED
  | reg-transition RESTRICTED RESTRICT  = None
  | reg-transition RESTRICTED SEIZE     = None
  | reg-transition RESTRICTED UNFREEZE  = None
  | reg-transition RESTRICTED RELEASE   = None
```

7 Fundamental Properties of the Transition Function

7.1 Invariant I1: CONFISCATED is terminal

lemma *confiscated-terminal*:
 reg-transition CONFISCATED a = None
 by (*cases a*) *auto*

7.2 Invariant I2: CONFISCATE reaches CONFISCATED from any non-terminal state

lemma *confiscate-universal*:
 $s \neq \text{CONFISCATED} \implies \text{reg-transition } s \text{ CONFISCATE} = \text{Some CONFISCATED}$
 by (*cases s*) *auto*

7.3 Invariant I3: Determinism (inherent from function definition)

Determinism follows directly from the **fun** definition.

7.4 SEIZED to FROZEN exclusion

SEIZED is a strictly stronger constraint than FROZEN. Direct transition from SEIZED to FROZEN is legally nonsensical (cannot "weaken" a court-ordered seizure to a mere freeze). The path is: RELEASE then ACTIVE then FREEZE.

lemma *seized-no-freeze*:
 reg-transition SEIZED FREEZE = None
 by *simp*

lemma *seized-no-unfreeze*:
 reg-transition SEIZED UNFREEZE = None
 by *simp*

lemma *seized-no-restrict*:
 reg-transition SEIZED RESTRICT = None
 by *simp*

7.5 FROZEN to RESTRICTED exclusion

Must go through ACTIVE: UNFREEZE then ACTIVE then RESTRICT.

lemma *frozen-no-restrict*:
 reg-transition FROZEN RESTRICT = None
 by *simp*

7.6 Transition completeness: every non-terminal state has at least one valid action

```

lemma non-terminal-has-action:
  assumes  $s \neq \text{CONFISCATED}$ 
  shows  $\exists a. \text{reg-transition } s \ a \neq \text{None}$ 
proof (cases  $s$ )
  case ACTIVE
    show ?thesis
    apply (rule exI[where  $x = \text{FREEZE}$ ])
    using ACTIVE by simp
  next
    case FROZEN
    show ?thesis
    apply (rule exI[where  $x = \text{UNFREEZE}$ ])
    using FROZEN by simp
  next
    case SEIZED
    show ?thesis
    apply (rule exI[where  $x = \text{RELEASE}$ ])
    using SEIZED by simp
  next
    case CONFISCATED
    then show ?thesis using assms by contradiction
  next
    case RESTRICTED
    show ?thesis
    apply (rule exI[where  $x = \text{UNRESTRICT}$ ])
    using RESTRICTED by simp
qed

```

7.7 Reachability from ACTIVE

Every state is reachable from ACTIVE via some sequence of actions.

```

lemma active-reaches-frozen: reg-transition ACTIVE FREEZE = Some FROZEN
  by simp

```

```

lemma active-reaches-seized: reg-transition ACTIVE SEIZE = Some SEIZED
  by simp

```

```

lemma active-reaches-confiscated: reg-transition ACTIVE CONFISCATE = Some
CONFISCATED
  by simp

```

```

lemma active-reaches-restricted: reg-transition ACTIVE RESTRICT = Some RE-
STRICTED
  by simp

```

7.8 Return paths to ACTIVE

lemma *frozen-returns: reg-transition FROZEN UNFREEZE = Some ACTIVE*
by *simp*

lemma *seized-returns: reg-transition SEIZED RELEASE = Some ACTIVE*
by *simp*

lemma *restricted-returns: reg-transition RESTRICTED UNRESTRICT = Some ACTIVE*
by *simp*

8 State Machine Instance

We show that `reg_state` / `reg_action` / `reg_transition` satisfy the `state_machine` locale assumptions.

definition *reg-states :: reg-state set where*
reg-states = {ACTIVE, FROZEN, SEIZED, CONFISCATED, RESTRICTED}

definition *reg-actions :: reg-action set where*
reg-actions = {FREEZE, SEIZE, CONFISCATE, RESTRICT, UNFREEZE, UNRESTRICT, RELEASE}

definition *reg-terminal :: reg-state set where*
reg-terminal = {CONFISCATED}

lemma *reg-states-UNIV: reg-states = UNIV*
unfolding *reg-states-def by (auto, case-tac x, auto)*

lemma *reg-actions-UNIV: reg-actions = UNIV*
unfolding *reg-actions-def by (auto, case-tac x, auto)*

lemma *reg-transition-closed:*
 $\llbracket \text{reg-transition } s \ a = \text{Some } s' \rrbracket \implies s' \in \text{reg-states}$
unfolding *reg-states-def by (cases s; cases a; auto)*

interpretation *reg-sm: state-machine reg-states reg-actions reg-transition reg-terminal*

proof *unfold-locales*

show *finite reg-states unfolding reg-states-def by auto*

next

show *finite reg-actions unfolding reg-actions-def by auto*

next

show *reg-terminal $\subseteq$ reg-states unfolding reg-terminal-def reg-states-def by auto*

next

fix *s a*

assume *s $\in$ reg-terminal a $\in$ reg-actions*

then show *reg-transition s a = None*

unfolding *reg-terminal-def by (auto simp: confiscated-terminal)*

```

next
  fix s a s'
  assume s ∈ reg-states a ∈ reg-actions reg-transition s a = Some s'
  then show s' ∈ reg-states
    using reg-transition-closed by auto
next
  fix s :: reg-state and a :: reg-action
  assume s ∉ reg-states
  then show reg-transition s a = None
    using reg-states-UNIV by auto
qed

```

9 Asset and Global State Modeling

```

type-synonym asset-id = nat
type-synonym chain-id = nat
type-synonym timestamp = nat

```

```

record asset-state =
  as-reg-state :: reg-state

```

```

type-synonym chain-state = asset-id ⇒ asset-state option

```

```

record global-state =
  gs-chains    :: chain-id ⇒ chain-state
  gs-locks     :: asset-id ⇒ bool

```

The synchronization message, reduced to the fields this model consumes: the regulatory action and its timestamp. The product's full message also carries routing and attribution data (asset, authority, source, targets); in this model routing lives at the global-state level (`connected_chains` and the target set of `update_all_chains`), and the D-quencer layer adds exactly the attribution fields its priority order consumes (see `dq_message`).

```

record oss-message =
  msg-action    :: reg-action
  msg-timestamp :: timestamp

```

10 State Accessors and Predicates

```

definition get-asset-state :: global-state ⇒ chain-id ⇒ asset-id ⇒ asset-state option
where
  get-asset-state gs cid aid = gs-chains gs cid aid

```

```

definition get-reg-state :: global-state ⇒ chain-id ⇒ asset-id ⇒ reg-state option
where
  get-reg-state gs cid aid =
    (case get-asset-state gs cid aid of
      None ⇒ None

```

| *Some ast* $\Rightarrow$ *Some (as-reg-state ast)*)

definition *asset-exists* :: *global-state* $\Rightarrow$ *chain-id* $\Rightarrow$ *asset-id* $\Rightarrow$ *bool* **where**
asset-exists *gs* *cid* *aid* = (*get-asset-state* *gs* *cid* *aid* $\neq$ *None*)

definition *is-locked* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *bool* **where**
is-locked *gs* *aid* = *gs-locks* *gs* *aid*

Connected chains: all chains holding a given asset.

definition *connected-chains* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *chain-id* *set* **where**
connected-chains *gs* *aid* = {*cid*. *asset-exists* *gs* *cid* *aid*}

11 Global State Validity

A global state is valid if all chains holding the same asset agree on its regulatory state. This is the cross-chain consistency invariant that synchronization must preserve.

definition *consistent-state* :: *global-state* $\Rightarrow$ *bool* **where**

consistent-state *gs* $\equiv$
 $\forall$ *c1* *c2* *aid* *s1* *s2*.
get-reg-state *gs* *c1* *aid* = *Some* *s1* $\longrightarrow$
get-reg-state *gs* *c2* *aid* = *Some* *s2* $\longrightarrow$
s1 = *s2*

definition *no-locked-without-reason* :: *global-state* $\Rightarrow$ *bool* **where**

no-locked-without-reason *gs* $\equiv$
 $\forall$ *aid*. $\neg$ *is-locked* *gs* *aid*

— Total absence of outstanding locks: in a quiescent valid state no asset is locked at all (locks are transient, held only inside a synchronization), not merely no unjustified lock.

definition *valid-state* :: *global-state* $\Rightarrow$ *bool* **where**

valid-state *gs* $\equiv$ *consistent-state* *gs* $\wedge$ *no-locked-without-reason* *gs*

12 Synchronization Protocol

12.1 Lock acquisition

definition *acquire-lock* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state* *option* **where**

acquire-lock *gs* *aid* =
 (if *is-locked* *gs* *aid* then *None*
 else *Some* (*gs* | *gs-locks* := (*gs-locks* *gs*)(*aid* := *True*) |))

definition *release-lock* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state* **where**

release-lock *gs* *aid* = *gs* | *gs-locks* := (*gs-locks* *gs*)(*aid* := *False*) |

12.2 State update across chains

Update all connected chains to a new regulatory state.

We define this directly rather than using `Finite_Set.fold`, because the update for a given asset on each chain is independent we simply construct a new global state where every chain's view of the asset is updated. This avoids the need for `comp_fun_commute` proofs.

definition *update-all-chains* ::

global-state $\Rightarrow$ *asset-id* $\Rightarrow$ *reg-state* $\Rightarrow$ *chain-id set* $\Rightarrow$ *global-state*

where

update-all-chains *gs* *aid* *new-st* *targets* =
gs | *gs-chains* := (λ *cid*.
 if *cid* $\in$ *targets* then
 (λ *aid'*. if *aid'* = *aid* then
 (case *gs-chains* *gs* *cid* *aid* of
 None $\Rightarrow$ None
 | Some *ast* $\Rightarrow$ Some (*ast* | *as-reg-state* := *new-st* |))
 else *gs-chains* *gs* *cid* *aid'*)
 else *gs-chains* *gs* *cid*) |)

12.3 The synchronization function

sync: the core synchronization operation.

Given a source chain, action, and `asset_id`: 1. Verify the asset exists on the source chain 2. Check the transition is valid 3. Acquire global lock (fails if already locked) 4. Update all connected chains to the new state 5. Release lock

Returns None if any step fails (invalid transition, asset already locked, etc.)

definition *sync* ::

chain-id $\Rightarrow$ *reg-action* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state* $\Rightarrow$ *global-state option*

where

sync *source* *action* *aid* *gs* =
 (case *get-reg-state* *gs* *source* *aid* of
 None $\Rightarrow$ None
 | Some *current-st* $\Rightarrow$
 (case *reg-transition* *current-st* *action* of
 None $\Rightarrow$ None
 | Some *new-st* $\Rightarrow$
 (case *acquire-lock* *gs* *aid* of
 None $\Rightarrow$ None
 | Some *gs-locked* $\Rightarrow$
 let *targets* = *connected-chains* *gs* *aid* in
 let *gs-updated* = *update-all-chains* *gs-locked* *aid* *new-st* *targets* in
 Some (*release-lock* *gs-updated* *aid*))))

13 Synchronization Properties

13.1 No self-loops

The transition table has no self-loops: a successful transition always changes the regulatory state, so a successful synchronization never rewrites an asset to the state it already carries.

lemma *no-self-loops*:

reg-transition $s \ a = \text{Some } s \implies \text{False}$
by (*cases* s ; *cases* a ; *auto*)

lemma *transition-changes-state*:

reg-transition $s \ a = \text{Some } s' \implies s \neq s'$
using *no-self-loops* **by** *auto*

13.2 Properties of `update_all_chains`

`update_all_chains` correctly updates any chain in the target set.

lemma *update-all-chains-in-targets*:

assumes $cid \in \text{targets}$
and *gs-chains* $gs \ cid \ aid = \text{Some } ast$
shows *gs-chains* (*update-all-chains* $gs \ aid \ \text{new-st} \ \text{targets}$) $cid \ aid =$
 $\text{Some } (ast \sqcup \text{as-reg-state} := \text{new-st} \sqcup)$
unfolding *update-all-chains-def* **using** *assms* **by** *auto*

lemma *update-all-chains-outside-targets*:

assumes $cid \notin \text{targets}$
shows *gs-chains* (*update-all-chains* $gs \ aid \ \text{new-st} \ \text{targets}$) $cid = \text{gs-chains } gs \ cid$
unfolding *update-all-chains-def* **using** *assms* **by** *auto*

lemma *update-all-chains-other-asset*:

assumes $aid' \neq aid$
shows *gs-chains* (*update-all-chains* $gs \ aid \ \text{new-st} \ \text{targets}$) $cid \ aid' =$
 $\text{gs-chains } gs \ cid \ aid'$
unfolding *update-all-chains-def* **using** *assms* **by** *auto*

lemma *update-all-chains-locks*:

gs-locks (*update-all-chains* $gs \ aid \ \text{new-st} \ \text{targets}$) $= \text{gs-locks } gs$
unfolding *update-all-chains-def* **by** *auto*

Key lemma: `get_reg_state` after `update_all_chains` for a chain in targets.

lemma *update-all-chains-reg-state*:

assumes $cid \in \text{targets}$
and *get-asset-state* $gs \ cid \ aid = \text{Some } ast$
shows *get-reg-state* (*update-all-chains* $gs \ aid \ \text{new-st} \ \text{targets}$) $cid \ aid = \text{Some}$
new-st
proof –

```

from assms(2) have chain: gs-chains gs cid aid = Some ast
  unfolding get-asset-state-def by simp
then have gs-chains (update-all-chains gs aid new-st targets) cid aid =
  Some (ast (as-reg-state := new-st))
  using update-all-chains-in-targets[OF assms(1)] by simp
then show ?thesis
  unfolding get-reg-state-def get-asset-state-def by simp
qed

```

13.3 Lock mutual exclusion

If a lock is held, a further lock acquisition on the same asset fails (*acquire_lock* returns *None*). In the atomic model this guard expresses the intended mutual exclusion of regulatory actions on an asset; there is no concurrent execution in the model to serialise.

```

lemma lock-acquire-success:
  assumes  $\neg$  is-locked gs aid
  shows  $\exists$  gs'. acquire-lock gs aid = Some gs'  $\wedge$  is-locked gs' aid
proof –
  from assms have  $\neg$  gs-locks gs aid
  unfolding is-locked-def by simp
  then have acquire-lock gs aid = Some (gs (gs-locks := (gs-locks gs)(aid := True)
  ))
  unfolding acquire-lock-def is-locked-def by simp
  moreover have is-locked (gs (gs-locks := (gs-locks gs)(aid := True) )) aid
  unfolding is-locked-def by simp
  ultimately show ?thesis by auto
qed

```

14 Main Theorem: Cross-Chain Regulatory Consistency

The central theorem of this theory. If a regulatory action is applied via *sync* — from an arbitrary global state — every connected chain reflects the new regulatory state; under a valid input state, validity is preserved as well (*valid_state_preservation* below).

This is the regulatory instantiation of the cross-domain consistency theorem from *State_Preservation.thy*.

The proof proceeds by: 1. Unfolding *sync* into *lock*, then *update_all_chains*, then *unlock* 2. Showing *acquire_lock* and *release_lock* preserve chain state 3. Showing *update_all_chains* updates every chain in the target set 4. Combining these facts to establish the conclusion

Auxiliary: *release_lock* does not change chains.

```

lemma release-lock-chains:
  gs-chains (release-lock gs aid) = gs-chains gs

```

unfolding *release-lock-def* **by** *simp*

lemma *release-lock-reg-state*:

get-reg-state (release-lock gs aid) cid aid' = get-reg-state gs cid aid'

unfolding *get-reg-state-def get-asset-state-def release-lock-chains* **by** *simp*

Auxiliary: **acquire_lock** does not change chains.

lemma *acquire-lock-chains*:

assumes *acquire-lock gs aid = Some gs'*

shows *gs-chains gs' = gs-chains gs*

using *assms* **unfolding** *acquire-lock-def is-locked-def*

by (*auto split: if-splits*)

Auxiliary: **connected_chains** uses original gs, not locked gs.

The cross-chain regulatory homomorphism theorem.

theorem *regulatory-homomorphism*:

assumes *current: get-reg-state gs source aid = Some s*

and *trans: reg-transition s action = Some s'*

and *synced: sync source action aid gs = Some gs'*

and *connected: c ∈ connected-chains gs aid*

shows *get-reg-state gs' c aid = Some s'*

proof —

— Step 1: Unfold sync to extract intermediate states

from *synced current trans*

obtain *gs-locked* **where**

lock: acquire-lock gs aid = Some gs-locked

and *gs'-def: gs' = release-lock*

(update-all-chains gs-locked aid s' (connected-chains gs aid)) aid

unfolding *sync-def*

by (*auto split: option.splits simp: Let-def*)

— Step 2: **acquire_lock** preserves chains

have *locked-chains: gs-chains gs-locked = gs-chains gs*

using *acquire-lock-chains[OF lock]* **by** *simp*

— Step 3: *c* is in **connected_chains**, so the asset exists on *c*

from *connected* **have** *asset-exists gs c aid*

unfolding *connected-chains-def* **by** *simp*

then obtain *ast-c* **where** *ast-c: gs-chains gs c aid = Some ast-c*

unfolding *asset-exists-def get-asset-state-def* **by** *auto*

— Step 4: Therefore asset also exists in **gs_locked** on chain *c*

have *ast-locked: gs-chains gs-locked c aid = Some ast-c*

using *ast-c locked-chains* **by** *simp*

— Step 5: **update_all_chains** updates chain *c*

have *updated: gs-chains (update-all-chains gs-locked aid s' (connected-chains gs aid)) c aid =*

Some (ast-c[| as-reg-state := s' |])

```

using update-all-chains-in-targets[OF connected ast-locked] by simp

— Step 6: release_lock does not change chains
have gs-chains gs' c aid = Some (ast-c( as-reg-state := s' ))
using updated unfolding gs'-def release-lock-chains by simp

— Step 7: Extract reg_state
then show ?thesis
unfolding get-reg-state-def get-asset-state-def by simp
qed

```

15 Valid State Preservation

The central safety property: synchronization preserves global state validity.

If the global state is valid (consistent and unlocked) before sync, and sync succeeds, then the resulting global state is also valid.

This closes the inductive invariant: valid initial state + *valid_state* preservation under sync = *valid_state* holds at all reachable states.

The proof decomposes into two parts: (1) *consistent_state* preservation: sync updates all connected chains to the same new state and leaves other assets untouched. (2) *no_locked_without_reason* preservation: sync acquires and then releases the lock within the same operation, so the output has no outstanding locks (assuming no other locks were held).

15.1 Part 1: *consistent_state* preservation

After sync, any two chains holding asset aid agree on its new regulatory state (it is *s'*), and any two chains holding a different asset aid' still agree (their states are unchanged from the consistent input).

lemma *sync-preserves-consistent-state*:

```

assumes valid: valid-state gs
and current: get-reg-state gs source aid = Some s
and trans: reg-transition s action = Some s'
and synced: sync source action aid gs = Some gs'
shows consistent-state gs'
proof —
  — Unfold sync to extract the structure
from synced current trans
obtain gs-locked where
  lock: acquire-lock gs aid = Some gs-locked
and gs'-def: gs' = release-lock
  (update-all-chains gs-locked aid s' (connected-chains gs aid)) aid
unfolding sync-def
by (auto split: option.splits simp: Let-def)

have locked-chains: gs-chains gs-locked = gs-chains gs

```

```

using acquire-lock-chains[OF lock] by simp

show ?thesis
unfolding consistent-state-def
proof (intro allI impI)
  fix c1 c2 aid' s1-final s2-final
  assume s1-eq: get-reg-state gs' c1 aid' = Some s1-final
    and s2-eq: get-reg-state gs' c2 aid' = Some s2-final

  show s1-final = s2-final
  proof (cases aid' = aid)
    case True
    — Case: the synchronized asset. Two sub-cases per chain.
    show ?thesis
    proof (cases c1 ∈ connected-chains gs aid)
      case c1-in: True
      then have ae1: asset-exists gs c1 aid
      unfolding connected-chains-def by simp
      then obtain ast1 where ast1: gs-chains gs c1 aid = Some ast1
      unfolding asset-exists-def get-asset-state-def by auto
      have ast1-locked: gs-chains gs-locked c1 aid = Some ast1
      using ast1 locked-chains by simp
      have upd1: gs-chains (update-all-chains gs-locked aid s' (connected-chains
gs aid)) c1 aid =
        Some (ast1() as-reg-state := s' ))
      using update-all-chains-in-targets[OF c1-in ast1-locked] by simp
      have reg1: get-reg-state gs' c1 aid = Some s'
      unfolding gs'-def release-lock-chains get-reg-state-def get-asset-state-def
      using upd1 by simp
      show ?thesis
      proof (cases c2 ∈ connected-chains gs aid)
        case c2-in: True
        then have ae2: asset-exists gs c2 aid
        unfolding connected-chains-def by simp
        then obtain ast2 where ast2: gs-chains gs c2 aid = Some ast2
        unfolding asset-exists-def get-asset-state-def by auto
        have ast2-locked: gs-chains gs-locked c2 aid = Some ast2
        using ast2 locked-chains by simp
        have reg2: get-reg-state gs' c2 aid = Some s'
        unfolding gs'-def release-lock-chains get-reg-state-def get-asset-state-def
        using update-all-chains-in-targets[OF c2-in ast2-locked] by simp
        from reg1 reg2 s1-eq s2-eq True show ?thesis by simp
      next
      case c2-out: False
      — c2 is not connected, so asset doesn't exist on c2 for aid
      have gs-chains (update-all-chains gs-locked aid s' (connected-chains gs
aid)) c2 =
        gs-chains gs-locked c2
      using update-all-chains-outside-targets[OF c2-out] by simp

```

```

    then have gs-chains gs' c2 aid = gs-chains gs c2 aid
      unfolding gs'-def release-lock-chains using locked-chains by simp
    moreover have gs-chains gs c2 aid = None
      using c2-out unfolding connected-chains-def asset-exists-def
get-asset-state-def
    by auto
    ultimately have get-reg-state gs' c2 aid = None
      unfolding get-reg-state-def get-asset-state-def by simp
    with s2-eq True show ?thesis by simp
  qed
next
case c1-out: False
  — c1 not connected: asset doesn't exist there
  have gs-chains (update-all-chains gs-locked aid s' (connected-chains gs aid))
c1 =
    gs-chains gs-locked c1
    using update-all-chains-outside-targets[OF c1-out] by simp
  then have gs-chains gs' c1 aid = gs-chains gs c1 aid
    unfolding gs'-def release-lock-chains using locked-chains by simp
  moreover have gs-chains gs c1 aid = None
    using c1-out unfolding connected-chains-def asset-exists-def
get-asset-state-def
  by auto
  ultimately have get-reg-state gs' c1 aid = None
    unfolding get-reg-state-def get-asset-state-def by simp
  with s1-eq True show ?thesis by simp
  qed
next
case diff: False
  — Case: a different asset. Sync does not touch it.
  have other1: gs-chains (update-all-chains gs-locked aid s' (connected-chains
gs aid)) c1 aid' =
    gs-chains gs-locked c1 aid'
    using update-all-chains-other-asset[OF diff] by simp
  have other2: gs-chains (update-all-chains gs-locked aid s' (connected-chains
gs aid)) c2 aid' =
    gs-chains gs-locked c2 aid'
    using update-all-chains-other-asset[OF diff] by simp
  have gs-chains gs' c1 aid' = gs-chains gs c1 aid'
    unfolding gs'-def release-lock-chains using other1 locked-chains by simp
  moreover have gs-chains gs' c2 aid' = gs-chains gs c2 aid'
    unfolding gs'-def release-lock-chains using other2 locked-chains by simp
  ultimately have get-reg-state gs' c1 aid' = get-reg-state gs c1 aid'
    and get-reg-state gs' c2 aid' = get-reg-state gs c2 aid'
    unfolding get-reg-state-def get-asset-state-def by simp-all
  with s1-eq s2-eq have
    get-reg-state gs c1 aid' = Some s1-final
    get-reg-state gs c2 aid' = Some s2-final
    by simp-all

```

```

    with valid show ?thesis
    unfolding valid-state-def consistent-state-def by blast
  qed
qed
qed

```

15.2 Part 2: no_locked_without_reason preservation

After sync, no assets are locked. The sync operation acquires a lock on the target asset and releases it before returning. For all other assets, the lock state is unchanged (and was False by `valid_state`).

lemma *sync-preserves-no-locks*:

```

  assumes valid: valid-state gs
    and current: get-reg-state gs source aid = Some s
    and trans: reg-transition s action = Some s'
    and synced: sync source action aid gs = Some gs'
  shows no-locked-without-reason gs'
proof -
  from synced current trans
  obtain gs-locked where
    lock: acquire-lock gs aid = Some gs-locked
    and gs'-def: gs' = release-lock
      (update-all-chains gs-locked aid s' (connected-chains gs aid)) aid
  unfolding sync-def
  by (auto split: option.splits simp: Let-def)

```

— Locks after acquire: only aid is locked

```

from lock have locked-locks: gs-locks gs-locked = (gs-locks gs)(aid := True)
  unfolding acquire-lock-def is-locked-def
  by (auto split: if-splits)

```

— `update_all_chains` does not change locks

```

have upd-locks: gs-locks (update-all-chains gs-locked aid s' (connected-chains gs
aid)) =
  gs-locks gs-locked
  using update-all-chains-locks by simp

```

— `release_lock` clears the lock on aid

```

have final-locks: gs-locks gs' = (gs-locks gs-locked)(aid := False)
  unfolding gs'-def release-lock-def using upd-locks by simp

```

show *?thesis*

unfolding *no-locked-without-reason-def is-locked-def*

proof

fix aid'

show $\neg$ *gs-locks gs' aid'*

proof (*cases aid' = aid*)

case True

then show *?thesis* using *final-locks* by simp

```

next
  case False
  then have gs-locks gs' aid' = gs-locks gs-locked aid'
    using final-locks by simp
  also have ... = gs-locks gs aid'
    using locked-locks False by simp
  also have ... = False
    using valid unfolding valid-state-def no-locked-without-reason-def
is-locked-def
    by simp
  finally show ?thesis by simp
qed
qed
qed

```

15.3 The `valid_state` preservation theorem

Combining the two parts: `sync` preserves `valid_state`. This is the inductive invariant that guarantees `valid_state` holds at every reachable global state.

theorem *valid-state-preservation:*

```

assumes valid: valid-state gs
  and current: get-reg-state gs source aid = Some s
  and trans: reg-transition s action = Some s'
  and synced: sync source action aid gs = Some gs'
shows valid-state gs'
unfolding valid-state-def
using sync-preserves-consistent-state[OF assms]
  sync-preserves-no-locks[OF assms]
by auto

```

16 Heterogeneous-Action State Preservation Instance

This section instantiates the `state_preservation` locale with a concrete heterogeneous-action scenario between two chains.

The scenario models the following operational situation. Two chains A and B share the same regulatory state space (ACTIVE, FROZEN, SEIZED, CONFISCATED, RESTRICTED), but their on-chain action vocabularies differ. Chain A supports the full seven-action set used elsewhere in this theory. Chain B is a chain whose jurisdiction handles de-escalation (UNFREEZE, UNRESTRICT, RELEASE) exclusively through separate judicial procedures rather than as on-chain actions; correspondingly, Chain B's on-chain action vocabulary is the four-action escalation subset only. The synchronisation map between the two chains is therefore non-trivial in two ways:

- The source action type is `reg_action` and the target action type is a separate datatype `chain_b_action` with four constructors. This exercises the locale's heterogeneous source / target action types ('a vs. 'b).
- The locale parameter `actions_s` is instantiated with a strict subset of the full `reg_action` set, namely the four escalation actions. De-escalation actions exist in `reg_action` as a datatype but are out of scope for this synchronisation instance.

The naturality assumption then has to hold only on the escalation subset, which is exactly the design intent of the locale's `actions_s` parameter: the user can scope structural preservation to a subset of source actions rather than to the full source action type.

datatype *chain-b-action* = *B-FREEZE* | *B-SEIZE* | *B-CONFISCATE* | *B-RESTRICT*

The escalation subset of `reg_action` that this instance synchronises across to Chain B.

definition *escalation-actions* :: *reg-action* set **where**
escalation-actions = {*FREEZE*, *SEIZE*, *CONFISCATE*, *RESTRICT*}

definition *chain-b-actions* :: *chain-b-action* set **where**
chain-b-actions = {*B-FREEZE*, *B-SEIZE*, *B-CONFISCATE*, *B-RESTRICT*}

Chain B's transition function. Its values mirror Chain A's transition function on the four escalation actions: when Chain A maps (`s`, `escalation action`) to `Some s'`, Chain B does the same; when Chain A rejects, Chain B rejects. Both chains share the regulatory state space, so no state translation is required (the state map is the identity).

fun *chain-b-transition* :: *reg-state* $\Rightarrow$ *chain-b-action* $\Rightarrow$ *reg-state option* **where**
chain-b-transition *s* *B-FREEZE* = *reg-transition* *s* *FREEZE*
| *chain-b-transition* *s* *B-SEIZE* = *reg-transition* *s* *SEIZE*
| *chain-b-transition* *s* *B-CONFISCATE* = *reg-transition* *s* *CONFISCATE*
| *chain-b-transition* *s* *B-RESTRICT* = *reg-transition* *s* *RESTRICT*

The escalation action map: a 1-to-1 correspondence between Chain A's four escalation actions and Chain B's four constructors.

fun *escalation-action-map* :: *reg-action* $\Rightarrow$ *chain-b-action* **where**
escalation-action-map *FREEZE* = *B-FREEZE*
| *escalation-action-map* *SEIZE* = *B-SEIZE*
| *escalation-action-map* *CONFISCATE* = *B-CONFISCATE*
| *escalation-action-map* *RESTRICT* = *B-RESTRICT*
| *escalation-action-map* *UNFREEZE* = *B-FREEZE*
| *escalation-action-map* *UNRESTRICT* = *B-FREEZE*
| *escalation-action-map* *RELEASE* = *B-FREEZE*

— The last three clauses are unused by the locale instance, since `actions_s = escalation_actions` excludes the de-escalation actions. They are present only to make the function total on `reg_action`.

Chain B is also a state machine over the same regulatory state space.

lemma *chain-b-transition-closed*:

chain-b-transition s a = Some s' $\implies$ s' $\in$ reg-states

proof —

assume *chain-b-transition s a = Some s'*

then have $\exists a'. \text{reg-transition } s \ a' = \text{Some } s'$

by (*cases a*) *auto*

then show $s' \in \text{reg-states}$ **using** *reg-transition-closed* **by** *auto*

qed

lemma *chain-b-terminal*:

s $\in$ reg-terminal $\implies$ chain-b-transition s a = None

unfolding *reg-terminal-def*

by (*cases a*) (*auto simp: confiscated-terminal*)

The naturality conditions of `state_preservation` hold for the escalation subset. Both directions (`Some` and `None` branches) follow by case analysis on the escalation action: by construction, Chain B's transition function on each `B_X` constructor mirrors Chain A's transition function on the corresponding `X` action.

lemma *escalation-naturality-some*:

assumes *a $\in$ escalation-actions*

and *reg-transition s a = Some s'*

shows *chain-b-transition s (escalation-action-map a) = Some s'*

proof —

from *assms(1)* **have** $a \in \{\text{FREEZE}, \text{SEIZE}, \text{CONFISCATE}, \text{RESTRICT}\}$

unfolding *escalation-actions-def* **by** *simp*

then consider $a = \text{FREEZE} \mid a = \text{SEIZE} \mid a = \text{CONFISCATE} \mid a = \text{RESTRICT}$

by *auto*

then show *?thesis* **using** *assms(2)* **by** (*cases*) *auto*

qed

lemma *escalation-naturality-none*:

assumes *a $\in$ escalation-actions*

and *reg-transition s a = None*

shows *chain-b-transition s (escalation-action-map a) = None*

proof —

from *assms(1)* **have** $a \in \{\text{FREEZE}, \text{SEIZE}, \text{CONFISCATE}, \text{RESTRICT}\}$

unfolding *escalation-actions-def* **by** *simp*

then consider $a = \text{FREEZE} \mid a = \text{SEIZE} \mid a = \text{CONFISCATE} \mid a = \text{RESTRICT}$

by *auto*

then show *?thesis* **using** *assms(2)* **by** (*cases*) *auto*

qed

The escalation instance: `state_preservation` with `actions_s` the escalation subset of `reg_action`, target action type `chain_b_action`, and identity state map (both chains share the regulatory state space).

Both state-machine sides of the heterogeneous-action instance — the source on the escalation subset of `reg_actions` and Chain B's target — have their obligations discharged inline by *unfold-locales* in the interpretation below, drawing on the closure and terminal lemmas above.

Heterogeneous-action instance: the source action set is the `escalation_actions` subset of `reg_action`, the target action set is `chain_b_actions`, and the state map is the identity. Both source and target state machines share the regulatory state space, so two of the state-machine obligations (`finite reg_states` and `reg_terminal \<subsepeq> reg_states`) collapse between source and target; the proof structure below reflects this by issuing each `show` exactly once for the merged obligation.

interpretation *escalation-preservation*:

```

state-preservation
  reg-states escalation-actions reg-transition reg-terminal
  reg-states chain-b-actions chain-b-transition reg-terminal
  id :: reg-state ⇒ reg-state escalation-action-map
proof unfold-locales
  show finite reg-states unfolding reg-states-def by auto
next
  show finite escalation-actions unfolding escalation-actions-def by auto
next
  show reg-terminal ⊆ reg-states unfolding reg-terminal-def reg-states-def by auto
next
  fix s a
  assume s ∈ reg-terminal a ∈ escalation-actions
  then show reg-transition s a = None
    unfolding reg-terminal-def by (auto simp: confiscated-terminal)
next
  fix s a s'
  assume s ∈ reg-states a ∈ escalation-actions reg-transition s a = Some s'
  then show s' ∈ reg-states using reg-transition-closed by auto
next
  fix s :: reg-state and a :: reg-action
  assume s ∉ reg-states
  then show reg-transition s a = None using reg-states-UNIV by auto
next
  show finite chain-b-actions unfolding chain-b-actions-def by auto
next
  fix s a
  assume s ∈ reg-terminal a ∈ chain-b-actions
  then show chain-b-transition s a = None using chain-b-terminal by auto
next
  fix s a s'
  assume s ∈ reg-states a ∈ chain-b-actions chain-b-transition s a = Some s'

```

```

    then show  $s' \in \text{reg-states}$  using chain-b-transition-closed by auto
next
  fix  $s :: \text{reg-state}$  and  $a :: \text{chain-b-action}$ 
  assume  $s \notin \text{reg-states}$ 
  then show  $\text{chain-b-transition } s \ a = \text{None}$  using reg-states-UNIV by auto
next
  fix  $s$ 
  assume  $s \in \text{reg-states}$ 
  then show  $\text{id } s \in \text{reg-states}$  by simp
next
  fix  $a$ 
  assume  $a \in \text{escalation-actions}$ 
  then show  $\text{escalation-action-map } a \in \text{chain-b-actions}$ 
    unfolding escalation-actions-def chain-b-actions-def by auto
next
  fix  $s$ 
  assume  $s \in \text{reg-terminal}$ 
  then show  $\text{id } s \in \text{reg-terminal}$  by simp
next
  fix  $s \ a \ s'$ 
  assume  $s \in \text{reg-states } a \in \text{escalation-actions } \text{reg-transition } s \ a = \text{Some } s'$ 
  then show  $\text{chain-b-transition } (\text{id } s) (\text{escalation-action-map } a) = \text{Some } (\text{id } s')$ 
    using escalation-naturality-some by simp
next
  fix  $s \ a$ 
  assume  $s \in \text{reg-states } a \in \text{escalation-actions } \text{reg-transition } s \ a = \text{None}$ 
  then show  $\text{chain-b-transition } (\text{id } s) (\text{escalation-action-map } a) = \text{None}$ 
    using escalation-naturality-none by simp
qed

```

As a corollary of the locale interpretation, sequential preservation (the locale's main theorem) applies to escalation action sequences: any valid sequence of escalation actions on Chain A produces, when mapped through `escalation_action_map`, a valid sequence on Chain B ending in the same regulatory state.

corollary *escalation-sequential-preservation*:

```

  assumes  $s \in \text{reg-states}$ 
    and  $\forall a \in \text{set } as. \ a \in \text{escalation-actions}$ 
    and  $\text{escalation-preservation.source.apply-actions } s \ as = \text{Some } s'$ 
  shows  $\text{escalation-preservation.target.apply-actions } s \ (\text{map } \text{escalation-action-map } as)$ 
    =  $\text{Some } s'$ 
  using assms escalation-preservation.sequential-preservation [where  $as = as$ ]
  by simp

```

17 Layer-Crossing Symmetric State Preservation Instance

This section instantiates the `symmetric_state_preservation` locale with a concrete bidirectional binding between two representations of the same regulatory state.

The on-chain representation is the five-element `reg_state` enum used throughout this theory. The off-chain representation is a structured permission record (`daml_perm`) modelling a DAML party-permission ledger entry: a status tag plus auxiliary fields that carry the party / scope metadata associated with a non-trivial regulatory status (the seizing party for `SEIZED`, the restriction scope for `RESTRICTED`). The identifier values themselves are representative placeholders; what the invariant tracks is their presence, and the layer-crossing content lives in the status tag. A type-level invariant (`valid_daml_perm`) ties the auxiliary fields to the status tag; the bijection domain itself is the image of the representation map (`daml_states`), every element of which satisfies the invariant.

The action vocabularies on the two layers coincide: both layers process the seven actions of `reg_action`, and the action map is the identity. All of the non-trivial content of the symmetric instance therefore lives in the layer-crossing state mapping. The roundtrip assumptions of `symmetric_state_preservation` become a pair of bijection conditions between `reg_state` and the image `daml_states` of the representation map.

```
datatype daml-status-tag =
  D-Active | D-Frozen | D-Seized | D-Confiscated | D-Restricted
```

```
record daml-perm =
  status-tag      :: daml-status-tag
  seized-by      :: nat option
  restriction-scope :: nat option
```

```
definition default-party-id :: nat where
  default-party-id = 0
```

```
definition default-scope-id :: nat where
  default-scope-id = 0
```

The well-formedness invariant: the auxiliary fields are non-None exactly on the status tags that semantically require them.

```
definition valid-daml-perm :: daml-perm  $\Rightarrow$  bool where
  valid-daml-perm p  $\longleftrightarrow$ 
    (status-tag p = D-Seized  $\longleftrightarrow$  seized-by p  $\neq$  None)  $\wedge$ 
    (status-tag p = D-Restricted  $\longleftrightarrow$  restriction-scope p  $\neq$  None)
```

The two state maps between the two representations.

```
fun reg-to-daml :: reg-state  $\Rightarrow$  daml-perm where
```

```

    reg-to-daml ACTIVE      = (| status-tag = D-Active,
                                seized-by = None,
                                restriction-scope = None |)
  | reg-to-daml FROZEN      = (| status-tag = D-Frozen,
                                seized-by = None,
                                restriction-scope = None |)
  | reg-to-daml SEIZED      = (| status-tag = D-Seized,
                                seized-by = Some default-party-id,
                                restriction-scope = None |)
  | reg-to-daml CONFISCATED = (| status-tag = D-Confiscated,
                                seized-by = None,
                                restriction-scope = None |)
  | reg-to-daml RESTRICTED = (| status-tag = D-Restricted,
                                seized-by = None,
                                restriction-scope = Some default-scope-id |)

```

```

fun daml-to-reg :: daml-perm  $\Rightarrow$  reg-state where
  daml-to-reg p = (case status-tag p of
    D-Active       $\Rightarrow$  ACTIVE
  | D-Frozen       $\Rightarrow$  FROZEN
  | D-Seized       $\Rightarrow$  SEIZED
  | D-Confiscated  $\Rightarrow$  CONFISCATED
  | D-Restricted   $\Rightarrow$  RESTRICTED)

```

The DAML target state space: the image of `reg_to_daml`.

```

definition daml-states :: daml-perm set where
  daml-states = range reg-to-daml

```

```

definition daml-terminal :: daml-perm set where
  daml-terminal = {reg-to-daml CONFISCATED}

```

The DAML transition function: lifts `reg_transition` through the representation map. By construction, `daml_transition` respects the `reg_to_daml` map.

```

definition daml-transition :: daml-perm  $\Rightarrow$  reg-action  $\Rightarrow$  daml-perm option where
  daml-transition p a =
    (if p  $\in$  daml-states then
      (case reg_transition (daml-to-reg p) a of
        None       $\Rightarrow$  None
      | Some s'  $\Rightarrow$  Some (reg-to-daml s'))
    else None)

```

Roundtrip lemmas establishing the layer-crossing bijection.

```

lemma daml-to-reg-to-daml-id:
  daml-to-reg (reg-to-daml s) = s
by (cases s) auto

```

```

lemma reg-to-daml-to-reg-on-image:
  assumes p  $\in$  daml-states

```

shows $\text{reg-to-daml} (\text{daml-to-reg } p) = p$
proof –
 from *assms* **obtain** s **where** $p = \text{reg-to-daml } s$
 unfolding *daml-states-def* **by** *auto*
 then show *?thesis* **using** *daml-to-reg-to-daml-id* **by** *simp*
qed

lemma *reg-to-daml-valid*:
valid-daml-perm ($\text{reg-to-daml } s$)
 unfolding *valid-daml-perm-def* **by** (*cases s*) *auto*

The DAML side is a state machine on the image of *reg_to_daml*.

lemma *daml-states-finite*: *finite daml-states*
proof –
 have *fin-reg*: *finite reg-states* **unfolding** *reg-states-def* **by** *auto*
 hence *fin-img*: *finite* ($\text{reg-to-daml} \text{ ` } \text{reg-states}$) **by** (*rule finite-imageI*)
 have *eq*: $\text{daml-states} = \text{reg-to-daml} \text{ ` } \text{reg-states}$
 unfolding *daml-states-def* **using** *reg-states-UNIV* **by** *simp*
 from *fin-img eq* **show** *?thesis* **by** *simp*
qed

lemma *daml-terminal-subset*: $\text{daml-terminal} \subseteq \text{daml-states}$
 unfolding *daml-terminal-def* *daml-states-def* **by** *blast*

lemma *daml-terminal-absorbing*:
 assumes $p \in \text{daml-terminal}$ **and** $a \in \text{reg-actions}$
 shows $\text{daml-transition } p \ a = \text{None}$
proof –
 from *assms*(1) **have** $p = \text{reg-to-daml } \text{CONFISCATED}$
 unfolding *daml-terminal-def* **by** *simp*
 then **have** $\text{daml-to-reg } p = \text{CONFISCATED}$ **using** *daml-to-reg-to-daml-id* **by** *simp*
 moreover **have** $p \in \text{daml-states}$
 using *assms*(1) *daml-terminal-subset* **by** *auto*
 ultimately **show** *?thesis*
 unfolding *daml-transition-def* **using** *confiscated-terminal* **by** *simp*
qed

lemma *daml-transition-closed*:
 assumes $p \in \text{daml-states}$ **and** $a \in \text{reg-actions}$
 and $\text{daml-transition } p \ a = \text{Some } p'$
 shows $p' \in \text{daml-states}$
proof –
 from *assms*(1,3) **obtain** s' **where**
 $s'\text{-eq}$: $\text{reg-transition } (\text{daml-to-reg } p) \ a = \text{Some } s'$
 and $p'\text{-def}$: $p' = \text{reg-to-daml } s'$
 unfolding *daml-transition-def*
 by (*auto split: option.splits if-splits*)
 then **show** *?thesis*

unfolding *daml-states-def* **by** *auto*
qed

lemma *daml-transition-outside-states*:
 $p \notin \text{daml-states} \implies \text{daml-transition } p \ a = \text{None}$
unfolding *daml-transition-def* **by** *simp*

Forward naturality: *reg_to_daml* commutes with transitions.

lemma *reg-to-daml-naturality-some*:
assumes $s \in \text{reg-states}$ **and** $a \in \text{reg-actions}$
and $\text{reg-transition } s \ a = \text{Some } s'$
shows $\text{daml-transition } (\text{reg-to-daml } s) \ a = \text{Some } (\text{reg-to-daml } s')$
proof –
have $\text{in-states: } \text{reg-to-daml } s \in \text{daml-states}$
unfolding *daml-states-def* **by** *auto*
have $\text{daml-to-reg } (\text{reg-to-daml } s) = s$ **using** *daml-to-reg-to-daml-id* **by** *simp*
with *assms(3)* **have**
 $\text{reg-transition } (\text{daml-to-reg } (\text{reg-to-daml } s)) \ a = \text{Some } s'$ **by** *simp*
with *in-states* **show** *?thesis*
unfolding *daml-transition-def* **by** *simp*
qed

lemma *reg-to-daml-naturality-none*:
assumes $s \in \text{reg-states}$ **and** $a \in \text{reg-actions}$
and $\text{reg-transition } s \ a = \text{None}$
shows $\text{daml-transition } (\text{reg-to-daml } s) \ a = \text{None}$
proof –
have $\text{in-states: } \text{reg-to-daml } s \in \text{daml-states}$
unfolding *daml-states-def* **by** *auto*
have $\text{daml-to-reg } (\text{reg-to-daml } s) = s$ **using** *daml-to-reg-to-daml-id* **by** *simp*
with *assms(3)* **have**
 $\text{reg-transition } (\text{daml-to-reg } (\text{reg-to-daml } s)) \ a = \text{None}$ **by** *simp*
with *in-states* **show** *?thesis*
unfolding *daml-transition-def* **by** *simp*
qed

Backward naturality: *daml_to_reg* on the image of *reg_to_daml* commutes with transitions in the opposite direction.

lemma *daml-to-reg-naturality-some*:
assumes $p \in \text{daml-states}$ **and** $a \in \text{reg-actions}$
and $\text{daml-transition } p \ a = \text{Some } p'$
shows $\text{reg-transition } (\text{daml-to-reg } p) \ a = \text{Some } (\text{daml-to-reg } p')$
proof –
from *assms(1,3)* **obtain** s' **where**
 $s'\text{-eq: } \text{reg-transition } (\text{daml-to-reg } p) \ a = \text{Some } s'$
and $p'\text{-def: } p' = \text{reg-to-daml } s'$
unfolding *daml-transition-def*
by (*auto split: option.splits if-splits*)
from $p'\text{-def}$ **have** $\text{daml-to-reg } p' = s'$ **using** *daml-to-reg-to-daml-id* **by** *simp*

with s' -eq **show** ?thesis **by** simp
qed

lemma *daml-to-reg-naturality-none*:
assumes $p \in \text{daml-states}$ **and** $a \in \text{reg-actions}$
and $\text{daml-transition } p \ a = \text{None}$
shows $\text{reg-transition } (\text{daml-to-reg } p) \ a = \text{None}$
proof (rule ccontr)
assume $\text{reg-transition } (\text{daml-to-reg } p) \ a \neq \text{None}$
then obtain s' **where** $\text{reg-transition } (\text{daml-to-reg } p) \ a = \text{Some } s'$ **by** auto
with $\text{assms}(1)$ **have** $\text{daml-transition } p \ a = \text{Some } (\text{reg-to-daml } s')$
unfolding *daml-transition-def* **by** simp
with $\text{assms}(3)$ **show** False **by** simp
qed

Forward state preservation: `reg_to_daml` as a structure-preserving map.

Forward state preservation. The source state machine is on `(reg_states, reg_actions, reg_transition, reg_terminal)`, which exactly matches the `reg_sm` interpretation; consequently `unfold_locales` auto-discharges all six source state-machine obligations. The target side is the DAML structured-permission record, for which only individual lemmas (not a registered interpretation) are available, so all six target state-machine obligations remain pending alongside the five preservation-specific axioms eleven obligations in total. The asymmetry with `backward_layer_preservation` below (which has only the five preservation axioms remaining) reflects the order in which the locale extension presents the two state-machine sub-locales.

interpretation *forward-layer-preservation*:
state-preservation
 $\text{reg-states } \text{reg-actions } \text{reg-transition } \text{reg-terminal}$
 $\text{daml-states } \text{reg-actions } \text{daml-transition } \text{daml-terminal}$
 $\text{reg-to-daml } \text{id}$
proof *unfold-locales*
show *finite daml-states* **by** (rule *daml-states-finite*)
next
show *finite reg-actions* **unfolding** *reg-actions-def* **by** auto
next
show $\text{daml-terminal} \subseteq \text{daml-states}$ **by** (rule *daml-terminal-subset*)
next
fix $s \ a$
assume $s \in \text{daml-terminal}$ $a \in \text{reg-actions}$
then show $\text{daml-transition } s \ a = \text{None}$ **by** (rule *daml-terminal-absorbing*)
next
fix $s \ a \ s'$
assume $s \in \text{daml-states}$ $a \in \text{reg-actions}$ $\text{daml-transition } s \ a = \text{Some } s'$
then show $s' \in \text{daml-states}$ **by** (rule *daml-transition-closed*)
next
fix $s :: \text{daml-perm}$ **and** $a :: \text{reg-action}$

```

    assume  $s \notin \text{daml-states}$ 
    then show  $\text{daml-transition } s \ a = \text{None}$  by (rule  $\text{daml-transition-outside-states}$ )
next
  fix  $s$ 
  assume  $s \in \text{reg-states}$ 
  then show  $\text{reg-to-daml } s \in \text{daml-states}$  unfolding  $\text{daml-states-def}$  by auto
next
  fix  $a$ 
  assume  $a \in \text{reg-actions}$ 
  then show  $\text{id } a \in \text{reg-actions}$  by simp
next
  fix  $s$ 
  assume  $s \in \text{reg-terminal}$ 
  then show  $\text{reg-to-daml } s \in \text{daml-terminal}$ 
    unfolding  $\text{reg-terminal-def}$   $\text{daml-terminal-def}$  by simp
next
  fix  $s \ a \ s'$ 
  assume  $s \in \text{reg-states} \ a \in \text{reg-actions} \ \text{reg-transition } s \ a = \text{Some } s'$ 
  then show  $\text{daml-transition } (\text{reg-to-daml } s) \ (\text{id } a) = \text{Some } (\text{reg-to-daml } s')$ 
    using  $\text{reg-to-daml-naturality-some}$  by simp
next
  fix  $s \ a$ 
  assume  $s \in \text{reg-states} \ a \in \text{reg-actions} \ \text{reg-transition } s \ a = \text{None}$ 
  then show  $\text{daml-transition } (\text{reg-to-daml } s) \ (\text{id } a) = \text{None}$ 
    using  $\text{reg-to-daml-naturality-none}$  by simp
qed

```

Backward state preservation: daml_to_reg as a structure-preserving map on the image of reg_to_daml .

Backward state preservation. The source side is DAML and the target side matches reg_sm . unfold_locales auto-discharges all twelve state-machine obligations from the previously registered interpretations, so only the five preservation axioms remain pending.

interpretation *backward-layer-preservation:*

```

  state-preservation
    daml-states reg-actions daml-transition daml-terminal
    reg-states reg-actions reg-transition reg-terminal
    daml-to-reg id

```

proof *unfold-locals*

```

  fix  $p$ 
  assume  $p \in \text{daml-states}$ 
  then show  $\text{daml-to-reg } p \in \text{reg-states}$  using  $\text{reg-states-UNIV}$  by auto
next
  fix  $a$ 
  assume  $a \in \text{reg-actions}$ 
  then show  $\text{id } a \in \text{reg-actions}$  by simp
next
  fix  $p$ 

```

```

    assume  $p \in \text{daml-terminal}$ 
    then have  $p = \text{reg-to-daml CONFISCATED unfolding daml-terminal-def by simp}$ 
    then have  $\text{daml-to-reg } p = \text{CONFISCATED using daml-to-reg-to-daml-id by simp}$ 
    then show  $\text{daml-to-reg } p \in \text{reg-terminal unfolding reg-terminal-def by simp}$ 
  next
    fix  $p \ a \ p'$ 
    assume  $p \in \text{daml-states } a \in \text{reg-actions daml-transition } p \ a = \text{Some } p'$ 
    then show  $\text{reg-transition } (\text{daml-to-reg } p) \ (\text{id } a) = \text{Some } (\text{daml-to-reg } p')$ 
      using  $\text{daml-to-reg-naturality-some by simp}$ 
  next
    fix  $p \ a$ 
    assume  $p \in \text{daml-states } a \in \text{reg-actions daml-transition } p \ a = \text{None}$ 
    then show  $\text{reg-transition } (\text{daml-to-reg } p) \ (\text{id } a) = \text{None}$ 
      using  $\text{daml-to-reg-naturality-none by simp}$ 
qed

```

The symmetric instance: forward and backward state preservation composed with roundtrip guarantees. The non-trivial content is the bijection between `reg_state` and the image of `reg_to_daml` in `daml_perm`; the action map is the identity because both layers share the same regulatory action vocabulary.

interpretation *onchain-daml-bridge:*

```

symmetric-state-preservation
  reg-states reg-actions reg-transition reg-terminal
  daml-states reg-actions daml-transition daml-terminal
  reg-to-daml id
  daml-to-reg id

```

proof *unfold-locales*

```

  fix  $s$ 
  assume  $s \in \text{reg-states}$ 
  then show  $\text{daml-to-reg } (\text{reg-to-daml } s) = s \text{ using daml-to-reg-to-daml-id by simp}$ 
next
  fix  $p$ 
  assume  $p \in \text{daml-states}$ 
  then show  $\text{reg-to-daml } (\text{daml-to-reg } p) = p \text{ using reg-to-daml-to-reg-on-image by simp}$ 
next
  fix  $a$ 
  assume  $a \in \text{reg-actions}$ 
  then show  $\text{id } (\text{id } a) = a \text{ by simp}$ 
qed

```

As a corollary of the symmetric interpretation, `reg_to_daml` is injective on `reg_states`: distinct on-chain states map to distinct off-chain DAML records, so the layer-crossing representation incurs no information loss.

corollary *reg-to-daml-injective-on-states:*

```

[[  $s1 \in \text{reg-states}; s2 \in \text{reg-states}; \text{reg-to-daml } s1 = \text{reg-to-daml } s2$  ]]

```

$\implies s1 = s2$
using *onchain-daml-bridge.state-map-injective* .

18 Multi-Domain Locale Instantiation

We now instantiate the `multi_domain_preservation` locale from `State_Preservation.thy` with the regulatory state machine.

The instantiation is parametric: given ANY finite set of domains and ANY initial `domain_state` function satisfying the consistency precondition, the locale is satisfied. This proves that the abstract framework applies to the concrete regulatory model without fixing a specific topology.

The key bridge: we construct a `domain_state` function from our `global_state`, and show that `valid_state` implies the consistency precondition required by `multi_domain_preservation`.

Bridge from our `global_state` model to the `multi_domain_preservation` locale's `domain_state` signature.

`multi_domain_preservation` expects: `domain_state :: 'd => 'id => 's option`

We instantiate with: `'d = chain_id, 'id = asset_id, 's = reg_state`

The bridge function extracts `reg_state` from our richer `asset_state`.

definition *reg-domain-state :: global-state $\Rightarrow$ chain-id $\Rightarrow$ asset-id $\Rightarrow$ reg-state option where*

reg-domain-state gs cid aid = get-reg-state gs cid aid

The parametric instantiation theorem: for any finite set of `chain_ids` and any valid `global_state`, we can interpret the `multi_domain_preservation` locale with the regulatory state machine.

The `state_machine` predicate as a standalone fact. This is needed for instantiating `multi_domain_preservation`, which takes `state_machine` as an opaque predicate assumption.

lemma *reg-state-machine-pred:*

state-machine reg-states reg-actions reg-transition reg-terminal

proof –

have *f1: finite reg-states unfolding reg-states-def by auto*

have *f2: finite reg-actions unfolding reg-actions-def by auto*

have *f3: reg-terminal $\subseteq$ reg-states*

unfolding *reg-terminal-def reg-states-def by auto*

have *f4: $\bigwedge s a. s \in \text{reg-terminal} \implies a \in \text{reg-actions} \implies \text{reg-transition } s a = \text{None}$*

unfolding *reg-terminal-def by (auto simp: confiscated-terminal)*

have *f5: $\bigwedge s a s'. s \in \text{reg-states} \implies a \in \text{reg-actions} \implies \text{reg-transition } s a = \text{Some } s' \implies s' \in \text{reg-states}$*

using *reg-transition-closed by auto*

have *f6: $\bigwedge s a. (s :: \text{reg-state}) \notin \text{reg-states} \implies \text{reg-transition } s a = \text{None}$*

using *reg-states-UNIV by auto*

```

show ?thesis
  by (intro state-machine.intro) (use f1 f2 f3 f4 f5 f6 in auto)
qed

theorem reg-multi-domain-instantiation:
  assumes fin-doms: finite (doms :: chain-id set)
  and valid: valid-state gs
  shows multi-domain-preservation doms reg-states reg-actions reg-transition
    reg-terminal (reg-domain-state gs)
proof (intro multi-domain-preservation.intro)
  show finite doms using fin-doms by simp
next
  show state-machine reg-states reg-actions reg-transition reg-terminal
    by (rule reg-state-machine-pred)
next
  fix d1 d2 :: chain-id and aid :: asset-id and s1 s2 :: reg-state
  assume d1 ∈ doms d2 ∈ doms
  and reg-domain-state gs d1 aid = Some s1
  and reg-domain-state gs d2 aid = Some s2
  then have get-reg-state gs d1 aid = Some s1
  and get-reg-state gs d2 aid = Some s2
  unfolding reg-domain-state-def by simp-all
  with valid show s1 = s2
  unfolding valid-state-def consistent-state-def by blast
qed

```

Corollary: the generic `cross_domain_consistency` theorem from `State_Preservation.thy` applies to regulatory synchronization.

This connects the abstract theory to the concrete model: the generic theorem guarantees that after `sync_all` (the abstract synchronization), all connected domains agree. Combined with our concrete `regulatory_homomorphism` theorem, we have both the abstract and concrete guarantees.

The instantiation above (`reg_multi_domain_instantiation`) enables using all theorems proven in `multi_domain_preservation` in particular `cross_domain_consistency` and `sync_isolation` for the regulatory model without reproving them. Any consumer of this theory can invoke:

```

multi_domain_preservation.cross_domain_consistency [OF
reg_multi_domain_instantiation[OF fin_doms valid_gs]]
to obtain the concrete guarantee for their domain set and global state.

```

Summary of what this theory establishes:

1. The regulatory state machine (`reg_state`, `reg_action`, `reg_transition`) satisfies the `state_machine` locale from `State_Preservation.thy`.
2. CONFISCATED is terminal (I1), CONFISCATE is universally reachable (I2), and the transition function is deterministic (I3).
3. SEIZED to FROZEN and FROZEN to RESTRICTED direct transitions are excluded by design, with legal and complexity-reduction justifica-

tions.

4. The synchronization protocol (lock, then update, then unlock) preserves cross-chain consistency for the same asset while isolating other assets.

5. The atomic lock guard protects the sync function against a second action while the lock is held; under the atomic model this expresses the intended exclusion of competing regulatory actions on the same asset (NOT double-spend prevention that is handled at a different layer).

6. The `regulatory_homomorphism` theorem establishes that after sync, all connected chains agree on the new regulatory state.

7. The `valid_state_preservation` theorem establishes that sync preserves the global validity invariant: `consistent_state` AND no outstanding locks. This closes the inductive invariant.

8. The heterogeneous-action instance (`escalation_preservation`) instantiates `state_preservation` with the four-action escalation subset of `reg_action` on the source side, the dedicated four-constructor datatype `chain_b_action` on the target side, and the identity state map. This exercises both the locale's `actions_s` subset parameter and the heterogeneous source / target action types.

9. The layer-crossing instance (`onchain_daml_bridge`) instantiates `symmetric_state_preservation` with the on-chain enum representation and a structured DAML permission record (`daml_perm`) carrying auxiliary fields. The action vocabularies coincide so the action map is the identity; the non-trivial content is the bijection between the enum and the image `daml_states` of the representation map, with a carrier predicate (`valid_daml_perm`) tying the auxiliary fields to the status tag.

10. The `multi_domain_preservation` locale is instantiated parametrically: for any finite domain set and valid global state, the abstract framework from `State_Preservation.thy` applies to the regulatory model.

Open work for subsequent entries: - Eventual consistency under the partially synchronous network model (with message delays, redelivery, and out-of-order arrival).

end

```
theory Priority-Resolution
  imports Main
begin
```

19 Priority-Based Deterministic Selection

A system where messages carry priorities from a linear order. Given a finite non-empty set of messages with injective priorities, there exists a unique highest-priority message, and selection is deterministic.

The injectivity assumption models tiebreaking: in practice, a compos-

ite key (e.g., authority level, timestamp, severity, node ID) ensures no two distinct messages share the same priority.

```

locale priority-system =
  fixes priority :: 'm  $\Rightarrow$  'k::linorder
  assumes priority-injective:
     $\llbracket \text{priority } m1 = \text{priority } m2 \rrbracket \implies m1 = m2$ 
begin

```

In a finite non-empty set with injective priorities, there exists a unique element with the maximum priority.

```

lemma highest-priority-exists:
  assumes finite S and  $S \neq \{\}$ 
  shows  $\exists! m. m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$ 
proof -
  have fin-img: finite (priority ' S) using assms(1) by simp
  have ne-img: priority ' S  $\neq \{\}$  using assms(2) by simp
  obtain m where m-in:  $m \in S$  and m-pri:  $\text{priority } m = \text{Max } (\text{priority } ' S)$ 
    using Max-in[OF fin-img ne-img] by auto
  have m-max:  $\forall m' \in S. \text{priority } m' \leq \text{priority } m$ 
proof
  fix m' assume  $m' \in S$ 
  hence priority m'  $\in \text{priority } ' S$  by simp
  hence  $\text{priority } m' \leq \text{Max } (\text{priority } ' S)$  using Max-ge[OF fin-img] by simp
  thus  $\text{priority } m' \leq \text{priority } m$  using m-pri by simp
qed
show ?thesis
proof (rule ex1I[of - m])
  show  $m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$ 
    using m-in m-max by auto
next
  fix m2 assume  $m2 \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m2)$ 
  then have m2-in:  $m2 \in S$  and m2-max:  $\forall m' \in S. \text{priority } m' \leq \text{priority } m2$ 
by auto
  from m-max m2-in have  $\text{priority } m2 \leq \text{priority } m$  by auto
  from m2-max m-in have  $\text{priority } m \leq \text{priority } m2$  by auto
  then have  $\text{priority } m = \text{priority } m2$  using  $\langle \text{priority } m2 \leq \text{priority } m \rangle$  by auto
  then show  $m2 = m$  using priority-injective by auto
qed
qed

```

```

definition select-highest :: 'm set  $\Rightarrow$  'm option where
  select-highest S =
    (if  $S = \{\}$  then None
     else Some (THE m.  $m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$ )))

```

```

theorem select-highest-deterministic:
  assumes fin: finite S and ne:  $S \neq \{\}$ 
  shows  $\exists! m. \text{select-highest } S = \text{Some } m$ 
     $\wedge m \in S$ 

```

$\wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$

proof –

from *highest-priority-exists*[*OF fin ne*] **have** *uniq*:

$\exists! m. m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)$.

then obtain *v* **where** *vmax*: $v \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } v)$

and *vuniq*: $\bigwedge w. w \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } w) \implies w = v$

by *blast*

have *the-v*: $(\text{THE } m. m \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } m)) = v$

proof (*rule the-equality*)

show $v \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } v)$

using *vmax* .

next

fix *w*

assume $w \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } w)$

then show $w = v$ **by** (*rule vuniq*)

qed

have *hv*: *select-highest* *S* = *Some v*

unfolding *select-highest-def* **using** *ne the-v* **by** *simp*

show *?thesis*

proof (*rule ex1I*[*of - v*])

show *select-highest* *S* = *Some v* $\wedge v \in S$

$\wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } v)$

using *hv vmax* **by** *blast*

next

fix *w*

assume *select-highest* *S* = *Some w* $\wedge w \in S$

$\wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } w)$

then show $w = v$ **using** *vuniq* **by** *blast*

qed

qed

theorem *select-highest-in-set*:

assumes *fin*: *finite S* **and** *sel*: *select-highest S* = *Some m*

shows $m \in S$

proof –

from *sel* **have** *ne*: $S \neq \{\}$

unfolding *select-highest-def* **by** (*simp split: if-splits*)

from *highest-priority-exists*[*OF fin ne*] **have** *uniq*:

$\exists! x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x)$.

from *theI*'[*OF uniq*] **have** *the-in*:

$(\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x)) \in S$ **by** *auto*

from *sel ne* **have** $m = (\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x))$

unfolding *select-highest-def* **by** *simp*

with *the-in* **show** *?thesis* **by** *simp*

qed

theorem *select-highest-is-max*:

assumes *fin*: *finite S* **and** *ne*: $S \neq \{\}$ **and** *sel*: *select-highest S* = *Some m*

shows $\forall m' \in S. \text{priority } m' \leq \text{priority } m$


```

proof –
  from highest-priority-exists[OF fin ne] have uniq:
     $\exists!x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x) .$ 
  from theI'[OF uniq] have the-max:
     $\forall m' \in S. \text{priority } m' \leq$ 
       $\text{priority } (\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x))$ 
  by auto
  from sel ne have  $m = (\text{THE } x. x \in S \wedge (\forall m' \in S. \text{priority } m' \leq \text{priority } x))$ 
  unfolding select-highest-def by simp
  with the-max show ?thesis by simp
qed

end

```

20 Fair-Leader Aggregate Progress

A leader-based system where an honest leader always processes at least one pending request, and a fairness assumption guarantees that honest leaders appear periodically.

The fairness assumption is deterministic: it requires an honest-tagged leader in every bounded window. This theory provides no probability space, independence assumption, VRF semantics, or derivation of that condition from a Byzantine threshold.

The key theorem: under the fairness assumption, a positive aggregate pending count strictly decreases within a bounded number of epochs. Individual request identities and their service order are not modelled.

```

locale fair-leader-system =
  fixes leader-at :: nat  $\Rightarrow$  'n
  and is-honest :: 'n  $\Rightarrow$  bool
  and pending :: nat  $\Rightarrow$  nat
  and fairness-bound :: nat
  assumes fair-leader:
     $\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge \text{is-honest } (\text{leader-at } e)$ 
  and honest-progress:
     $\llbracket \text{is-honest } (\text{leader-at } e); \text{pending } e > 0 \rrbracket \implies \text{pending } (\text{Suc } e) < \text{pending } e$ 
  and non-honest-bounded:
     $\text{pending } (\text{Suc } e) \leq \text{pending } e$ 
begin

```

Positivity of the fairness bound is implied by fairness itself: the window starting at any epoch must be non-empty to contain an honest slot. It is therefore derived rather than assumed.

lemma *fairness-bound-positive*: *fairness-bound* > 0

```

proof –
  obtain e where  $(0::\text{nat}) \leq e$  and  $e < 0 + \text{fairness-bound}$ 
  using fair-leader by blast

```

then show *?thesis* by *simp*
qed

Pending count is monotonically non-increasing.

lemma *pending-monotone*:
 assumes $e1 \leq e2$
 shows $\text{pending } e2 \leq \text{pending } e1$
 using *assms*
proof (*induction rule: inc-induct*)
 case *base*
 then show *?case* by *simp*
next
 case (*step e*)
 have $\text{pending } e2 \leq \text{pending } (\text{Suc } e)$ by (*rule step.IH*)
 also have $\text{pending } (\text{Suc } e) \leq \text{pending } e$ using *non-honest-bounded* by *auto*
 finally show *?case* .
 qed

If there are pending requests, within *fairness-bound* epochs at least one will be processed (the pending count strictly decreases).

theorem *starvation-bound*:
 assumes *pos: pending epoch > 0*
 shows $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge \text{pending } (\text{Suc } e) < \text{pending } e$
proof –
 from *fair-leader* obtain *h* where
 h-range: $\text{epoch} \leq h < \text{epoch} + \text{fairness-bound}$ and
 h-honest: *is-honest* (*leader-at h*)
 by *auto*
 show *?thesis*
proof (*cases pending h > 0*)
 case *True*
 with *h-honest* have $\text{pending } (\text{Suc } h) < \text{pending } h$ using *honest-progress* by *auto*
 with *h-range* show *?thesis* by *auto*
next
 case *False*
 then have *ph0: pending h = 0* by *simp*
 — pending dropped from >0 (at epoch) to 0 (at h): find a strict decrease step
 have *key*: $\forall n. 0 < n \longrightarrow \text{pending } (\text{epoch} + n) = 0 \longrightarrow$
 $(\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + n \wedge \text{pending } (\text{Suc } e) < \text{pending } e)$
proof (*rule allI*)
 fix *n*
 show $0 < n \longrightarrow \text{pending } (\text{epoch} + n) = 0 \longrightarrow$
 $(\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + n \wedge \text{pending } (\text{Suc } e) < \text{pending } e)$
proof (*induction n*)
 case 0 show *?case* by *simp*
next
 case (*Suc k*)

```

show ?case
proof (intro impI)
  assume hpk: pending (epoch + Suc k) = 0
  show  $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{Suc } k \wedge \text{pending} (\text{Suc } e) < \text{pending } e$ 
  proof (cases k = 0)
    case True
      from hpk True have pending (Suc epoch) < pending epoch using pos by
simp
      thus ?thesis by (intro exI[of - epoch]) auto
    next
      case False
      then have kpos:  $0 < k$  by simp
      show ?thesis
      proof (cases pending (epoch + k) = 0)
        case True
          then have pk0: pending (epoch + k) = 0 .
          from Suc.IH[THEN mp, OF kpos, THEN mp, OF pk0]
          obtain e' where epoch  $\leq e'$   $e' < \text{epoch} + k$  pending (Suc e') < pending
e'
            by auto
            thus ?thesis by auto
          next
            case False
            from False hpk have pending (epoch + Suc k) < pending (epoch + k)
by simp
            thus ?thesis by (intro exI[of - epoch + k]) auto
          qed
        qed
      qed
    qed
  qed
  have dpos:  $0 < h - \text{epoch}$ 
  proof -
    have  $h \neq \text{epoch}$  using ph0 pos by auto
    with h-range(1) show ?thesis by arith
  qed
  have pd: pending (epoch + (h - epoch)) = 0
  proof -
    have epoch + (h - epoch) = h using h-range(1) by arith
    with ph0 show ?thesis by simp
  qed
  from key[rule-format, OF dpos, OF pd]
  obtain e' where he': epoch  $\leq e'$   $e' < \text{epoch} + (h - \text{epoch})$  pending (Suc e') <
pending e'
    by auto
    have epoch + (h - epoch)  $\leq h$  using h-range(1) by arith
    with he' h-range(2) show ?thesis by auto
  qed
qed

```

Corollary: the aggregate pending count eventually reaches zero in this closed count model. By induction on the pending count (which is a natural number and thus a well-founded decreasing measure), repeated application of `starvation_bound` drives the count to zero. The statement does not identify requests or cover continuous arrivals.

theorem *eventual-completion*:

shows $\exists e\text{-final}. \text{pending } e\text{-final} = 0$

proof –

define *P* **where**

$P\ n \equiv \forall \text{epoch}. \text{pending epoch} = n \longrightarrow (\exists e\text{-final}. \text{pending } e\text{-final} = 0)$

for $n :: \text{nat}$

have *all-P*: $\bigwedge n. P\ n$

proof –

fix *n* **show** *P n*

unfolding *P-def*

proof (*induction n rule: less-induct*)

case (*less n*)

show $\forall \text{epoch}. \text{pending epoch} = n \longrightarrow (\exists e\text{-final}. \text{pending } e\text{-final} = 0)$

proof (*intro allI impI*)

fix *epoch* **assume** *eq*: $\text{pending epoch} = n$

show $\exists e\text{-final}. \text{pending } e\text{-final} = 0$

proof (*cases n = 0*)

case *True* **thus** *?thesis* **using** *eq* **by** *auto*

next

case *False*

have *pos*: $\text{pending epoch} > 0$ **using** *eq* *False* **by** *auto*

from *starvation-bound*[*OF pos*] **obtain** *e* **where**

$e\text{-ge}: \text{epoch} \leq e$ **and** $e\text{-dec}: \text{pending } (\text{Suc } e) < \text{pending } e$

by *auto*

have *lt*: $\text{pending } (\text{Suc } e) < n$

using *pending-monotone*[*OF e-ge*] *eq e-dec* **by** *simp*

from *less.IH*[*OF lt*] **show** *?thesis*

unfolding *P-def* **by** *blast*

qed

qed

qed

from *all-P*[*of pending 0*] **show** *?thesis*

unfolding *P-def* **by** *blast*

qed

end

end

theory *DQuencer-Instance*

imports *Priority-Resolution Regulatory-Instance HOL-Library.Product-Lexorder*

begin

21 Authority Level and Action Severity

Regulatory authority hierarchy. International authorities (e.g., FATF) take precedence over national (e.g., SEC), which take precedence over regional authorities. This reflects the RCP framework’s jurisdictional priority model.

datatype *authority-level* = *Regional* | *National* | *International*

fun *authority-rank* :: *authority-level* $\Rightarrow$ *nat* **where**

authority-rank *Regional* = 1
| *authority-rank* *National* = 2
| *authority-rank* *International* = 3

lemma *authority-rank-injective*:

authority-rank *a1* = *authority-rank* *a2* $\implies$ *a1* = *a2*
 by (*cases a1*; *cases a2*; *auto*)

Action severity determines priority among actions from the same authority level and timestamp (it is the third component of the lexicographic key). Stronger enforcement actions (CONFISCATE, SEIZE) take precedence over weaker ones (RESTRICT, UNFREEZE). This reflects the legal principle that more conservative (asset-protective) actions prevail when conflicting orders must be ranked against one another.

fun *action-severity* :: *reg-action* $\Rightarrow$ *nat* **where**

action-severity *UNRESTRICT* = 1
| *action-severity* *UNFREEZE* = 2
| *action-severity* *RELEASE* = 3
| *action-severity* *RESTRICT* = 4
| *action-severity* *FREEZE* = 5
| *action-severity* *SEIZE* = 6
| *action-severity* *CONFISCATE* = 7

lemma *action-severity-injective*:

action-severity *a1* = *action-severity* *a2* $\implies$ *a1* = *a2*
 by (*cases a1*; *cases a2*; *auto*)

22 Priority Key

The priority key is a 4-tuple of natural numbers, using Isabelle’s built-in lexicographic order on products. No manual *linorder* instance registration is needed.

Components (highest to lowest significance):

1. **authority_rank**: higher authority = higher priority
2. **inverted_timestamp**: earlier timestamp = higher priority (inverted)

3. **action_severity**: stronger action = higher priority
4. **node_id**: deterministic tiebreaker (lower node ID = higher priority, so we invert to make larger = higher in the nat order)

For timestamp and **node_id** inversion, we use $max-val - actual-val$ to convert "smaller is better" to "larger is better" for compatibility with the nat product order where larger = higher.

type-synonym $priority-key = nat \times nat \times nat \times nat$

23 Extended Message Type

Extends **oss_message** from **Regulatory_Instance** with authority level and source node ID. Uses Isabelle record inheritance so that existing functions on **oss_message** remain applicable.

record $dq-message = oss-message +$
 $\quad dqm-authority-level :: authority-level$
 $\quad dqm-source-node \quad :: nat$

24 Priority Construction

Construct a priority key from message fields. Uses two upper bounds for inversion: **max_time** for timestamps and **max_node** for node IDs.

definition $make-priority-key ::$
 $\quad nat \Rightarrow nat \Rightarrow dq-message \Rightarrow priority-key$
where
 $make-priority-key\ max-time\ max-node\ msg =$
 $\quad (authority-rank\ (dqm-authority-level\ msg),$
 $\quad max-time - msg-timestamp\ msg,$
 $\quad action-severity\ (msg-action\ msg),$
 $\quad max-node - dqm-source-node\ msg)$

25 Node and System Model

datatype $node-behavior = Honest \mid Byzantine$

record $node-info =$
 $\quad ni-id \quad :: nat$
 $\quad ni-behavior :: node-behavior$

definition $honest-nodes :: node-info\ set \Rightarrow node-info\ set$ **where**
 $honest-nodes\ ns = \{n \in ns. ni-behavior\ n = Honest\}$

definition $byzantine-nodes :: node-info\ set \Rightarrow node-info\ set$ **where**
 $byzantine-nodes\ ns = \{n \in ns. ni-behavior\ n = Byzantine\}$

lemma *honest-byzantine-partition*:
 $honest-nodes\ ns \cup byzantine-nodes\ ns = ns$
proof (rule *set-eqI*)
 fix x
 show $x \in honest-nodes\ ns \cup byzantine-nodes\ ns \longleftrightarrow x \in ns$
 unfolding *honest-nodes-def byzantine-nodes-def*
 by (cases *ni-behavior x*; auto)
qed

lemma *honest-byzantine-disjoint*:
 $honest-nodes\ ns \cap byzantine-nodes\ ns = \{\}$
 unfolding *honest-nodes-def byzantine-nodes-def* **by** *auto*

26 D-quencer System Locale

The D-quencer system locale encapsulates the BFT assumption $((3::'a) * f + 1 \leq n)$ and system parameters.

locale *dquencer-system* =
 fixes $nodes :: node-info\ set$
 and $f-max :: nat$
 and $fairness-bound :: nat$
 and $max-time :: nat$
 and $max-node :: nat$
 assumes *finite-nodes*: $finite\ nodes$
 and *bft-threshold*: $card\ nodes \geq 3 * f-max + 1$
 and *byzantine-bound*: $card\ (byzantine-nodes\ nodes) \leq f-max$
 and *fairness-positive*: $fairness-bound > 0$
begin

lemma *honest-majority*:
 $card\ (honest-nodes\ nodes) > 2 * f-max$
proof –
 have $card\ nodes = card\ (honest-nodes\ nodes) + card\ (byzantine-nodes\ nodes)$
 using *honest-byzantine-partition honest-byzantine-disjoint finite-nodes*
 by (metis *card-Un-disjoint finite-Un*)
 with *bft-threshold byzantine-bound* **show** *?thesis* **by** *linarith*
qed

lemma *honest-nonempty*:
 $honest-nodes\ nodes \neq \{\}$
proof –
 have *fin*: $finite\ (honest-nodes\ nodes)$
 unfolding *honest-nodes-def* **using** *finite-nodes* **by** *simp*
 have *pos*: $0 < card\ (honest-nodes\ nodes)$
 using *honest-majority* **by** *linarith*
 with *fin pos* **show** *?thesis* **by** (auto *simp: card-eq-0-iff*)
qed

end

27 Priority System Instantiation

We instantiate the `priority_system` locale from `Priority_Resolution.thy` with D-quencer message priorities.

The priority function maps messages to `priority_key` (which is `nat \<times> nat \<times> nat \<times> nat` with the built-in lexicographic order).

Injectivity follows from: distinct messages have either different authority levels, different timestamps, different actions, or different source nodes. The source node ID serves as the final tiebreaker ensuring no two distinct messages share the same key.

For a concrete message set with the injectivity precondition, the `priority_system` locale provides deterministic selection.

lemma *priority-key-injectivity*:

assumes *make-priority-key* *mt mn m1 = make-priority-key mt mn m2*
and *msg-timestamp* *m1 ≤ mt and msg-timestamp m2 ≤ mt*
and *dqm-source-node* *m1 ≤ mn and dqm-source-node m2 ≤ mn*
shows *dqm-authority-level* *m1 = dqm-authority-level m2*
 $\wedge$ *msg-timestamp* *m1 = msg-timestamp m2*
 $\wedge$ *msg-action* *m1 = msg-action m2*
 $\wedge$ *dqm-source-node* *m1 = dqm-source-node m2*

proof –

from *assms(1)* **have**

eq1: authority-rank (dqm-authority-level m1) = authority-rank (dqm-authority-level m2) **and**

eq2: mt – msg-timestamp m1 = mt – msg-timestamp m2 **and**

eq3: action-severity (msg-action m1) = action-severity (msg-action m2) **and**

eq4: mn – dqm-source-node m1 = mn – dqm-source-node m2

unfolding *make-priority-key-def* **by** *auto*

from *eq1* **have** *al: dqm-authority-level m1 = dqm-authority-level m2*

using *authority-rank-injective* **by** *auto*

from *eq2* *assms(2,3)* **have** *ts: msg-timestamp m1 = msg-timestamp m2*

by *linarith*

from *eq3* **have** *act: msg-action m1 = msg-action m2*

using *action-severity-injective* **by** *auto*

from *eq4* *assms(4,5)* **have** *nd: dqm-source-node m1 = dqm-source-node m2*

by *linarith*

from *al ts act nd* **show** *?thesis* **by** *auto*

qed

Auxiliary corollary: under the well-formedness bounds, equal priority keys imply equal messages on every priority-relevant field.

The generic `priority_system` locale assumes priority injectivity unconditionally on its carrier type. We instantiate it with the identity function on

`priority_key` as the carrier; injectivity holds by reflexivity on the carrier. The recovery of the unique D-quencer message corresponding to a chosen priority key is provided as a corollary inside the `dquencer_priority_concrete` locale, using the well-formedness bounds and the additional priority-key distinctness assumption.

interpretation *dq-priority*:

priority-system id :: priority-key $\Rightarrow$ priority-key
by *unfold-locales auto*

The `dquencer_priority_context` locale extends `dquencer_system` with a fixed message set whose elements satisfy the well-formedness bounds. Together with the bound-preservation, this set induces the priority key set `msg_priority_keys` on which priority-based selection operates.

locale *dquencer-priority-context* = *dquencer-system* +
fixes *msg-set :: dq-message set*
assumes *msg-set-nonempty*: *msg-set* $\neq \{\}$
and *msg-set-well-formed*:
 $\bigwedge m. m \in \text{msg-set} \implies \text{msg-timestamp } m \leq \text{max-time}$
 $\quad \wedge \text{dqm-source-node } m \leq \text{max-node}$

begin

definition *msg-priority-keys :: priority-key set where*

msg-priority-keys = *make-priority-key max-time max-node ' msg-set*

lemma *msg-priority-keys-nonempty*: *msg-priority-keys* $\neq \{\}$

using *msg-set-nonempty* **unfolding** *msg-priority-keys-def* **by** *simp*

lemma *msg-priority-keys-finite*:

finite msg-set $\implies$ finite msg-priority-keys

unfolding *msg-priority-keys-def* **by** *simp*

The well-formedness bounds are what make the priority key faithful to the message fields: under them, equal keys force equal priority-relevant fields. This lifts *priority-key-injectivity* into the context, consuming the bounds.

lemma *msg-priority-key-field-injective*:

assumes *m1* $\in$ *msg-set* **and** *m2* $\in$ *msg-set*

and *make-priority-key max-time max-node m1* = *make-priority-key max-time max-node m2*

shows *dqm-authority-level m1* = *dqm-authority-level m2*

$\wedge$ *msg-timestamp m1* = *msg-timestamp m2*

$\wedge$ *msg-action m1* = *msg-action m2*

$\wedge$ *dqm-source-node m1* = *dqm-source-node m2*

using *priority-key-injectivity*[*OF assms(3)*]

msg-set-well-formed[*OF assms(1)*] *msg-set-well-formed*[*OF assms(2)*]

by *blast*

end

The `dquencer_priority_concrete` locale strengthens the context with the priority-key distinctness assumption on `msg_set`. This is the natural BFT-consensus precondition: distinct authority / timestamp / severity / source-node tuples ensure determinism. Under this assumption the priority key uniquely identifies a message in `msg_set`, so selecting the highest priority key in `msg_priority_keys` is equivalent to selecting the highest-priority message in `msg_set`.

```

locale dquencer-priority-concrete = dquencer-priority-context +
  assumes msg-set-priority-distinct:
     $\bigwedge m1\ m2. m1 \in \text{msg\_set} \implies m2 \in \text{msg\_set}$ 
     $\implies \text{make-priority-key max-time max-node } m1$ 
     $= \text{make-priority-key max-time max-node } m2$ 
     $\implies m1 = m2$ 

```

begin

Recovery: given a priority key from `msg_priority_keys`, return the unique `msg_set` message with that key.

definition *recover-msg* :: *priority-key* $\Rightarrow$ *dq-message* **where**
recover-msg $k = (\text{THE } m. m \in \text{msg_set} \wedge \text{make-priority-key max-time max-node } m = k)$

lemma *recover-msg-unique-existence*:

```

assumes  $k \in \text{msg-priority-keys}$ 
shows  $\exists! m. m \in \text{msg\_set} \wedge \text{make-priority-key max-time max-node } m = k$ 

```

proof –

```

from assms obtain  $m$ 
  where  $m\text{-in}: m \in \text{msg\_set}$ 
    and  $m\text{-key}: \text{make-priority-key max-time max-node } m = k$ 
  unfolding msg-priority-keys-def by auto
show ?thesis
proof (rule ex1I[of - m])
  show  $m \in \text{msg\_set} \wedge \text{make-priority-key max-time max-node } m = k$ 
    using  $m\text{-in } m\text{-key}$  by auto
next
  fix  $m'$ 
  assume  $m' \in \text{msg\_set} \wedge \text{make-priority-key max-time max-node } m' = k$ 
  then have  $m'\text{-in}: m' \in \text{msg\_set}$ 
    and  $m'\text{-key}: \text{make-priority-key max-time max-node } m' = k$ 
    by auto
  from  $m\text{-key } m'\text{-key}$ 
  have  $\text{make-priority-key max-time max-node } m'$ 
     $= \text{make-priority-key max-time max-node } m$ 
    by simp
  then show  $m' = m$ 
    using msg-set-priority-distinct[OF m'-in m-in] by simp
qed
qed

```

lemma *recover-msg-correct*:
assumes $k \in \text{msg-priority-keys}$
shows $\text{recover-msg } k \in \text{msg-set}$
 $\wedge \text{make-priority-key max-time max-node } (\text{recover-msg } k) = k$
proof –
from *recover-msg-unique-existence*[*OF assms*]
have $\text{ex1}: \exists!m. m \in \text{msg-set} \wedge \text{make-priority-key max-time max-node } m = k$.
show *?thesis*
using *theI*'[*OF ex1*] **unfolding** *recover-msg-def*.
qed

Concrete consequence: deterministic selection of the highest-priority key from a non-empty finite `msg_priority_keys` set, and recovery of the unique well-formed D-quencer message that produced it.

corollary *dq-select-highest-deterministic*:
assumes *finite msg-set*
shows $\exists!k. \text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$
 $\wedge k \in \text{msg-priority-keys}$
 $\wedge (\forall k' \in \text{msg-priority-keys}. k' \leq k)$
using *dq-priority.select-highest-deterministic*
[*OF msg-priority-keys-finite*[*OF assms*] *msg-priority-keys-nonempty*]
by *simp*

corollary *dq-select-highest-in-set*:
assumes *finite msg-set*
and $\text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$
shows $k \in \text{msg-priority-keys}$
using *dq-priority.select-highest-in-set*
[*OF msg-priority-keys-finite*[*OF assms*(1)] *assms*(2)] .

corollary *dq-select-highest-message*:
assumes *finite msg-set*
and $\text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$
shows $\text{recover-msg } k \in \text{msg-set}$
 $\wedge \text{make-priority-key max-time max-node } (\text{recover-msg } k) = k$
using *recover-msg-correct*[*OF dq-select-highest-in-set*[*OF assms*]] .

The recovered message is maximal: no well-formed message in the set carries a strictly higher priority key. This projects the key-level maximality of `select_highest` back onto messages, closing the loop of the “highest-priority message” reading.

corollary *dq-select-highest-message-maximal*:
assumes *finite msg-set*
and $\text{dq-priority.select-highest msg-priority-keys} = \text{Some } k$
shows $\forall m \in \text{msg-set}. \text{make-priority-key max-time max-node } m$
 $\leq \text{make-priority-key max-time max-node } (\text{recover-msg } k)$
proof –
have $\text{max}: \forall k' \in \text{msg-priority-keys}. k' \leq k$
using *dq-priority.select-highest-is-max*

```

    [OF msg-priority-keys-finite[OF assms(1)] msg-priority-keys-nonempty
assms(2)]
  by simp
  have rk: make-priority-key max-time max-node (recover-msg k) = k
    using recover-msg-correct[OF dq-select-highest-in-set[OF assms]] by simp
  show ?thesis
  proof
    fix m assume m ∈ msg-set
    then have make-priority-key max-time max-node m ∈ msg-priority-keys
      unfolding msg-priority-keys-def by simp
    with max rk show make-priority-key max-time max-node m
      ≤ make-priority-key max-time max-node (recover-msg k) by simp
  qed
qed
end

```

28 Fair-Leader Aggregate-Progress Instantiation

We instantiate the `fair_leader_system` locale with an in-roster leader schedule and an aggregate pending-count evolution.

The fairness assumption requires that within any `fairness_bound` consecutive epochs, at least one epoch has an honest-tagged leader and that every scheduled leader belongs to the locale's roster. No probability distribution, independence assumption, VRF mechanism, or network execution is formalized.

For concrete instantiation, we need a leader schedule and pending count function that satisfy the locale assumptions. These are provided as parameters in the instantiation context.

```

locale dquencer-liveness = dquencer-system +
  fixes leader-schedule :: nat ⇒ node-info
  and pending-count :: nat ⇒ nat
  assumes fair-leader:
    range leader-schedule ⊆ nodes ∧
    (∀ epoch. ∃ e. epoch ≤ e ∧ e < epoch + fairness-bound ∧
      ni-behavior (leader-schedule e) = Honest)
  and honest-processes:
    [ ni-behavior (leader-schedule e) = Honest; pending-count e > 0 ]
    ⇒ pending-count (Suc e) < pending-count e
  and pending-non-increasing:
    pending-count (Suc e) ≤ pending-count e
begin

```

```

interpretation dq-fair: fair-leader-system
  leader-schedule λn. ni-behavior n = Honest pending-count fairness-bound
proof unfold-locales
  show ∀ epoch. ∃ e. epoch ≤ e ∧ e < epoch + fairness-bound ∧

```

```

      ni-behavior (leader-schedule e) = Honest
    using fair-leader by blast
  next
    fix e
    assume ni-behavior (leader-schedule e) = Honest 0 < pending-count e
    then show pending-count (Suc e) < pending-count e
      using honest-processes by auto
  next
    fix e
    show pending-count (Suc e) ≤ pending-count e
      using pending-non-increasing .
qed

```

Concrete aggregate progress for the D-quencer abstraction: if the pending count is positive, it strictly decreases within **fairness_bound** epochs. This does not identify which request is discharged.

```

corollary dq-starvation-bound:
  assumes pending-count epoch > 0
  shows ∃ e. epoch ≤ e ∧ e < epoch + fairness-bound ∧
    pending-count (Suc e) < pending-count e
  using dq-fair.starvation-bound[OF assms] .

```

end

29 Conditional Safety alongside the Liveness Results

This section records how the safety result of Property 1 (cross-domain state preservation) sits alongside the liveness results of Property 2 (deterministic selection and fair scheduling).

The theorem below, **conditional_safety_preservation**, is a *conditional safety* statement: from a valid global state and an enabled transition, validity itself implies that the target asset is unlocked; the synchronization function succeeds and preserves the global validity invariant. Its proof uses only the safety side (**lock_acquire_success** and **valid_state_preservation**); it does not invoke the liveness interpretations (**dq_priority**, **dq_fair**) and makes no Byzantine, determinism, deadlock, or starvation claim of its own. Those liveness properties are the separate theorems of this theory (**dq_select_highest_deterministic** and **dq_starvation_bound**), each stated in its own right.

The genuine fusion of the two sides — safety freed of its initial-validity hypothesis by a well-founded progress measure on cross-chain inconsistency, under the fair-leader assumption — is **oraclizer_guarded_bounded_convergence** in **Functor_Laws.thy**, which drops the initial-validity hypothesis assumed here.

The statement below is a leaf corollary: it restates `valid_state_preservation` under the explicit precondition that the asset is unlocked — the precondition the lock discipline is designed to establish. It is deliberately not the place where liveness and safety are fused (see `Functor_Laws.thy`).

```
theorem conditional-safety-preservation:
  assumes valid: valid-state gs
    and current: get-reg-state gs source aid = Some s
    and trans: reg-transition s action = Some s'
  shows  $\exists gs'. \text{sync source action aid gs} = \text{Some } gs' \wedge \text{valid-state } gs'$ 
proof –
  have not-locked:  $\neg \text{is-locked } gs \text{ aid}$ 
    using valid unfolding valid-state-def no-locked-without-reason-def by blast
  from not-locked obtain gs-locked where
    lock: acquire-lock gs aid = Some gs-locked
  using lock-acquire-success by auto
  from valid current trans lock have
     $\exists gs'. \text{sync source action aid gs} = \text{Some } gs'$ 
  unfolding sync-def
  using current trans lock
  by (auto simp: get-reg-state-def get-asset-state-def
    split: option.splits
    intro!: exI)
  then obtain gs' where synced: sync source action aid gs = Some gs'
  by auto
  have valid-state gs'
    using valid-state-preservation[OF valid current trans synced] .
  with synced show ?thesis by auto
qed
```

This entry groups its regulatory results into two properties: Property 1, cross-domain state preservation (the safety side), and Property 2, deterministic selection and fair scheduling (the liveness side). Summary of what they together establish:

1. **Safety** (Property 1): After synchronization, all connected chains agree on the regulatory state. The global validity invariant is preserved.
2. **Determinism** (Property 2): Conflicting regulatory actions are resolved by a total order on priority keys. The consensus output — abstracted here as deterministic priority selection — is unique. The `priority_system` locale is instantiated on the `priority_key` type as carrier (with the corresponding messages recovered inside `dquencer_priority_concrete` via `recover_msg`), yielding deterministic selection from any finite non-empty set of valid candidate messages. Priority here resolves conflicts *among regulatory messages*; precedence of regulatory actions over ordinary (non-regulatory) transactions is not modelled — the message universe of this theory contains regulatory actions only.

3. **Deadlock** (Property 2, scope note): deadlock is a concurrency phenomenon. The atomic sync model has no concurrent lock contention (lock holding is a boolean without contended holders), so deadlock does not arise within the model's scope; forced lock release under contention is out of scope, deferred to a future concurrent refinement. The theory states no proved deadlock-freedom theorem.
4. **Aggregate pending-count progress** (Property 2): Under the fair-leader assumption, a positive pending count strictly decreases within a bounded number of epochs. The pending-count assumptions encode a closed system — no new requests arrive within the analysis horizon; individual request fairness, continuous arrivals, and a partially synchronous network lift are not established here.

Open work for subsequent entries: - Compositional assurance across heterogeneous verification regimes. - Refinement: formal correspondence between these models and the deployed implementation.

end

```
theory Composition
  imports State-Preservation
begin
```

30 Guarded Invariants

A guarded transition system: a carrier set closed under stepping, together with an invariant that is preserved by every *guarded* step. The guard isolates exactly the operations that may fire while maintaining the invariant.

```
locale guarded-invariant =
  fixes carrier :: 's set
  and step :: 's ⇒ 'op ⇒ 's option
  and ops :: 'op set
  and inv :: 's ⇒ bool
  and guard :: 's ⇒ 'op ⇒ bool
  assumes step-closed:
     $\llbracket s \in \text{carrier}; \text{opn} \in \text{ops}; \text{step } s \text{ opn} = \text{Some } s' \rrbracket \implies s' \in \text{carrier}$ 
  and guarded-preservation:
     $\llbracket s \in \text{carrier}; \text{opn} \in \text{ops}; \text{inv } s; \text{guard } s \text{ opn}; \text{step } s \text{ opn} = \text{Some } s' \rrbracket$ 
     $\implies \text{inv } s'$ 
```

31 Eventual Dischargers

A scheduler that, within every window of length *window*, produces at least one discharging event. This is a bounded-fairness abstraction: each bounded

window contains at least one progress-bearing event. This condition does not identify queued requests or establish per-request service.

```

locale eventual-discharger =
  fixes schedule :: nat  $\Rightarrow$  'event
    and discharges :: 'event  $\Rightarrow$  bool
    and window :: nat
  assumes bounded-occurrence:
     $\forall t. \exists t'. t \leq t' \wedge t' < t + \text{window} \wedge \text{discharges } (\text{schedule } t')$ 
begin

```

Window positivity is implied by bounded occurrence: the window starting at any time index must be non-empty to contain a discharging slot. It is therefore derived rather than assumed.

```

lemma window-positive: window > 0
proof –
  obtain t' where  $(0::\text{nat}) \leq t'$  and  $t' < 0 + \text{window}$ 
  using bounded-occurrence by blast
  then show ?thesis by simp
qed

end

```

32 Compositional Safety and Liveness

The composition couples a guarded invariant with an eventual discharger through a realization map *realize* that turns a discharging event, in the current state, into an operation. The coupling assumption states that any realized operation is admissible and guarded; invariant preservation along a trajectory then follows from the guard, and progress is supplied by the measure discharge of the converging extension below.

```

locale safety-liveness-composition =
  safe: guarded-invariant carrier step ops inv guard +
  live: eventual-discharger schedule discharges window
for carrier :: 's set
  and step :: 's  $\Rightarrow$  'op  $\Rightarrow$  's option
  and ops :: 'op set
  and inv :: 's  $\Rightarrow$  bool
  and guard :: 's  $\Rightarrow$  'op  $\Rightarrow$  bool
  and schedule :: nat  $\Rightarrow$  'event
  and discharges :: 'event  $\Rightarrow$  bool
  and window :: nat +
fixes realize :: 'event  $\Rightarrow$  's  $\Rightarrow$  'op option
assumes realize-discharges:
   $\llbracket s \in \text{carrier}; \text{discharges } \text{ev}; \text{realize } \text{ev } s = \text{Some } \text{opn} \rrbracket$ 
   $\implies \text{opn} \in \text{ops} \wedge \text{guard } s \text{ opn}$ 
begin

```


One evolution step at absolute time index t : a scheduled discharging event is realized and applied; any non-discharging event (or a realization that does not enable a step) stutters, leaving the state unchanged.

definition $evolve\text{-}step :: nat \Rightarrow 's \Rightarrow 's$ **where**

```

evolve-step t s =
  (if discharges (schedule t) then
    (case realize (schedule t) s of
      None  $\Rightarrow$  s
    | Some opn  $\Rightarrow$  (case step s opn of None  $\Rightarrow$  s | Some s'  $\Rightarrow$  s'))
  else s)

```

The trajectory of length n started at time index k .

fun $run\text{-}from :: nat \Rightarrow 's \Rightarrow nat \Rightarrow 's$ **where**

```

run-from k s 0 = s
| run-from k s (Suc n) = evolve-step (k + n) (run-from k s n)

```

The trajectory started at time 0 , and the induced evolution relation.

definition $run :: 's \Rightarrow nat \Rightarrow 's$ **where**

```

run s n = run-from 0 s n

```

definition $evolves\text{-}to :: 's \Rightarrow nat \Rightarrow 's \Rightarrow bool$ **where**

```

evolves-to s t s'  $\longleftrightarrow$  run s t = s'

```

Trajectories compose: continuing for b more steps re-roots the schedule index by a .

lemma $run\text{-}from\text{-}add$:

```

run-from k s (a + b) = run-from (k + a) (run-from k s a) b

```

proof ($induction\ b$)

case 0

show $?case$ **by** $simp$

next

case $(Suc\ b)$

have $run\text{-}from\ k\ s\ (a + Suc\ b) = evolve\text{-}step\ (k + (a + b))\ (run\text{-}from\ k\ s\ (a + b))$

by $simp$

also have $\dots = evolve\text{-}step\ (k + (a + b))\ (run\text{-}from\ (k + a)\ (run\text{-}from\ k\ s\ a)\ b)$

using $Suc.IH$ **by** $simp$

also have $\dots = run\text{-}from\ (k + a)\ (run\text{-}from\ k\ s\ a)\ (Suc\ b)$

by ($simp\ add$: $add.assoc$)

finally show $?case$.

qed

lemma $evolve\text{-}step\text{-}stutter$:

```

 $\neg discharges\ (schedule\ t) \implies evolve\text{-}step\ t\ s = s$ 

```

by ($simp\ add$: $evolve\text{-}step\text{-}def$)

A single evolution step keeps the carrier.

lemma $evolve\text{-}step\text{-}carrier$:

```

    assumes  $s \in \text{carrier}$ 
    shows  $\text{evolve-step } t \ s \in \text{carrier}$ 
  proof (cases  $\text{discharges } (\text{schedule } t)$ )
    case False
    then show ?thesis using  $\text{assms}$  by (simp add:  $\text{evolve-step-def}$ )
  next
    case True
    show ?thesis
    proof (cases  $\text{realize } (\text{schedule } t) \ s$ )
      case None
      with  $\text{assms True}$  show ?thesis by (simp add:  $\text{evolve-step-def}$ )
    next
      case (Some  $\text{opn}$ )
      note  $r = \text{this}$ 
      show ?thesis
      proof (cases  $\text{step } s \ \text{opn}$ )
        case None
        with  $\text{assms True } r$  show ?thesis by (simp add:  $\text{evolve-step-def}$ )
      next
        case (Some  $s'$ )
        note  $st = \text{this}$ 
        have  $\text{op-in: } \text{opn} \in \text{ops}$  using  $\text{realize-discharges}[OF \ \text{assms True } r]$  by simp
        have  $s' \in \text{carrier}$  using  $\text{safe.step-closed}[OF \ \text{assms op-in } st]$  .
        with  $\text{True } r \ st$  show ?thesis by (simp add:  $\text{evolve-step-def}$ )
      qed
    qed
  qed

```

A single evolution step preserves the invariant on the carrier.

```

lemma  $\text{evolve-step-inv}$ :
  assumes  $s \in \text{carrier}$  and  $\text{inv } s$ 
  shows  $\text{inv } (\text{evolve-step } t \ s)$ 
  proof (cases  $\text{discharges } (\text{schedule } t)$ )
    case False
    then show ?thesis using  $\text{assms}$  by (simp add:  $\text{evolve-step-def}$ )
  next
    case True
    show ?thesis
    proof (cases  $\text{realize } (\text{schedule } t) \ s$ )
      case None
      with  $\text{assms True}$  show ?thesis by (simp add:  $\text{evolve-step-def}$ )
    next
      case (Some  $\text{opn}$ )
      note  $r = \text{this}$ 
      show ?thesis
      proof (cases  $\text{step } s \ \text{opn}$ )
        case None
        with  $\text{assms True } r$  show ?thesis by (simp add:  $\text{evolve-step-def}$ )
      next
        case (Some  $s'$ )
        note  $st = \text{this}$ 
        have  $\text{op-in: } \text{opn} \in \text{ops}$  using  $\text{realize-discharges}[OF \ \text{assms True } r]$  by simp
        have  $s' \in \text{carrier}$  using  $\text{safe.step-closed}[OF \ \text{assms op-in } st]$  .
        with  $\text{True } r \ st$  show ?thesis by (simp add:  $\text{evolve-step-def}$ )
      qed
    qed
  qed

```

```

      case (Some s')
      note st = this
      have c:  $opn \in ops \wedge guard\ s\ opn$  using realize-discharges[OF assms(1) True
r] .
      have inv s'
        using safe.guarded-preservation[OF assms(1) conjunct1[OF c] assms(2)
conjunct2[OF c] st] .
      with True r st show ?thesis by (simp add: evolve-step-def)
    qed
  qed
qed

```

```

lemma run-from-carrier:
  assumes  $s \in carrier$ 
  shows run-from  $k\ s\ n \in carrier$ 
proof (induction n)
  case 0
  show ?case using assms by simp
next
  case (Suc n)
  show ?case using evolve-step-carrier[OF Suc.IH] by simp
qed

```

```

lemma run-from-inv:
  assumes  $s \in carrier$  and inv s
  shows inv (run-from  $k\ s\ n$ )
proof (induction n)
  case 0
  show ?case using assms(2) by simp
next
  case (Suc n)
  have car: run-from  $k\ s\ n \in carrier$  using run-from-carrier[OF assms(1)] .
  have inv (evolve-step ( $k + n$ ) (run-from  $k\ s\ n$ )) using evolve-step-inv[OF car
Suc.IH] .
  then show ?case by simp
qed

```

If no discharging event occurs in the first n slots from k , the state is unchanged.

```

lemma run-from-stutter:
  assumes  $\forall j < n. \neg discharges\ (schedule\ (k + j))$ 
  shows run-from  $k\ s\ n = s$ 
  using assms
proof (induction n)
  case 0
  show ?case by simp
next
  case (Suc n)
  have  $\forall j < n. \neg discharges\ (schedule\ (k + j))$  using Suc.prem by simp

```

```

then have ih: run-from k s n = s using Suc.IH by simp
have  $\neg$  discharges (schedule (k + n)) using Suc.prem by simp
then have evolve-step (k + n) (run-from k s n) = run-from k s n
  by (simp add: evolve-step-stutter)
then show ?case using ih by simp
qed

```

Invariant preservation along the whole trajectory: from an invariant carrier state, every reachable state still satisfies the invariant and stays in the carrier.

```

lemma invariant-preserved:
  assumes s  $\in$  carrier and inv s
  shows inv (run s t)  $\wedge$  run s t  $\in$  carrier
proof -
  have inv (run-from 0 s t) using run-from-inv[OF assms] .
  moreover have run-from 0 s t  $\in$  carrier using run-from-carrier[OF assms(1)] .
  ultimately show ?thesis by (simp add: run-def)
qed
end

```

33 Guarded Bounded Convergence

safety-liveness-composition is extended with a natural-number progress measure whose zero set is contained in the invariant (*measure-zero-inv*) and which every discharging step strictly decreases while the invariant fails (*discharge-progresses*). This is exactly the “well-founded measure connecting non-invariant states to invariant states via discharging events” that drives convergence; well-foundedness is supplied by the natural-number order (*measure progress-measure*).

Together with bounded fairness this yields convergence to the invariant within *progress-measure s * window* steps from an arbitrary carrier state, with *no* assumption that the invariant holds initially.

```

locale converging-composition =
  safety-liveness-composition carrier step ops inv guard schedule discharges window
  realize
  for carrier :: 's set
  and step :: 's  $\Rightarrow$  'op  $\Rightarrow$  's option
  and ops :: 'op set
  and inv :: 's  $\Rightarrow$  bool
  and guard :: 's  $\Rightarrow$  'op  $\Rightarrow$  bool
  and schedule :: nat  $\Rightarrow$  'event
  and discharges :: 'event  $\Rightarrow$  bool
  and window :: nat
  and realize :: 'event  $\Rightarrow$  's  $\Rightarrow$  'op option +
  fixes progress-measure :: 's  $\Rightarrow$  nat

```

assumes *discharge-progresses*:
 $\llbracket s \in \text{carrier}; \neg \text{inv } s; \text{discharges } ev \rrbracket$
 $\implies \exists \text{opn } s'. \text{realize } ev \ s = \text{Some } \text{opn} \wedge \text{step } s \ \text{opn} = \text{Some } s'$
 $\wedge \text{progress-measure } s' < \text{progress-measure } s$
begin

The zero set of the measure lies inside the invariant — derived rather than assumed: at measure zero a failing invariant would admit a discharging step whose target measure lies strictly below zero.

lemma *measure-zero-inv*:
assumes $s \in \text{carrier}$ **and** $\text{progress-measure } s = 0$
shows $\text{inv } s$
proof (*rule ccontr*)
assume $\text{ninvs}: \neg \text{inv } s$
obtain t **where** $\text{discharges } (\text{schedule } t)$
using *live.bounded-occurrence* **by** *blast*
then obtain $\text{opn } s'$ **where** $\text{progress-measure } s' < \text{progress-measure } s$
using $\text{discharge-progresses}[\text{OF } \text{assms}(1) \ \text{ninvs}]$ **by** *blast*
with $\text{assms}(2)$ **show** *False* **by** *simp*
qed

definition *convergence-bound* $:: 's \Rightarrow \text{nat}$ **where**
 $\text{convergence-bound } s = \text{progress-measure } s * \text{window}$

The core convergence lemma, by strong induction on the measure. From any carrier state with measure m , an invariant state is reached within $m * \text{window}$ steps, regardless of the starting time index k .

lemma *convergence-bound-reachable*:
 $\text{progress-measure } s = m \implies s \in \text{carrier}$
 $\implies \exists t \leq m * \text{window}. \text{inv } (\text{run-from } k \ s \ t)$
proof (*induction m arbitrary: s k rule: less-induct*)
case (*less m*)
show *?case*
proof (*cases inv s*)
case *True*
have $\text{inv } (\text{run-from } k \ s \ 0)$ **using** *True* **by** *simp*
moreover have $(0::\text{nat}) \leq m * \text{window}$ **by** *simp*
ultimately show *?thesis* **by** *blast*
next
case *False*
have $m\text{-eq}: \text{progress-measure } s = m$ **using** *less.prem1* .
have $\text{car}: s \in \text{carrier}$ **using** *less.prem2* .
— Locate the least discharging offset within the fairness window.
obtain t' **where** $t'1: k \leq t'$ **and** $t'2: t' < k + \text{window}$
and $t'3: \text{discharges } (\text{schedule } t')$
using *live.bounded-occurrence* **by** *blast*
define Q **where** $Q = (\lambda j. j < \text{window} \wedge \text{discharges } (\text{schedule } (k + j)))$
have $w: t' - k < \text{window}$ **using** $t'1 \ t'2$ **by** *linarith*
have $kk: k + (t' - k) = t'$ **using** $t'1$ **by** *simp*

have $Qwit$: $Q (t' - k)$ **unfolding** $Q\text{-def}$ **using** $w \ t'3 \ kk$ **by** *simp*
 define $j0$ **where** $j0 = (LEAST \ j. \ Q \ j)$
 have $Qj0$: $Q \ j0$ **unfolding** $j0\text{-def}$ **using** $Qwit$ **by** (rule $LeastI[\text{where } P = Q]$)
 have $j0w$: $j0 < window$ **using** $Qj0$ **unfolding** $Q\text{-def}$ **by** *simp*
 have $disc0$: $discharges \ (schedule \ (k + j0))$ **using** $Qj0$ **unfolding** $Q\text{-def}$ **by**
simp
 have $before$: $\forall j < j0. \neg discharges \ (schedule \ (k + j))$
proof (intro *allI impI*)
 fix j **assume** jl : $j < j0$
 have lt : $j < Least \ Q$ **using** jl **by** (*simp add: j0-def*)
 have nQ : $\neg Q \ j$ **using** *not-less-Least[OF lt]* .
 have jw : $j < window$ **using** $jl \ j0w$ **by** *simp*
 show $\neg discharges \ (schedule \ (k + j))$ **using** $nQ \ jw$ **by** (*simp add: Q-def*)
qed
 have $stut$: $run\text{-from} \ k \ s \ j0 = s$ **using** *run-from-stutter[OF before]* .
 — The discharging step at offset $j0$ strictly decreases the measure.
 obtain $opn \ s'$ **where** r : $realize \ (schedule \ (k + j0)) \ s = Some \ opn$
 and st : $step \ s \ opn = Some \ s'$
 and dec : $progress\text{-measure} \ s' < progress\text{-measure} \ s$
 using *discharge-progresses[OF car False disc0]* **by** *blast*
 have $op\text{-in}$: $opn \in ops$ **using** *realize-discharges[OF car disc0 r]* **by** *simp*
 have $s'\text{-car}$: $s' \in carrier$ **using** *safe.step-closed[OF car op-in st]* .
 have ev : $run\text{-from} \ k \ s \ (Suc \ j0) = s'$
proof –
 have $run\text{-from} \ k \ s \ (Suc \ j0) = evolve\text{-step} \ (k + j0) \ (run\text{-from} \ k \ s \ j0)$ **by** *simp*
 also have $\dots = evolve\text{-step} \ (k + j0) \ s$ **using** $stut$ **by** *simp*
 also have $\dots = s'$ **using** $disc0 \ r \ st$ **by** (*simp add: evolve-step-def*)
 finally show *?thesis* .
qed
 have $decm$: $progress\text{-measure} \ s' < m$ **using** $dec \ m\text{-eq}$ **by** *simp*
 — Apply the induction hypothesis to the smaller-measure state s' .
 have $\exists t \leq progress\text{-measure} \ s' * window. inv \ (run\text{-from} \ (k + Suc \ j0) \ s' \ t)$
 using *less.IH[OF decm refl s'-car]* **by** *blast*
then obtain $t2$ **where** $t2b$: $t2 \leq progress\text{-measure} \ s' * window$
 and $inv2$: $inv \ (run\text{-from} \ (k + Suc \ j0) \ s' \ t2)$ **by** *blast*
 have $comp$: $run\text{-from} \ k \ s \ (Suc \ j0 + t2) = run\text{-from} \ (k + Suc \ j0) \ s' \ t2$
proof –
 have $run\text{-from} \ k \ s \ (Suc \ j0 + t2)$
 $= run\text{-from} \ (k + Suc \ j0) \ (run\text{-from} \ k \ s \ (Suc \ j0)) \ t2$
 by (rule *run-from-add*)
 also have $\dots = run\text{-from} \ (k + Suc \ j0) \ s' \ t2$ **by** (*simp only: ev*)
 finally show *?thesis* .
qed
 have $inv\text{-fin}$: $inv \ (run\text{-from} \ k \ s \ (Suc \ j0 + t2))$ **using** $comp \ inv2$ **by** *simp*
 have $Suc \ j0 + t2 \leq window + progress\text{-measure} \ s' * window$
 using $j0w \ t2b$ **by** *linarith*
 also have $\dots = (1 + progress\text{-measure} \ s') * window$
 by (*simp add: add-mult-distrib*)
 also have $\dots \leq m * window$

```

proof –
  have  $1 + \text{progress-measure } s' \leq m$  using decn by linarith
  then show ?thesis by (rule mult-le-mono1)
qed
finally have  $\text{Suc } j0 + t2 \leq m * \text{window}$  .
then show ?thesis using inv-fin by blast
qed
qed

```

Guarded bounded convergence. From an arbitrary carrier state with no assumption that the invariant holds the system reaches, within *convergence-bound* s evolution steps, a state satisfying the invariant.

```

theorem bounded-convergence-from-arbitrary:
  assumes  $s \in \text{carrier}$ 
  shows  $\exists t \leq \text{convergence-bound } s. \exists s'. \text{evolves-to } s \ t \ s' \wedge \text{inv } s'$ 
proof –
  obtain  $t$  where  $tb: t \leq \text{progress-measure } s * \text{window}$ 
    and  $\text{inv-}t: \text{inv } (\text{run-from } 0 \ s \ t)$ 
    using convergence-bound-reachable[OF refl assms] by blast
  have  $\text{evolves-to } s \ t \ (\text{run } s \ t)$  by (simp add: evolves-to-def)
  with  $tb \ \text{inv-}t$  show ?thesis
    by (auto simp: convergence-bound-def run-def evolves-to-def)
qed

end

end

```

```

theory Proof-Automation
  imports State-Preservation HOL-Eisbach.Eisbach
begin

```

34 Discharge Methods for Instance Obligations

Two named theorem collections feed the methods. *discharge-simps* holds the equational content of a concrete instance: defining equations of the state/action/terminal sets, finiteness facts, terminal-absorption equations, and naturality equations of structured state maps (conditional rewrites whose premises the simplifier discharges from the goal's assumptions). *discharge-intros* holds introduction rules whose extra premises must be instantiated by unification against the goal's assumptions — typically closure lemmas, whose condition variables do not all occur in the conclusion and which are therefore out of reach of conditional rewriting.

```

named-theorems discharge-simps
   $\langle \text{defining equations and conditional rewrites consumed by the discharge methods} \rangle$ 

```

named-theorems *discharge-intros*
⟨introduction rules consumed by the discharge methods⟩

named-theorems *discharge-dels*
⟨default simplification rules suppressed by the discharge methods ---
typically the defining equations of structured state maps introduced with
⟨fun⟩, whose eager unfolding would destroy the redexes that the
conditional naturality rewrites of ⟨discharge-simps⟩ match on⟩

The common skeleton: unfold the locale predicate into its atomic obligations (*unfold-locales* also discharges sub-locale obligations already available from registered interpretations), then close each remaining obligation by simplifier-driven exhaustive case analysis — membership of an action in an enumerated set splits into finitely many cases, after which the defining equations of the transition functions decide the goal, with *option/if* splits for partially defined transitions.

discharge-state-machine is the entry point for *state-machine* obligations (finiteness, terminal containment, terminal absorption, closure, domain); *discharge-preservation* is the entry point for *state-preservation* obligations, whose unfolding contains the two sub-machines' obligations together with the morphism axioms (well-definedness, terminal preservation, and the two naturality directions).

method *discharge-state-machine* **declares** *discharge-simps discharge-intros discharge-dels* =
 (*unfold-locales*;
 (*auto simp add: discharge-simps simp del: discharge-dels*
 intro: discharge-intros
 split: option.splits if-splits))

method *discharge-preservation* **declares** *discharge-simps discharge-intros discharge-dels* =
 (*unfold-locales*;
 (*auto simp add: discharge-simps simp del: discharge-dels*
 intro: discharge-intros
 split: option.splits if-splits))

The two entry points share their skeleton deliberately: a *state-preservation* goal unfolds into the union of the two obligation families, so the preservation method must subsume the machine method. Keeping both names separates the reusable surface by intent — an importer discharging a bare machine instance is not invited to reach for the morphism-level method — and lets the two evolve independently if the obligation families diverge in a future revision.

end

theory *Functor-Laws*


```

imports
  Composition
  Regulatory-Instance
  DQuencer-Instance
  Proof-Automation
  ADS-Functor.Merkle-Interface
begin

```

35 Functor Laws on State-Preservation Morphisms

We regard the locale *state-preservation* as a morphism between two objects of a category. An object is a state machine (*states*, *actions*, *transition*, *terminal*); a morphism from one object to another is a pair (*state-map*, *action-map*) satisfying the well-definedness, terminal-preservation and naturality conditions of *state-preservation*. The following three theorems are the category axioms for this notion of morphism.

35.1 Identity morphism

The identity pair (*id*, *id*) is a state-preservation morphism from any object to itself: *F* sends identities to identities.

```

theorem preservation-id:
  assumes state-machine states actions transition terminal
  shows state-preservation states actions transition terminal
      states actions transition terminal id id

  using assms
  unfolding state-preservation-def state-preservation-axioms-def
  by simp

```

35.2 Composition of morphisms

The composite of two state-preservation morphisms is a state-preservation morphism: the naturality square of the second morphism is stacked on that of the first. This is the cross-domain analogue of "closed under composition".

```

theorem preservation-compose:
  assumes f: state-preservation Sa Aa Ta Fa Sb Ab Tb Fb f f'
    and g: state-preservation Sb Ab Tb Fb Sc Ac Tc Fc g g'
  shows state-preservation Sa Aa Ta Fa Sc Ac Tc Fc (g ∘ f) (g' ∘ f')

proof –
  interpret f: state-preservation Sa Aa Ta Fa Sb Ab Tb Fb f f' by (rule f)
  interpret g: state-preservation Sb Ab Tb Fb Sc Ac Tc Fc g g' by (rule g)
  show ?thesis
proof (unfold-locales)
  fix s assume s ∈ Sa
  then show (g ∘ f) s ∈ Sc

```

```

    by (simp add: f.state-map-well-defined g.state-map-well-defined)
next
  fix a assume a ∈ Aa
  then show (g' ∘ f') a ∈ Ac
    by (simp add: f.action-map-well-defined g.action-map-well-defined)
next
  fix s assume s ∈ Fa
  then show (g ∘ f) s ∈ Fc
    by (simp add: f.terminal-preservation g.terminal-preservation)
next
  fix s a s'
  assume A: s ∈ Sa and B: a ∈ Aa and C: Ta s a = Some s'
  have fs: f s ∈ Sb using A f.state-map-well-defined by simp
  have fa: f' a ∈ Ab using B f.action-map-well-defined by simp
  have Tb (f s) (f' a) = Some (f s') using f.naturality A B C by simp
  then have Tc (g (f s)) (g' (f' a)) = Some (g (f s'))
    using g.naturality fs fa by simp
  then show Tc ((g ∘ f) s) ((g' ∘ f') a) = Some ((g ∘ f) s')
    by (simp add: comp-def)
next
  fix s a
  assume A: s ∈ Sa and B: a ∈ Aa and C: Ta s a = None
  have fs: f s ∈ Sb using A f.state-map-well-defined by simp
  have fa: f' a ∈ Ab using B f.action-map-well-defined by simp
  have Tb (f s) (f' a) = None using f.naturality-none A B C by simp
  then have Tc (g (f s)) (g' (f' a)) = None
    using g.naturality-none fs fa by simp
  then show Tc ((g ∘ f) s) ((g' ∘ f') a) = None
    by (simp add: comp-def)
qed
qed

```

35.3 Associativity of morphism composition

Composition of state-preservation morphisms is associative on both the state and the action components. Since the components are functions, this is exactly associativity of function composition; the three morphism hypotheses fix the objects across which the equation is read, completing the category axioms (identity, composition, associativity).

theorem *preservation-assoc:*

```

  assumes state-preservation Sa Aa Ta Fa Sb Ab Tb Fb f f'
    and state-preservation Sb Ab Tb Fb Sc Ac Tc Fc g g'
    and state-preservation Sc Ac Tc Fc Sd Ad Td Fd k k'
  shows state-preservation Sa Aa Ta Fa Sd Ad Td Fd
    ((k ∘ g) ∘ f) ((k' ∘ g') ∘ f')
    ∧ state-preservation Sa Aa Ta Fa Sd Ad Td Fd
    (k ∘ (g ∘ f)) (k' ∘ (g' ∘ f'))
    ∧ ((k ∘ g) ∘ f = k ∘ (g ∘ f))

```

$$\wedge ((k' \circ g') \circ f' = k' \circ (g' \circ f'))$$

proof –

have *kg*: *state-preservation* $S_b A_b T_b F_b S_d A_d T_d F_d$
 $(k \circ g) (k' \circ g')$
using *preservation-compose*[*OF* *assms*(2) *assms*(3)] .

have *gf*: *state-preservation* $S_a A_a T_a F_a S_c A_c T_c F_c$
 $(g \circ f) (g' \circ f')$
using *preservation-compose*[*OF* *assms*(1) *assms*(2)] .

have *left*: *state-preservation* $S_a A_a T_a F_a S_d A_d T_d F_d$
 $((k \circ g) \circ f) ((k' \circ g') \circ f')$
using *preservation-compose*[*OF* *assms*(1) *kg*] .

have *right*: *state-preservation* $S_a A_a T_a F_a S_d A_d T_d F_d$
 $(k \circ (g \circ f)) (k' \circ (g' \circ f'))$
using *preservation-compose*[*OF* *gf* *assms*(3)] .

from *left right* **show** *?thesis* **by** (*simp add: comp-assoc*)

qed

36 Composition Exercised: A Heterogeneous Three-Domain Chain

preservation-compose is exercised on a concrete chain of three distinct domains: the off-chain DAML permission ledger (structured records), the on-chain regulatory enum, and Chain B with its dedicated four-action vocabulary. In HOL the morphisms of a heterogeneous chain cannot be packed into one object — the domains carry genuinely different state and action types — so the correct formal expression of an N -domain chain is exactly the composite of pairwise morphisms, with *preservation-assoc* making longer path composites unambiguous. Both legs are scoped to the escalation subset of the action vocabulary: the DAML machine restricts a fortiori (a machine restricted to a subset of its action set is again a machine), and the layer-crossing naturality of *daml-to-reg* holds on the subset because it holds on the full vocabulary.

lemma *daml-escalation-state-machine*:

state-machine daml-states escalation-actions daml-transition daml-terminal

proof *unfold-locales*

show *finite daml-states* **by** (*rule daml-states-finite*)

next

show *finite escalation-actions* **unfolding** *escalation-actions-def* **by** *simp*

next

show *daml-terminal* $\subseteq$ *daml-states* **by** (*rule daml-terminal-subset*)

next

fix $p a$

assume $p \in \text{daml-terminal}$ **and** $a \in \text{escalation-actions}$

then show *daml-transition* $p a = \text{None}$

using *daml-terminal-absorbing reg-actions-UNIV* **by** *auto*

next

```

fix  $p \ a \ p'$ 
assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
  and  $\text{daml-transition } p \ a = \text{Some } p'$ 
then show  $p' \in \text{daml-states}$ 
  using daml-transition-closed reg-actions-UNIV by auto
next
  fix  $p :: \text{daml-perm}$  and  $a :: \text{reg-action}$ 
  assume  $p \notin \text{daml-states}$ 
  then show  $\text{daml-transition } p \ a = \text{None}$  by (rule daml-transition-outside-states)
qed

```

The escalation-scoped backward bridge: *daml-to-reg* as a morphism from the DAML machine to the regulatory machine, both restricted to the escalation actions.

lemma *daml-to-reg-escalation-bridge*:

state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states escalation-actions reg-transition reg-terminal
daml-to-reg id

proof *unfold-locales*

— Source state-machine obligations (the DAML machine on the escalation subset is not registered, so its six obligations remain pending); the target machine coincides with the source machine of the registered *escalation-preservation* interpretation and is discharged automatically.

```

  show finite daml-states by (rule daml-states-finite)
next
  show finite escalation-actions unfolding escalation-actions-def by simp
next
  show  $\text{daml-terminal} \subseteq \text{daml-states}$  by (rule daml-terminal-subset)
next
  fix  $p \ a$ 
  assume  $p \in \text{daml-terminal}$  and  $a \in \text{escalation-actions}$ 
  then show  $\text{daml-transition } p \ a = \text{None}$ 
    using daml-terminal-absorbing reg-actions-UNIV by auto
next
  fix  $p \ a \ p'$ 
  assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
    and  $\text{daml-transition } p \ a = \text{Some } p'$ 
  then show  $p' \in \text{daml-states}$ 
    using daml-transition-closed reg-actions-UNIV by auto
next
  fix  $p :: \text{daml-perm}$  and  $a :: \text{reg-action}$ 
  assume  $p \notin \text{daml-states}$ 
  then show  $\text{daml-transition } p \ a = \text{None}$  by (rule daml-transition-outside-states)
next
  fix  $p$ 
  assume  $p \in \text{daml-states}$ 
  then show  $\text{daml-to-reg } p \in \text{reg-states}$  using reg-states-UNIV by auto
next
  fix  $a$ 

```

```

    assume  $a \in \text{escalation-actions}$ 
    then show  $\text{id } a \in \text{escalation-actions}$  by simp
next
  fix  $p$ 
  assume  $p \in \text{daml-terminal}$ 
  then have  $p = \text{reg-to-daml CONFISCATED unfolding daml-terminal-def}$  by
simp
  then have  $\text{daml-to-reg } p = \text{CONFISCATED using daml-to-reg-to-daml-id}$  by
simp
  then show  $\text{daml-to-reg } p \in \text{reg-terminal unfolding reg-terminal-def}$  by simp
next
  fix  $p \ a \ p'$ 
  assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
  and  $\text{daml-transition } p \ a = \text{Some } p'$ 
  then show  $\text{reg-transition } (\text{daml-to-reg } p) (\text{id } a) = \text{Some } (\text{daml-to-reg } p')$ 
  using  $\text{daml-to-reg-naturality-some reg-actions-UNIV}$  by auto
next
  fix  $p \ a$ 
  assume  $p \in \text{daml-states}$  and  $a \in \text{escalation-actions}$ 
  and  $\text{daml-transition } p \ a = \text{None}$ 
  then show  $\text{reg-transition } (\text{daml-to-reg } p) (\text{id } a) = \text{None}$ 
  using  $\text{daml-to-reg-naturality-none reg-actions-UNIV}$  by auto
qed

```

The escalation morphism of *Cross-Domain-State-Preservation.Regulatory-Instance* in predicate (bundle) form, so that it can be fed to *preservation-compose*.

lemma *escalation-preservation-bundle*:

```

state-preservation reg-states escalation-actions reg-transition reg-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
id escalation-action-map

```

— This very instance is registered as the *escalation-preservation* interpretation of *Cross-Domain-State-Preservation.Regulatory-Instance*, so every obligation is discharged from the registration.

by *unfold-locales*

The composed three-domain morphism: DAML permission records, synchronized through the regulatory enum, arrive on Chain B’s vocabulary. The composite is literally *preservation-compose* applied to the two legs; no new naturality argument is needed, which is the point of the functor laws. Arbitrary finite heterogeneous chains are expressed the same way, as composites of pairwise morphisms.

theorem *daml-to-chain-b-composed*:

```

state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
(id o daml-to-reg) (escalation-action-map o id)
using preservation-compose[OF daml-to-reg-escalation-bridge
escalation-preservation-bundle] .

```

The domain guard of the layer-crossing bridge is load-bearing, not a

notational convenience. The record below carries the *D-Seized* tag with no seizing party: it violates *valid-daml-perm* and lies outside *daml-states*. The guarded DAML transition rejects *RELEASE* on it, while the bare enum projection of the same record would accept it — so dropping the guard would break naturality on exactly such records. Preservation holds on the carved-out domain and only there; the guard is the boundary of the preservation guarantee, recorded here permanently as a machine-checked witness.

definition *invalid-daml-perm* :: *daml-perm* **where**

invalid-daml-perm =
 (| *status-tag* = *D-Seized*, *seized-by* = *None*, *restriction-scope* = *None* |)

lemma *guard-is-load-bearing*:

¬ *valid-daml-perm invalid-daml-perm*
invalid-daml-perm ∉ *daml-states*
daml-transition invalid-daml-perm RELEASE = *None*
reg-transition (daml-to-reg invalid-daml-perm) RELEASE = *Some ACTIVE*

proof –

show ¬ *valid-daml-perm invalid-daml-perm*
by (*simp add: valid-daml-perm-def invalid-daml-perm-def*)

have ∧*s. invalid-daml-perm* ≠ *reg-to-daml s*

proof –

fix *s* **show** *invalid-daml-perm* ≠ *reg-to-daml s*
by (*cases s*) (*auto simp: invalid-daml-perm-def*)

qed

then show *notin: invalid-daml-perm* ∉ *daml-states*
by (*auto simp: daml-states-def*)

from *notin* **show** *daml-transition invalid-daml-perm RELEASE* = *None*
by (*rule daml-transition-outside-states*)

show *reg-transition (daml-to-reg invalid-daml-perm) RELEASE* = *Some ACTIVE*

by (*simp add: invalid-daml-perm-def*)

qed

37 Proof Automation Exercised on the Regulatory Instances

The discharge methods of *Cross-Domain-State-Preservation.Proof-Automation* are fed here with the equational content of the regulatory, Chain B and DAML domains, and then exercised: each *-via-method* lemma below restates a preservation or machine instance — proved manually above or registered in *Cross-Domain-State-Preservation.Regulatory-Instance* — and discharges it with a single method invocation, giving a one-to-one comparison between the manual proofs and the automated ones. The instance obligations are not weakened in any way: the statements are identical to their manual counterparts.

Finiteness of the enumerated regulatory types, in the normal form the simplifier reaches after rewriting the set definitions to *UNIV*.

lemma *finite-reg-state-UNIV* [*discharge-simps*]: *finite* (*UNIV* :: *reg-state set*)
using *reg-sm.finite-states* **by** (*simp add: reg-states-UNIV*)

lemma *finite-reg-action-UNIV* [*discharge-simps*]: *finite* (*UNIV* :: *reg-action set*)
using *reg-sm.finite-actions* **by** (*simp add: reg-actions-UNIV*)

Terminal and well-definedness helpers for the layer-crossing maps, in conditional-rewrite form.

lemma *daml-to-reg-terminal-val* [*discharge-simps*]:
 $p \in \text{daml-terminal} \implies \text{daml-to-reg } p = \text{CONFISCATED}$
by (*auto simp: daml-terminal-def daml-to-reg-to-daml-id*)

lemma *reg-to-daml-terminal* [*discharge-simps*]:
 $s \in \text{reg-terminal} \implies \text{reg-to-daml } s \in \text{daml-terminal}$
by (*auto simp: reg-terminal-def daml-terminal-def*)

lemma *reg-to-daml-in-states* [*discharge-simps*]:
 $\text{reg-to-daml } s \in \text{daml-states}$
by (*simp add: daml-states-def*)

The collections. Equational and conditional-rewrite content goes to *discharge-simps*; closure rules, whose condition variables do not all occur in their conclusions, go to *discharge-intros*.

lemmas [*discharge-simps*] =
reg-states-UNIV reg-actions-UNIV
escalation-actions-def chain-b-actions-def reg-terminal-def
confiscated-terminal chain-b-terminal
daml-states-finite daml-terminal-subset
daml-terminal-absorbing daml-transition-outside-states
reg-to-daml-naturality-some reg-to-daml-naturality-none
daml-to-reg-naturality-some daml-to-reg-naturality-none
escalation-naturality-some escalation-naturality-none

lemmas [*discharge-intros*] =
reg-transition-closed chain-b-transition-closed daml-transition-closed

The layer-crossing maps are introduced with *fun*, so their defining equations are default simplification rules; the methods must keep the maps opaque so that the conditional naturality and terminal rewrites above can match their redexes.

lemmas [*discharge-dels*] =
daml-to-reg.simps reg-to-daml.simps

The comparison lemmas. The first three restate instances proved manually in this theory; none of these instances is itself registered, so the goal

is not discharged wholesale by a registration — sub-machine obligations already available from prior registrations are absorbed by *unfold-locales*, and the method closes every remaining obligation from the collections alone.

lemma *daml-escalation-state-machine-via-method:*

state-machine daml-states escalation-actions daml-transition daml-terminal
by *discharge-state-machine*

lemma *daml-to-reg-escalation-bridge-via-method:*

state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states escalation-actions reg-transition reg-terminal
daml-to-reg id
by *discharge-preservation*

lemma *daml-to-chain-b-composed-via-method:*

state-preservation daml-states escalation-actions daml-transition daml-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
(id ∘ daml-to-reg) (escalation-action-map ∘ id)
by *discharge-preservation*

The forward escalation-scoped bridge is a new instance with no manual counterpart: the method constructs it outright, which is the reuse claim in its sharpest form.

lemma *reg-to-daml-escalation-bridge-via-method:*

state-preservation reg-states escalation-actions reg-transition reg-terminal
daml-states escalation-actions daml-transition daml-terminal
reg-to-daml id
by *discharge-preservation*

The remaining three restate instances registered as interpretations in *Cross-Domain-State-Preservation.Regulatory-Instance*; for these, *unfold-locales* discharges the obligations from the registrations, so the one-line proofs are registration-backed rather than search-backed. The search-backed counterparts above show that the method does not depend on the registrations.

lemma *escalation-preservation-via-method:*

state-preservation reg-states escalation-actions reg-transition reg-terminal
reg-states chain-b-actions chain-b-transition reg-terminal
id escalation-action-map
by *discharge-preservation*

lemma *forward-layer-preservation-via-method:*

state-preservation reg-states reg-actions reg-transition reg-terminal
daml-states reg-actions daml-transition daml-terminal
reg-to-daml id
by *discharge-preservation*

lemma *backward-layer-preservation-via-method:*

state-preservation daml-states reg-actions daml-transition daml-terminal

$\text{reg-states reg-actions reg-transition reg-terminal}$
 daml-to-reg id
 by *discharge-preservation*

38 State Glue: Join and Refinement

The two glue predicates lift the authenticated-data-structure operations to the global-state level. *state-join* $sa\ sb\ sab$ says that sab is the consistent combination of the partial views sa and sb : on every chain connected in either view, sab carries the value of whichever view knows the chain (preferring sa), the two views agree wherever both are defined, and sab reveals no chain beyond the union of the two views (the containment clause). This is the state-level reading of the ADS *join*: on revealed holdings sab is the least consistent combination of sa and sb — the lock components are unconstrained — and the agreement clause is the state-level reading of equal hashes (mergeability). Without the containment clause a candidate join could carry an arbitrary holding on a chain outside both views and still be accepted; the regression witness *rogue-join-excluded* below records that such a candidate is now rejected.

state-refines $sa\ sb$ says that sa is a partial view of sb : every chain known to sa is known to sb with the same regulatory state. This is the state-level reading of the ADS blinding order *bo*: a more-blinded value agrees with a less-blinded one on everything it can observe.

definition *state-join* :: $\text{global-state} \Rightarrow \text{global-state} \Rightarrow \text{global-state} \Rightarrow \text{bool}$ **where**

$\text{state-join } sa\ sb\ sab \equiv$
 $(\forall \text{aid } c.$
 $(c \in \text{connected-chains } sa\ \text{aid} \vee c \in \text{connected-chains } sb\ \text{aid})$
 $\longrightarrow \text{get-reg-state } sab\ c\ \text{aid} =$
 $(\text{if } c \in \text{connected-chains } sa\ \text{aid} \text{ then } \text{get-reg-state } sa\ c\ \text{aid}$
 $\text{else } \text{get-reg-state } sb\ c\ \text{aid}))$
 $\wedge (\forall \text{aid } c1\ c2.$
 $c1 \in \text{connected-chains } sa\ \text{aid} \wedge c2 \in \text{connected-chains } sb\ \text{aid}$
 $\longrightarrow \text{get-reg-state } sa\ c1\ \text{aid} = \text{get-reg-state } sb\ c2\ \text{aid})$
 $\wedge (\forall \text{aid } c.$
 $c \in \text{connected-chains } sab\ \text{aid}$
 $\longrightarrow c \in \text{connected-chains } sa\ \text{aid} \vee c \in \text{connected-chains } sb\ \text{aid})$

definition *state-refines* :: $\text{global-state} \Rightarrow \text{global-state} \Rightarrow \text{bool}$ **where**

$\text{state-refines } sa\ sb \equiv$
 $\forall \text{aid } c. c \in \text{connected-chains } sa\ \text{aid}$
 $\longrightarrow c \in \text{connected-chains } sb\ \text{aid} \wedge \text{get-reg-state } sa\ c\ \text{aid} = \text{get-reg-state } sb\ c\ \text{aid}$

Refinement is reflexive and transitive: a partial-view preorder on states.

lemma *state-refines-refl*: $\text{state-refines } sa\ sa$

unfolding *state-refines-def* **by** *simp*

lemma *state-refines-trans*:

state-refines sa sb $\implies$ state-refines sb sc $\implies$ state-refines sa sc

unfolding *state-refines-def* **by** *metis*

A consistent state refined by another transports its consistency upward:
if *sa* refines a consistent *sb*, then *sa* is consistent.

lemma *state-refines-preserves-consistency*:

assumes *state-refines sa sb* **and** *consistent-state sb*

shows *consistent-state sa*

unfolding *consistent-state-def*

proof (*intro allI impI*)

fix *c1 c2 aid s1 s2*

assume *get-reg-state sa c1 aid = Some s1* **and** *get-reg-state sa c2 aid = Some s2*

moreover have *c1 $\in$ connected-chains sa aid* **and** *c2 $\in$ connected-chains sa aid*

using *calculation* **unfolding** *connected-chains-def asset-exists-def*
get-reg-state-def get-asset-state-def

by (*auto split: option.splits*)

ultimately have *get-reg-state sb c1 aid = Some s1* **and** *get-reg-state sb c2 aid = Some s2*

using *assms(1)* **unfolding** *state-refines-def* **by** *metis+*

then show *s1 = s2* **using** *assms(2)* **unfolding** *consistent-state-def* **by** *blast*
qed

39 Authenticated Cross-Domain State

An authenticated state structure equips a Merkle interface (*h*, *bo*, *m*) with an extraction map *extract-map* that reads a global state out of an authenticated value. The three coherence assumptions tie the authenticated layer to the state-glue layer:

* *extract-respects-merging*: merging hash-compatible authenticated values yields a state that is the join of the extracted views; * *extract-under-blinding*: a blinding of an authenticated value extracts to a refinement of the extracted state; * *extract-preserves-validity*: every extracted state is valid.

The merging and blinding assumptions are guarded by the hash-equality condition *h a = h b*, which is the ADS notion of mergeability. The two soundness theorems below discharge this condition from the ADS lemmas *merge-respects-hashes* and *hash*, and use *join* to exhibit the join as a refinement of each input so the Merkle interface is used in the proofs, not merely imported.

Because *extract-map* is partial, this construction is functorial on the extractable values and the blinding morphisms whose relevant endpoints extract. It is not a total object map over every value in the ambient ADS carrier unless a concrete instance separately establishes total extraction.

Reuse beyond the regulatory instance: the glue layer commits only to a single authoritative value per object identifier, so any system of that shape is a potential transplant target — for instance Merkle-replicated registries, authenticated cross-shard asset records, or the public/witness layering of zero-knowledge proofs, where *state-refines* would express that a public view is a blinding refinement of a full witness. The present entry instantiates the layer only on the regulatory state model; those other domains are not formalized here.

The extraction map is named *extract-map* rather than *extract*, because *extract* is a reserved Isabelle command keyword and cannot be used as a locale parameter name.

```
locale authenticated-state =
  mk: merkle-interface h bo m
  for h :: 'd  $\Rightarrow$  'e
    and bo :: 'd  $\Rightarrow$  'd  $\Rightarrow$  bool
    and m :: 'd  $\Rightarrow$  'd  $\Rightarrow$  'd option
    and extract-map :: 'd  $\Rightarrow$  global-state option +
  assumes extract-respects-merging:
     $\llbracket h\ a = h\ b; m\ a\ b = \text{Some}\ ab; \text{extract-map}\ a = \text{Some}\ sa; \text{extract-map}\ b = \text{Some}\ sb \rrbracket$ 
     $\implies \exists\ sab. \text{extract-map}\ ab = \text{Some}\ sab \wedge \text{state-join}\ sa\ sb\ sab$ 
  and extract-under-blinding:
     $\llbracket h\ a = h\ b; bo\ a\ b; \text{extract-map}\ b = \text{Some}\ sb \rrbracket$ 
     $\implies \exists\ sa. \text{extract-map}\ a = \text{Some}\ sa \wedge \text{state-refines}\ sa\ sb$ 
  and extract-preserves-validity:
     $\text{extract-map}\ a = \text{Some}\ s \implies \text{valid-state}\ s$ 
begin
```

Blinding preserves hashes: the explicit state-level use of the *mk.hash* lemma of the Merkle interface.

```
lemma bo-hash-eq:
  assumes bo x y
  shows h x = h y
proof –
  have vimage2p h h (=) x y using mk.hash assms by (rule predicate2D)
  then show ?thesis by (simp add: vimage2p-def)
qed
```

Soundness of authenticated preservation under merging. If two authenticated values merge, then their extracted states have a valid join, and that join refines each input view. The proof calls *merge-respects-hashes* (to certify mergeability as hash-equality), *join* (to obtain the blinding relation of each input to the merge), and *hash* (via *bo-hash-eq*, to transport that blinding relation to the extracted states).

```
theorem authenticated-preservation-soundness:
  assumes mab: m a b = Some ab
```

and ea : $\text{extract-map } a = \text{Some } sa$
and eb : $\text{extract-map } b = \text{Some } sb$
shows $\exists sab. \text{extract-map } ab = \text{Some } sab \wedge \text{valid-state } sab$
 $\wedge \text{state-join } sa \ sb \ sab$
 $\wedge \text{state-refines } sa \ sab \wedge \text{state-refines } sb \ sab$
proof –
 — ADS lemma 1: a merge exists, so the hashes agree (mergeability).
have $ex\text{-merge}$: $\exists x. m \ a \ b = \text{Some } x$ **using** mab **by** $blast$
have hab : $h \ a = h \ b$ **using** $mk.\text{merge-respects-hashes } ex\text{-merge}$ **by** $blast$
 — ADS lemma 2: the merge ab is the least upper bound of a and b (join).
have lub : $bo \ a \ ab \wedge bo \ b \ ab \wedge (\forall u. bo \ a \ u \longrightarrow bo \ b \ u \longrightarrow bo \ ab \ u)$
using $mk.\text{join } mab$ **by** $blast$
then have boa : $bo \ a \ ab$ **and** bob : $bo \ b \ ab$ **by** simp-all
 — State-level join from the merging assumption, using the hash equality hab .
obtain sab **where** $esab$: $\text{extract-map } ab = \text{Some } sab$ **and** sj : $\text{state-join } sa \ sb \ sab$
using $\text{extract-respects-merging}[OF \ hab \ mab \ ea \ eb]$ **by** $blast$
have $vsab$: $\text{valid-state } sab$ **using** $\text{extract-preserves-validity}[OF \ esab]$.
 — ADS lemma 3 (hash): each input blinds to ab , hence shares its hash, so the
 extracted join refines each input view.
have $h \ a = h \ ab$ **using** $bo\text{-hash-eq}[OF \ boa]$.
then have ra : $\text{state-refines } sa \ sab$
using $\text{extract-under-blinding}[OF \ \langle h \ a = h \ ab \rangle \ boa \ esab]$ ea **by** $auto$
have $h \ b = h \ ab$ **using** $bo\text{-hash-eq}[OF \ bob]$.
then have rb : $\text{state-refines } sb \ sab$
using $\text{extract-under-blinding}[OF \ \langle h \ b = h \ ab \rangle \ bob \ esab]$ eb **by** $auto$
from $esab \ vsab \ sj \ ra \ rb$ **show** $?thesis$ **by** $blast$
qed

Blinded views preserve validity. A blinding a of an authenticated value b extracts to a valid state that refines the extraction of b . This is the need-to-know guarantee: a partial (blinded) view is still a valid, consistency-respecting view. The proof calls *hash* (via *bo-hash-eq*) to certify the hash equality that licenses the blinding coherence assumption.

theorem *blinded-view-preserves-validity*:
assumes $boab$: $bo \ a \ b$ **and** eb : $\text{extract-map } b = \text{Some } sb$
shows $\exists sa. \text{extract-map } a = \text{Some } sa \wedge \text{valid-state } sa \wedge \text{state-refines } sa \ sb$
proof –
have hab : $h \ a = h \ b$ **using** $bo\text{-hash-eq}[OF \ boab]$.
obtain sa **where** ea : $\text{extract-map } a = \text{Some } sa$ **and** sr : $\text{state-refines } sa \ sb$
using $\text{extract-under-blinding}[OF \ hab \ boab \ eb]$ **by** $blast$
have $\text{valid-state } sa$ **using** $\text{extract-preserves-validity}[OF \ ea]$.
with $ea \ sr$ **show** $?thesis$ **by** $blast$
qed

end

40 Authenticated Cross-Domain State: the Oraculizer Instance

We discharge the model obligation of *authenticated-state* by exhibiting a concrete, non-degenerate instance. Without it the two soundness theorems above would range over a possibly empty class of structures; the instance below pins them to an extraction map that reads a genuine cross-domain state out of authenticated data.

An authenticated datum is a pair (r, P) : a consensus regulatory state r together with the set P of chains that have revealed — are known to hold — the shared asset in that state. The hash *auth-hash* commits to the consensus state r alone, so this is a state-keyed reveal-set witness, not a commitment to the chain set itself. Two data are mergeable exactly when they agree on r ; the merge *auth-merge* then unions the revealed-chain sets, and the blinding order *auth-bo* forgets some of them. This is the state-level reading the glue layer asks for: *auth-merge* realises the ADS *join*, *auth-bo* the ADS blinding order, and *auth-extract* sends (r, P) to the global state in which every chain of P holds the asset in state r . Because the hash fixes r , merging never forces two chains to disagree, so the extracted join is consistent and the three obligations hold with no weakening of the hypotheses.

definition *auth-hash* :: *reg-state* $\times$ *chain-id set* $\Rightarrow$ *reg-state* **where**
auth-hash $a = \text{fst } a$

definition *auth-bo* :: *reg-state* $\times$ *chain-id set* $\Rightarrow$ *reg-state* $\times$ *chain-id set* $\Rightarrow$ *bool* **where**
auth-bo $a \ b \longleftrightarrow \text{fst } a = \text{fst } b \wedge \text{snd } a \subseteq \text{snd } b$

definition *auth-merge* ::
reg-state $\times$ *chain-id set* $\Rightarrow$ *reg-state* $\times$ *chain-id set* $\Rightarrow$ (*reg-state* $\times$ *chain-id set*) *option* **where**
auth-merge $a \ b = (\text{if } \text{fst } a = \text{fst } b \text{ then } \text{Some } (\text{fst } a, \text{snd } a \cup \text{snd } b) \text{ else } \text{None})$

The merge is the union of revealed-chain sets, guarded by agreement on the consensus state. Idempotence, commutativity and associativity descend from the corresponding laws of set union, and the blinding order is exactly the subset order on revealed chains; hence the triple is a Merkle interface.

lemma *merkle-interface-auth* [*locale-witness*]:
merkle-interface *auth-hash* *auth-bo* *auth-merge*

proof (*rule merkle-interface.intro*)

show $\bigwedge a \ b. (\text{auth-hash } a = \text{auth-hash } b) = (\exists ab. \text{auth-merge } a \ b = \text{Some } ab)$

by (*auto simp: auth-hash-def auth-merge-def*)

show $\bigwedge a. \text{auth-merge } a \ a = \text{Some } a$

by (*simp add: auth-merge-def*)

show $\bigwedge a \ b. \text{auth-merge } a \ b = \text{auth-merge } b \ a$

by (*auto simp: auth-merge-def Un-commute*)

show $\bigwedge a \ b \ c. \text{auth-merge } a \ b \ggg \text{auth-merge } c = \text{auth-merge } b \ c \ggg \text{auth-merge}$

a
by (*auto simp: auth-merge-def ac-simps split: if-splits*)
show $\bigwedge a b. \text{auth-bo } a b = (\text{auth-merge } a b = \text{Some } b)$
by (*auto simp: auth-bo-def auth-merge-def prod-eq-iff subset-Un-eq*)
qed

The extraction map. *auth-state* *r P* is the global state in which each chain of *P* holds asset *0* in regulatory state *r*, with nothing locked.

definition *auth-state* :: *reg-state* $\Rightarrow$ *chain-id set* $\Rightarrow$ *global-state* **where**
auth-state *r P* =

$$\begin{aligned} &(\text{gs-chains} = (\lambda c a. \text{if } c \in P \wedge a = 0 \\ &\quad \text{then Some } (\text{as-reg-state} = r) \\ &\quad \text{else None}), \\ &\text{gs-locks} = (\lambda a. \text{False}) \end{aligned}$$

definition *auth-extract* :: *reg-state* $\times$ *chain-id set* $\Rightarrow$ *global-state option* **where**
auth-extract *a* = *Some* (*auth-state* (*fst a*) (*snd a*))

Accessors of the extracted state, read off the definition.

lemma *auth-state-get-reg*:
get-reg-state (*auth-state* *r P*) *c a* = (*if* *c* $\in$ *P* $\wedge$ *a* = 0 *then* *Some r* *else* *None*)
by (*simp add: auth-state-def get-reg-state-def get-asset-state-def split: if-splits*)

lemma *auth-state-connected*:
connected-chains (*auth-state* *r P*) *a* = (*if* *a* = 0 *then* *P* *else* $\{\}$)
by (*auto simp: connected-chains-def asset-exists-def get-asset-state-def auth-state-def split: if-splits*)

Every extracted state is valid: all chains carry the same consensus state *r*, so the state is consistent, and no asset is locked.

lemma *auth-state-valid*: *valid-state* (*auth-state* *r P*)

proof –

have *consistent-state* (*auth-state* *r P*)
unfolding *consistent-state-def* **by** (*auto simp: auth-state-get-reg split: if-splits*)
moreover have *no-locked-without-reason* (*auth-state* *r P*)
by (*simp add: no-locked-without-reason-def is-locked-def auth-state-def*)
ultimately show *?thesis* **by** (*simp add: valid-state-def*)
qed

A concrete global witness for the multi-domain locale: the two-chain authenticated state instantiates it outright, with no ambient hypotheses.

lemma *reg-multi-domain-concrete*:
multi-domain-preservation $\{0, \text{Suc } 0\}$ *reg-states* *reg-actions* *reg-transition*
reg-terminal (*reg-domain-state* (*auth-state* *ACTIVE* $\{0, \text{Suc } 0\}$))
by (*rule reg-multi-domain-instantiation[OF - auth-state-valid]*) *simp*

The merge of two views with the same consensus state extracts to the join of the two extracted states: the union of revealed chains, agreeing on *r*.

lemma *state-join-auth*:
state-join (*auth-state* *r* *Pa*) (*auth-state* *r* *Pb*) (*auth-state* *r* (*Pa* $\cup$ *Pb*))
unfolding *state-join-def*
by (*auto simp: auth-state-connected auth-state-get-reg split: if-splits*)

A view over fewer chains refines a view over more chains with the same consensus state.

lemma *state-refines-auth*:
 $Pa \subseteq Pb \implies \text{state-refines } (\text{auth-state } r \ Pa) (\text{auth-state } r \ Pb)$
unfolding *state-refines-def*
by (*auto simp: auth-state-connected auth-state-get-reg subset-iff split: if-splits*)

The instance. The three coherence obligations are discharged from the lemmas above: merging extracts to the join (*state-join-auth*), blinding extracts to a refinement (*state-refines-auth*), and every extracted state is valid (*auth-state-valid*). The Merkle interface obligation is *merkle-interface-auth*.

interpretation *oss-authenticated*:

authenticated-state auth-hash auth-bo auth-merge auth-extract

proof *unfold-locales*

fix *a b ab sa sb*
assume *hab: auth-hash a = auth-hash b* **and** *mab: auth-merge a b = Some ab*
and *ea: auth-extract a = Some sa* **and** *eb: auth-extract b = Some sb*
from *hab* **have** *rfst: fst a = fst b* **by** (*simp add: auth-hash-def*)
with *mab* **have** *ab-eq: ab = (fst a, snd a $\cup$ snd b)* **by** (*simp add: auth-merge-def*)
have *sa-eq: sa = auth-state (fst a) (snd a)* **using** *ea* **by** (*simp add: auth-extract-def*)
have *sb-eq: sb = auth-state (fst a) (snd b)* **using** *eb rfst* **by** (*simp add: auth-extract-def*)
have *auth-extract ab = Some (auth-state (fst a) (snd a $\cup$ snd b))*
by (*simp add: auth-extract-def ab-eq*)
moreover **have** *state-join sa sb (auth-state (fst a) (snd a $\cup$ snd b))*
unfolding *sa-eq sb-eq* **by** (*rule state-join-auth*)
ultimately show $\exists sab. \text{auth-extract } ab = \text{Some } sab \wedge \text{state-join } sa \ sb \ sab$
by *blast*

next

fix *a b sb*
assume *bo: auth-bo a b* **and** *eb: auth-extract b = Some sb*
from *bo* **have** *rfst: fst a = fst b* **and** *sub: snd a $\subseteq$ snd b*
by (*simp-all add: auth-bo-def*)
have *sb-eq: sb = auth-state (fst a) (snd b)* **using** *eb rfst* **by** (*simp add: auth-extract-def*)
have *auth-extract a = Some (auth-state (fst a) (snd a))* **by** (*simp add: auth-extract-def*)
moreover **have** *state-refines (auth-state (fst a) (snd a)) sb*
unfolding *sb-eq* **using** *sub* **by** (*rule state-refines-auth*)
ultimately show $\exists sa. \text{auth-extract } a = \text{Some } sa \wedge \text{state-refines } sa \ sb$
by *blast*

next

fix *a s*

```

assume auth-extract a = Some s
then have s = auth-state (fst a) (snd a) by (simp add: auth-extract-def)
then show valid-state s by (simp add: auth-state-valid)
qed

```

The model is non-degenerate: the extraction map returns rich states, and the merging law is exercised by genuinely distinct partial views. The witnesses below record that a two-chain view extracts to a state with two connected chains carrying *ACTIVE*, and that merging the single-chain views over chains *0* and *Suc 0* joins them into the two-chain view.

lemma *oss-authenticated-extract-nontrivial*:

```

auth-extract (ACTIVE, {0, Suc 0}) = Some (auth-state ACTIVE {0, Suc 0})
connected-chains (auth-state ACTIVE {0, Suc 0}) 0 = {0, Suc 0}
get-reg-state (auth-state ACTIVE {0, Suc 0}) 0 0 = Some ACTIVE
by (simp-all add: auth-extract-def auth-state-connected auth-state-get-reg)

```

lemma *oss-authenticated-merge-nontrivial*:

```

auth-merge (ACTIVE, {0}) (ACTIVE, {Suc 0}) = Some (ACTIVE, {0, Suc 0})
auth-extract (ACTIVE, {0}) = Some (auth-state ACTIVE {0})
auth-extract (ACTIVE, {Suc 0}) = Some (auth-state ACTIVE {Suc 0})
state-join (auth-state ACTIVE {0}) (auth-state ACTIVE {Suc 0}) (auth-state ACTIVE {0, Suc 0})
using state-join-auth[of ACTIVE {0} {Suc 0}]
by (simp-all add: auth-merge-def auth-extract-def insert-commute)

```

Regression witness for the containment clause of *state-join*. The rogue candidate extends the union of the two single-chain *ACTIVE* views with a *FROZEN* holding on chain γ — a chain revealed by neither view. The candidate is itself inconsistent, and the containment clause rejects it as a join of the two views.

definition *rogue-join-state* :: *global-state* **where**

```

rogue-join-state =
  (| gs-chains = ( $\lambda c$  a. if a = 0  $\wedge$  (c = 0  $\vee$  c = Suc 0)
    then Some (| as-reg-state = ACTIVE |)
    else if a = 0  $\wedge$  c =  $\gamma$ 
    then Some (| as-reg-state = FROZEN |)
    else None),
   gs-locks = ( $\lambda a$ . False) |)

```

lemma *rogue-join-state-get-reg*:

```

get-reg-state rogue-join-state c a =
  (if a = 0  $\wedge$  (c = 0  $\vee$  c = Suc 0) then Some ACTIVE
   else if a = 0  $\wedge$  c =  $\gamma$  then Some FROZEN else None)
by (simp add: rogue-join-state-def get-reg-state-def get-asset-state-def)

```

lemma *rogue-join-excluded*:

```

 $\neg$  consistent-state rogue-join-state

```



```

    ¬ state-join (auth-state ACTIVE {0}) (auth-state ACTIVE {Suc 0})
  rogue-join-state
proof –
    have a: get-reg-state rogue-join-state 0 0 = Some ACTIVE
    and f: get-reg-state rogue-join-state 7 0 = Some FROZEN
    by (simp-all add: rogue-join-state-get-reg)
    show ¬ consistent-state rogue-join-state
  proof
    assume consistent-state rogue-join-state
    then have ACTIVE = FROZEN using a f unfolding consistent-state-def by
  blast
    then show False by simp
  qed
  have c7: (7::nat) ∈ connected-chains rogue-join-state 0
  by (simp add: connected-chains-def asset-exists-def get-asset-state-def
    rogue-join-state-def)
  have (7::nat) ∉ connected-chains (auth-state ACTIVE {0}) 0
  and (7::nat) ∉ connected-chains (auth-state ACTIVE {Suc 0}) 0
  by (simp-all add: auth-state-connected)
  then show ¬ state-join (auth-state ACTIVE {0}) (auth-state ACTIVE {Suc
0}) rogue-join-state
    using c7 unfolding state-join-def by blast
  qed

```

41 Inconsistency Measure on Global States

The convergence argument needs a well-founded progress measure that a synchronization strictly decreases while the global state is inconsistent. We use the number of unordered pairs of connected chains that disagree on the regulatory state of a shared asset. On a state with finitely many asset holdings this count is finite, it is zero exactly when the state is consistent, and a synchronization of an inconsistent asset strictly decreases it.

definition *finite-domain* :: *global-state* ⇒ *bool* **where**
finite-domain *gs* ⇔ *finite* {(*c*, *aid*). *asset-exists* *gs* *c* *aid*}

definition *inconsistency-set* ::
global-state ⇒ (*chain-id* × *chain-id* × *asset-id*) *set* **where**
inconsistency-set *gs* =
 {(*c1*, *c2*, *aid*). *c1* ∈ *connected-chains* *gs* *aid* ∧ *c2* ∈ *connected-chains* *gs* *aid*
 ∧ *c1* < *c2* ∧ *get-reg-state* *gs* *c1* *aid* ≠ *get-reg-state* *gs* *c2* *aid*}

definition *inconsistency-pairs* :: *global-state* ⇒ *nat* **where**
inconsistency-pairs *gs* = *card* (*inconsistency-set* *gs*)

A connected chain always carries a defined regulatory state for the asset.

lemma *connected-chains-get-reg-state*:
assumes *c* ∈ *connected-chains* *gs* *aid*
shows ∃ *s*. *get-reg-state* *gs* *c* *aid* = *Some* *s*

using *assms*
unfolding *connected-chains-def asset-exists-def get-reg-state-def get-asset-state-def*
by (*auto split: option.splits*)

On a finite-domain state, the inconsistency set is finite.

lemma *inconsistency-set-finite*:
assumes *finite-domain gs*
shows *finite (inconsistency-set gs)*
proof –
let $?E = \{(c, a). \text{asset-exists } gs \ c \ a\}$
have $\text{inconsistency-set } gs \subseteq (\lambda(p, q). (fst \ p, fst \ q, snd \ p)) \ ' \ (?E \times ?E)$
proof
fix t **assume** $t \in \text{inconsistency-set } gs$
then obtain $c1 \ c2 \ aid$ **where** $t = (c1, c2, aid)$
and $m1: c1 \in \text{connected-chains } gs \ aid$ **and** $m2: c2 \in \text{connected-chains } gs \ aid$
unfolding *inconsistency-set-def* **by** *auto*
from $m1$ **have** $e1: (c1, aid) \in ?E$ **by** (*simp add: connected-chains-def*)
from $m2$ **have** $e2: (c2, aid) \in ?E$ **by** (*simp add: connected-chains-def*)
have $t = (\lambda(p, q). (fst \ p, fst \ q, snd \ p)) \ ((c1, aid), (c2, aid))$ **using** t **by** *simp*
then show $t \in (\lambda(p, q). (fst \ p, fst \ q, snd \ p)) \ ' \ (?E \times ?E)$
using $e1 \ e2$ **by** *blast*
qed
moreover have $\text{finite } ((\lambda(p, q). (fst \ p, fst \ q, snd \ p)) \ ' \ (?E \times ?E))$
using *assms* **unfolding** *finite-domain-def* **by** *simp*
ultimately show *?thesis* **by** (*rule finite-subset*)
qed

The measure is zero exactly on consistent states.

lemma *inconsistency-pairs-zero-iff-consistent*:
assumes *finite-domain gs*
shows $\text{inconsistency-pairs } gs = 0 \longleftrightarrow \text{consistent-state } gs$
proof
assume $\text{inconsistency-pairs } gs = 0$
then have $\text{empty: inconsistency-set } gs = \{\}$
using *inconsistency-set-finite[OF assms]* **unfolding** *inconsistency-pairs-def* **by** *simp*
show *consistent-state gs*
unfolding *consistent-state-def*
proof (*intro allI impI*)
fix $c1 \ c2 \ aid \ s1 \ s2$
assume $a1: \text{get-reg-state } gs \ c1 \ aid = \text{Some } s1$ **and** $a2: \text{get-reg-state } gs \ c2 \ aid = \text{Some } s2$
have $in1: c1 \in \text{connected-chains } gs \ aid$ **and** $in2: c2 \in \text{connected-chains } gs \ aid$
using $a1 \ a2$ **unfolding** *connected-chains-def asset-exists-def*
get-reg-state-def get-asset-state-def **by** (*auto split: option.splits*)
show $s1 = s2$
proof (*rule ccontr*)
assume $ne: s1 \neq s2$
then have $rne: \text{get-reg-state } gs \ c1 \ aid \neq \text{get-reg-state } gs \ c2 \ aid$ **using** $a1 \ a2$
by *simp*

```

have  $c1 \neq c2$  using  $a1\ a2\ ne$  by auto
then consider  $c1 < c2 \mid c2 < c1$  by linarith
then show False
proof cases
  case 1
  then have  $(c1, c2, aid) \in inconsistency\text{-}set\ gs$ 
    using  $in1\ in2\ rne$  unfolding inconsistency-set-def by simp
  with empty show False by simp
next
  case 2
  then have  $(c2, c1, aid) \in inconsistency\text{-}set\ gs$ 
    using  $in1\ in2\ rne$  unfolding inconsistency-set-def by auto
  with empty show False by simp
qed
qed
qed
next
assume cons: consistent-state gs
have inconsistency-set gs = {}
proof (rule ccontr)
  assume inconsistency-set gs  $\neq$  {}
  then obtain  $c1\ c2\ aid$  where
     $m1: c1 \in connected\text{-}chains\ gs\ aid$  and  $m2: c2 \in connected\text{-}chains\ gs\ aid$ 
    and  $rne: get\text{-}reg\text{-}state\ gs\ c1\ aid \neq get\text{-}reg\text{-}state\ gs\ c2\ aid$ 
    unfolding inconsistency-set-def by auto
  obtain  $s1$  where  $s1: get\text{-}reg\text{-}state\ gs\ c1\ aid = Some\ s1$ 
    using connected-chains-get-reg-state[OF  $m1$ ] by blast
  obtain  $s2$  where  $s2: get\text{-}reg\text{-}state\ gs\ c2\ aid = Some\ s2$ 
    using connected-chains-get-reg-state[OF  $m2$ ] by blast
  have  $s1 = s2$  using cons  $s1\ s2$  unfolding consistent-state-def by blast
  with  $s1\ s2\ rne$  show False by simp
qed
then show inconsistency-pairs gs = 0 unfolding inconsistency-pairs-def by
simp
qed

```

42 Structural Effect of Synchronization

The following lemmas isolate the effect of a successful synchronization on the regulatory states and on asset existence. They are proved without assuming the input state is valid (consistent and unlocked), because the convergence argument starts from an arbitrary state. The existing theorem *regulatory-homomorphism* establishes the analogous fact under the *valid-state* hypothesis; here we re-establish the parts needed for convergence with no such hypothesis.

lemma *sync-components*:
 assumes $sync\ src\ act\ aid0\ gs = Some\ gs'$

shows $\exists$ *current-st new-st gs-locked*.
 $\text{get-reg-state } gs \text{ src } aid0 = \text{Some } current\text{-}st$
 $\wedge \text{reg-transition } current\text{-}st \text{ act} = \text{Some } new\text{-}st$
 $\wedge \text{acquire-lock } gs \text{ aid0} = \text{Some } gs\text{-}locked$
 $\wedge gs' = \text{release-lock}$
 $(\text{update-all-chains } gs\text{-}locked \text{ aid0 } new\text{-}st (\text{connected-chains } gs \text{ aid0}))$

aid0
using *assms unfolding sync-def* **by** (*auto split: option.splits simp: Let-def*)

Synchronizing asset *aid0* leaves the chain map of every other asset unchanged.

lemma *sync-chains-other-asset*:
assumes *synced: sync src act aid0 gs = Some gs' and ne: aid $\neq$ aid0*
shows *gs-chains gs' c aid = gs-chains gs c aid*
proof –
from *sync-components[OF synced]* **obtain** *current-st new-st gs-locked* **where**
 $lock: \text{acquire-lock } gs \text{ aid0} = \text{Some } gs\text{-}locked$
and $gs': gs' = \text{release-lock}$
 $(\text{update-all-chains } gs\text{-}locked \text{ aid0 } new\text{-}st (\text{connected-chains } gs \text{ aid0}))$

aid0
by *blast*
have $lc: gs\text{-}chains \text{ gs-}locked = gs\text{-}chains \text{ gs}$ **using** *acquire-lock-chains[OF lock]* .
have $gs\text{-}chains (\text{update-all-chains } gs\text{-}locked \text{ aid0 } new\text{-}st (\text{connected-chains } gs \text{ aid0})) \text{ c } aid$
 $= gs\text{-}chains \text{ gs-}locked \text{ c } aid$
using *update-all-chains-other-asset[OF ne]* **by** *simp*
then show *?thesis unfolding gs' release-lock-chains* **using** *lc* **by** *simp*
qed

lemma *sync-reg-other-asset*:
assumes *synced: sync src act aid0 gs = Some gs' and ne: aid $\neq$ aid0*
shows *get-reg-state gs' c aid = get-reg-state gs c aid*
unfolding *get-reg-state-def get-asset-state-def*
using *sync-chains-other-asset[OF synced ne]* **by** *simp*

After synchronizing asset *aid0*, all chains connected to *aid0* agree on its new regulatory state.

lemma *sync-synced-asset-uniform*:
assumes *synced: sync src act aid0 gs = Some gs'*
shows $\exists ns. \forall c' \in \text{connected-chains } gs \text{ aid0}. \text{get-reg-state } gs' \text{ c' aid0} = \text{Some } ns$
proof –
from *sync-components[OF synced]* **obtain** *current-st new-st gs-locked* **where**
 $lock: \text{acquire-lock } gs \text{ aid0} = \text{Some } gs\text{-}locked$
and $gs': gs' = \text{release-lock}$
 $(\text{update-all-chains } gs\text{-}locked \text{ aid0 } new\text{-}st (\text{connected-chains } gs \text{ aid0}))$

aid0
by *blast*
have $lc: gs\text{-}chains \text{ gs-}locked = gs\text{-}chains \text{ gs}$ **using** *acquire-lock-chains[OF lock]* .
have $\forall c' \in \text{connected-chains } gs \text{ aid0}. \text{get-reg-state } gs' \text{ c' aid0} = \text{Some } new\text{-}st$

```

proof
  fix  $c'$  assume  $c'conn: c' \in \text{connected-chains } gs \text{ aid0}$ 
  then have  $\text{asset-exists } gs \text{ } c' \text{ aid0}$  by (simp add: connected-chains-def)
  then obtain  $ast$  where  $ast: \text{get-asset-state } gs \text{ } c' \text{ aid0} = \text{Some } ast$ 
    unfolding  $\text{asset-exists-def}$  by auto
  then have  $astl: \text{get-asset-state } gs\text{-locked } c' \text{ aid0} = \text{Some } ast$ 
    unfolding  $\text{get-asset-state-def}$  using  $lc$  by simp
  have  $\text{get-reg-state}$ 
    ( $\text{update-all-chains } gs\text{-locked } aid0 \text{ new-st } (\text{connected-chains } gs \text{ aid0})) \text{ } c'$ 
 $\text{aid0} = \text{Some new-st}$ 
    using  $\text{update-all-chains-reg-state}[OF \text{ } c'conn \text{ } astl]$  .
  then show  $\text{get-reg-state } gs' \text{ } c' \text{ aid0} = \text{Some new-st}$ 
    unfolding  $gs'$  by (simp add: release-lock-reg-state)
  qed
  then show ?thesis by blast
qed

Synchronization preserves asset existence, hence the connected-chain
sets.

lemma sync-asset-exists-preserved:
  assumes  $\text{synced}: \text{sync } src \text{ act } aid0 \text{ } gs = \text{Some } gs'$ 
  shows  $\text{asset-exists } gs' \text{ } c \text{ aid} = \text{asset-exists } gs \text{ } c \text{ aid}$ 
proof (cases aid = aid0)
  case False
  then have  $gs\text{-chains } gs' \text{ } c \text{ aid} = gs\text{-chains } gs \text{ } c \text{ aid}$ 
    using  $\text{sync-chains-other-asset}[OF \text{ } synced]$  by simp
  then show ?thesis unfolding  $\text{asset-exists-def}$   $\text{get-asset-state-def}$  by simp
next
  case True
  show ?thesis
  proof (cases c ∈ connected-chains gs aid0)
    case in-conn: True
    then have  $e\text{-gs}: \text{asset-exists } gs \text{ } c \text{ aid0}$  by (simp add: connected-chains-def)
    obtain  $ns$  where  $\forall c' \in \text{connected-chains } gs \text{ aid0}. \text{get-reg-state } gs' \text{ } c' \text{ aid0} =$ 
 $\text{Some } ns$ 
      using  $\text{sync-synced-asset-uniform}[OF \text{ } synced]$  by blast
    then have  $\text{get-reg-state } gs' \text{ } c \text{ aid0} = \text{Some } ns$  using in-conn by blast
    then have  $\text{asset-exists } gs' \text{ } c \text{ aid0}$ 
      unfolding  $\text{asset-exists-def}$   $\text{get-reg-state-def}$   $\text{get-asset-state-def}$ 
      by (auto split: option.splits)
    with  $e\text{-gs}$  True show ?thesis by simp
  next
    case out-conn: False
    then have  $ngs: \neg \text{asset-exists } gs \text{ } c \text{ aid0}$  by (simp add: connected-chains-def)
    from  $\text{sync-components}[OF \text{ } synced]$  obtain  $\text{current-st new-st } gs\text{-locked}$  where
       $\text{lock}: \text{acquire-lock } gs \text{ aid0} = \text{Some } gs\text{-locked}$ 
      and  $gs': gs' = \text{release-lock}$ 
      ( $\text{update-all-chains } gs\text{-locked } aid0 \text{ new-st } (\text{connected-chains } gs \text{ aid0}))$ 
 $\text{aid0}$ 

```

```

    by blast
  have lc: gs-chains gs-locked = gs-chains gs using acquire-lock-chains[OF lock]
  .
  have gs-chains gs' c = gs-chains gs c
    unfolding gs' release-lock-chains
    using update-all-chains-outside-targets[OF out-conn] lc by simp
  then have asset-exists gs' c aid0 = asset-exists gs c aid0
    unfolding asset-exists-def get-asset-state-def by simp
  with True show ?thesis by simp
qed
qed

```

```

lemma sync-connected-chains-preserved:
  assumes sync src act aid0 gs = Some gs'
  shows connected-chains gs' aid = connected-chains gs aid
  unfolding connected-chains-def using sync-asset-exists-preserved[OF assms] by
  simp

```

```

lemma sync-preserves-finite-domain:
  assumes finite-domain gs and sync src act aid0 gs = Some gs'
  shows finite-domain gs'
proof -
  have {(c, aid). asset-exists gs' c aid} = {(c, aid). asset-exists gs c aid}
    using sync-asset-exists-preserved[OF assms(2)] by simp
  then show ?thesis using assms(1) unfolding finite-domain-def by simp
qed

```

Synchronization preserves the unlocked invariant. This is the lock half of *valid-state-preservation*, re-established without the *consistent-state* hypothesis.

```

lemma sync-preserves-unlocked:
  assumes nl: no-locked-without-reason gs and synced: sync src act aid0 gs = Some
  gs'
  shows no-locked-without-reason gs'
proof -
  from sync-components[OF synced] obtain current-st new-st gs-locked where
    lock: acquire-lock gs aid0 = Some gs-locked
    and gs': gs' = release-lock
    (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0))
  aid0
  by blast
  from lock have locked-locks: gs-locks gs-locked = (gs-locks gs)(aid0 := True)
    unfolding acquire-lock-def is-locked-def by (auto split: if-splits)
  have upd-locks: gs-locks (update-all-chains gs-locked aid0 new-st (connected-chains
  gs aid0))
    = gs-locks gs-locked
    using update-all-chains-locks by simp
  have final-locks: gs-locks gs' = (gs-locks gs-locked)(aid0 := False)
    unfolding gs' release-lock-def using upd-locks by simp

```

```

show ?thesis
unfolding no-locked-without-reason-def is-locked-def
proof
  fix aid'
  show  $\neg$  gs-locks gs' aid'
  proof (cases aid' = aid0)
    case True then show ?thesis using final-locks by simp
  next
    case False
    then have gs-locks gs' aid' = gs-locks gs aid'
      using final-locks locked-locks by simp
    also have ... = False
      using nl unfolding no-locked-without-reason-def is-locked-def by simp
    finally show ?thesis by simp
  qed
qed
qed

```

43 The Critical Path: Synchronization Reduces Inconsistency

A connected chain witnesses asset existence, so a defined regulatory state places the chain in the connected set.

```

lemma get-reg-state-connected:
  assumes get-reg-state gs c aid = Some s
  shows c  $\in$  connected-chains gs aid
  using assms unfolding connected-chains-def asset-exists-def
    get-reg-state-def get-asset-state-def by (auto split: option.splits)

```

The critical convergence step. Synchronizing an asset *aid0* on which two connected chains disagree strictly decreases the inconsistency measure: every inconsistent pair of *aid0* is removed (all its connected chains are driven to one regulatory state), pairs of other assets are untouched, and the witness pair certifies that at least one pair really disappears.

```

lemma sync-reduces-inconsistency:
  assumes fin: finite-domain gs
  and synced: sync src act aid0 gs = Some gs'
  and ca: ca  $\in$  connected-chains gs aid0
  and cb: cb  $\in$  connected-chains gs aid0
  and dis: get-reg-state gs ca aid0  $\neq$  get-reg-state gs cb aid0
  shows inconsistency-pairs gs' < inconsistency-pairs gs
proof –
  obtain ns where uniform:  $\forall c' \in \text{connected-chains } gs \text{ aid0. } \text{get-reg-state } gs' \ c' \text{ aid0}$ 
    = Some ns
  using sync-synced-asset-uniform[OF synced] by blast
  have conn-eq:  $\bigwedge \text{aid. } \text{connected-chains } gs' \text{ aid} = \text{connected-chains } gs \text{ aid}$ 
    using sync-connected-chains-preserved[OF synced] by simp

```

— Step 1: the inconsistency set of gs' is contained in that of gs .

have *subset*: *inconsistency-set* $gs' \subseteq$ *inconsistency-set* gs

proof

fix t **assume** $t \in$ *inconsistency-set* gs'

then obtain $c1\ c2\ aid$ **where** $t: t = (c1, c2, aid)$

and $i1: c1 \in$ *connected-chains* $gs'\ aid$ **and** $i2: c2 \in$ *connected-chains* $gs'\ aid$

and $lt: c1 < c2$ **and** $rne: \text{get-reg-state } gs'\ c1\ aid \neq \text{get-reg-state } gs'\ c2\ aid$

unfolding *inconsistency-set-def* **by** *auto*

have $ane: aid \neq aid0$

proof

assume $eq: aid = aid0$

have $c1 \in$ *connected-chains* $gs\ aid0\ c2 \in$ *connected-chains* $gs\ aid0$

using $i1\ i2\ conn\ eq$ **by** *auto*

then have $\text{get-reg-state } gs'\ c1\ aid0 = \text{Some } ns\ \text{get-reg-state } gs'\ c2\ aid0 =$

Some ns

using *uniform* **by** *auto*

with $rne\ eq$ **show** *False* **by** *simp*

qed

have $c1 \in$ *connected-chains* $gs\ aid\ c2 \in$ *connected-chains* $gs\ aid$

using $i1\ i2\ conn\ eq$ **by** *auto*

moreover have $\text{get-reg-state } gs\ c1\ aid \neq \text{get-reg-state } gs\ c2\ aid$

using $rne\ sync\ reg\ other\ asset[OF\ synced\ ane]$ **by** *simp*

ultimately have $(c1, c2, aid) \in$ *inconsistency-set* gs

using lt **unfolding** *inconsistency-set-def* **by** *simp*

then show $t \in$ *inconsistency-set* gs **using** t **by** *simp*

qed

— Step 2: the witness pair on $aid0$ is in gs but not in gs' .

have $caneb: ca \neq cb$ **using** *dis* **by** *auto*

define $c1$ **where** $c1 = \min\ ca\ cb$

define $c2$ **where** $c2 = \max\ ca\ cb$

have $lt: c1 < c2$ **using** $caneb$ **unfolding** $c1\text{-def}\ c2\text{-def}$ **by** *auto*

have $c1conn: c1 \in$ *connected-chains* $gs\ aid0$ **and** $c2conn: c2 \in$ *connected-chains* $gs\ aid0$

using $ca\ cb$ **unfolding** $c1\text{-def}\ c2\text{-def}$ **by** (*auto simp: min-def max-def*)

have $wdis: \text{get-reg-state } gs\ c1\ aid0 \neq \text{get-reg-state } gs\ c2\ aid0$

using *dis* **unfolding** $c1\text{-def}\ c2\text{-def}$ **by** (*auto simp: min-def max-def*)

have $wit\text{-in}: (c1, c2, aid0) \in$ *inconsistency-set* gs

using $c1conn\ c2conn\ lt\ wdis$ **unfolding** *inconsistency-set-def* **by** *simp*

have $wit\text{-out}: (c1, c2, aid0) \notin$ *inconsistency-set* gs'

proof

assume $(c1, c2, aid0) \in$ *inconsistency-set* gs'

then have $rne: \text{get-reg-state } gs'\ c1\ aid0 \neq \text{get-reg-state } gs'\ c2\ aid0$

and $c1 \in$ *connected-chains* $gs'\ aid0$ **and** $c2 \in$ *connected-chains* $gs'\ aid0$

unfolding *inconsistency-set-def* **by** *auto*

then have $c1 \in$ *connected-chains* $gs\ aid0\ c2 \in$ *connected-chains* $gs\ aid0$

using *conn-eq* **by** *auto*

then have $\text{get-reg-state } gs'\ c1\ aid0 = \text{Some } ns\ \text{get-reg-state } gs'\ c2\ aid0 = \text{Some}$

ns

using *uniform* **by** *auto*


```

    with rne show False by simp
qed
— Step 3: a strict subset of a finite set has strictly smaller cardinality.
have psub: inconsistency-set gs'  $\subset$  inconsistency-set gs
  using subset wit-in wit-out by blast
have card (inconsistency-set gs') < card (inconsistency-set gs)
  using psubset-card-mono[OF inconsistency-set-finite[OF fin] psub] .
then show ?thesis unfolding inconsistency-pairs-def .
qed

```

44 Existence of a Reducing Synchronization

From any inconsistent, finite-domain, unlocked global state there is a synchronization that strictly reduces the inconsistency measure. An inconsistency exhibits two chains disagreeing on a shared asset; at least one carries a non-terminal (non-*CONFISCATED*) regulatory state, from which the *CONFISCATE* action is always enabled (*confiscate-universal*), so a synchronization succeeds and, by *sync-reduces-inconsistency*, reduces the measure.

definition *reducing-sync* ::

```

global-state  $\Rightarrow$  (chain-id  $\times$  reg-action  $\times$  asset-id)  $\Rightarrow$  bool where
reducing-sync gs p  $\longleftrightarrow$  (case p of (src, act, aid)  $\Rightarrow$ 
  ( $\exists$  gs'. sync src act aid gs = Some gs'  $\wedge$  inconsistency-pairs gs' < inconsistency-pairs gs))

```

lemma *inconsistent-has-reducing-sync*:

```

assumes fin: finite-domain gs and nl: no-locked-without-reason gs
and inc:  $\neg$  consistent-state gs
shows  $\exists$  p. reducing-sync gs p

```

proof —

```

from inc obtain c1 c2 aid0 s1 s2 where
  g1: get-reg-state gs c1 aid0 = Some s1 and g2: get-reg-state gs c2 aid0 = Some
s2
and ne: s1  $\neq$  s2
unfolding consistent-state-def by blast
have conn1: c1  $\in$  connected-chains gs aid0 using get-reg-state-connected[OF g1]
.
have conn2: c2  $\in$  connected-chains gs aid0 using get-reg-state-connected[OF g2]
.
— One of the two disagreeing chains is non-terminal.
obtain src ssrc where src-conn: src  $\in$  connected-chains gs aid0
and src-reg: get-reg-state gs src aid0 = Some ssrc
and src-nonterm: ssrc  $\neq$  CONFISCATED
proof (cases s1 = CONFISCATED)
  case True
    then have s2  $\neq$  CONFISCATED using ne by auto
    then show thesis using that conn2 g2 by blast
next
  case False

```

```

    then show thesis using that conn1 g1 by blast
qed
— CONFISCATE is enabled from any non-terminal state, and the asset is un-
locked.
have ctrans: reg-transition ssrc CONFISCATE = Some CONFISCATED
  using confiscate-universal[OF src-nonterm] .
have notlocked:  $\neg$  is-locked gs aid0
  using nl unfolding no-locked-without-reason-def by simp
obtain gsl where acq: acquire-lock gs aid0 = Some gsl
  using lock-acquire-success[OF notlocked] by blast
have sync src CONFISCATE aid0 gs
  = Some (release-lock
    (update-all-chains gsl aid0 CONFISCATED (connected-chains gs
aid0)) aid0)
  unfolding sync-def using src-reg ctrans acq by (simp add: Let-def)
then obtain gs' where synced: sync src CONFISCATE aid0 gs = Some gs' by
blast
have dis: get-reg-state gs c1 aid0  $\neq$  get-reg-state gs c2 aid0 using g1 g2 ne by
simp
have inconsistency-pairs gs' < inconsistency-pairs gs
  using sync-reduces-inconsistency[OF fin synced conn1 conn2 dis] .
then have reducing-sync gs (src, CONFISCATE, aid0)
  unfolding reducing-sync-def using synced by auto
then show ?thesis by blast
qed

```

45 Terminal-Faithful Safe Recovery

A measure-reducing synchronization alone leaves the realization map two degrees of freedom that no recovery procedure should have. First, it may resolve a mere *ACTIVE/FROZEN* disagreement by broadcasting *CONFISCATED*: an indiscriminate-confiscation implementation would refine the model. Second, in the reverse direction: the broadcast overwrites every connected chain without consulting the target chain's own transition relation, so it may overwrite a recorded *CONFISCATED* holding with a non-terminal value, erasing a confiscation. The predicate *safe-recovery* closes both: a recovery is safe when it reduces the measure *and* uses *CONFISCATE* exactly on the assets that already carry the terminal state on some chain. The equivalence is needed in both directions: left-to-right forbids fresh confiscations, and right-to-left forbids completing an inconsistency on a terminal-bearing asset with anything weaker than the terminal value.

definition *terminal-present* :: *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *bool* **where**
terminal-present gs aid $\longleftrightarrow$ ($\exists c.$ *get-reg-state gs c aid* = *Some CONFISCATED*)

definition *safe-recovery* ::
global-state $\Rightarrow$ (*chain-id* $\times$ *reg-action* $\times$ *asset-id*) $\Rightarrow$ *bool* **where**
safe-recovery gs p $\longleftrightarrow$ *reducing-sync gs p* $\wedge$

(case p of $(src, act, aid) \Rightarrow (act = CONFISCATE) = terminal-present\ gs\ aid$)

lemma *safe-recovery-reducing*:

safe-recovery $gs\ p \Rightarrow reducing-sync\ gs\ p$

by (*simp add: safe-recovery-def*)

The only regulatory action whose result is the terminal state is *CONFISCATE* itself: the transition table admits no other entry to *CONFISCATED*.

lemma *to-confiscated-only-confiscate*:

reg-transition $s\ a = Some\ CONFISCATED \Rightarrow a = CONFISCATE$

by (*cases s; cases a*) *simp-all*

From every non-terminal regulatory state some non-*CONFISCATE* action is enabled; the witnesses are the de-escalation returns and, from *ACTIVE*, an escalation that is not a confiscation.

lemma *non-terminal-non-confiscate-step*:

assumes $s \neq CONFISCATED$

shows $\exists a\ s'. a \neq CONFISCATE \wedge reg-transition\ s\ a = Some\ s'$

proof (*cases s*)

case *ACTIVE*

then have $FREEZE \neq CONFISCATE \wedge reg-transition\ s\ FREEZE = Some\ FROZEN$ **by** *simp*

then show *?thesis* **by** *blast*

next

case *FROZEN*

then have $UNFREEZE \neq CONFISCATE \wedge reg-transition\ s\ UNFREEZE = Some\ ACTIVE$ **by** *simp*

then show *?thesis* **by** *blast*

next

case *SEIZED*

then have $RELEASE \neq CONFISCATE \wedge reg-transition\ s\ RELEASE = Some\ ACTIVE$ **by** *simp*

then show *?thesis* **by** *blast*

next

case *CONFISCATED*

then show *?thesis* **using** *assms* **by** *contradiction*

next

case *RESTRICTED*

then have $UNRESTRICT \neq CONFISCATE \wedge reg-transition\ s\ UNRESTRICT = Some\ ACTIVE$ **by** *simp*

then show *?thesis* **by** *blast*

qed

Existence of a safe recovery from any inconsistent carrier state. The proof refines *inconsistent-has-reducing-sync* by selecting the synchronization action according to the terminal census of the inconsistent asset: if the terminal state is already present on some chain, the recovery completes it by confiscating from a non-terminal disagreeing chain; if it is absent, the recovery synchronizes with a non-*CONFISCATE* action enabled on a disagreeing

chain. Either way the synchronization succeeds and strictly reduces the inconsistency measure, and the chosen action satisfies the terminal-faithfulness equivalence.

lemma *inconsistent-has-safe-recovery*:

assumes *fin*: *finite-domain* *gs* **and** *nl*: *no-locked-without-reason* *gs*
and *inc*: $\neg$ *consistent-state* *gs*
shows $\exists p$. *safe-recovery* *gs* *p*

proof —

from *inc* **obtain** *c1 c2 aid0 s1 s2* **where**

g1: *get-reg-state* *gs c1 aid0* = *Some s1* **and** *g2*: *get-reg-state* *gs c2 aid0* = *Some s2*

and *ne*: *s1* $\neq$ *s2*

unfolding *consistent-state-def* **by** *blast*

have *conn1*: *c1* $\in$ *connected-chains* *gs aid0* **using** *get-reg-state-connected*[*OF g1*]

.

have *conn2*: *c2* $\in$ *connected-chains* *gs aid0* **using** *get-reg-state-connected*[*OF g2*]

.

have *dis*: *get-reg-state* *gs c1 aid0* $\neq$ *get-reg-state* *gs c2 aid0* **using** *g1 g2 ne* **by** *simp*

have *notlocked*: $\neg$ *is-locked* *gs aid0*

using *nl* **unfolding** *no-locked-without-reason-def* **by** *simp*

obtain *gsl* **where** *acq*: *acquire-lock* *gs aid0* = *Some gsl*

using *lock-acquire-success*[*OF notlocked*] **by** *blast*

show *?thesis*

proof (*cases terminal-present* *gs aid0*)

case *True*

— Terminal completion: one of the disagreeing chains is non-terminal, and confiscating from it completes the recorded confiscation.

obtain *src ssrc* **where** *src-reg*: *get-reg-state* *gs src aid0* = *Some ssrc*

and *src-nonterm*: *ssrc* $\neq$ *CONFISCATED*

proof (*cases s1* = *CONFISCATED*)

case *True*

then have *s2* $\neq$ *CONFISCATED* **using** *ne* **by** *simp*

then show *thesis* **using** *that g2* **by** *blast*

next

case *False*

then show *thesis* **using** *that g1* **by** *blast*

qed

have *ctrans*: *reg-transition* *ssrc CONFISCATE* = *Some CONFISCATED*

using *confiscate-universal*[*OF src-nonterm*] .

have *sync src CONFISCATE aid0 gs*

= *Some* (*release-lock*

(*update-all-chains* *gsl aid0 CONFISCATED* (*connected-chains* *gs aid0*)) *aid0*)

unfolding *sync-def* **using** *src-reg ctrans acq* **by** (*simp add*: *Let-def*)

then obtain *gs'* **where** *synced*: *sync src CONFISCATE aid0 gs* = *Some gs'*

by *blast*

have *inconsistency-pairs* *gs'* < *inconsistency-pairs* *gs*

using *sync-reduces-inconsistency*[*OF fin synced conn1 conn2 dis*] .

```

then have reducing-sync gs (src, CONFISCATE, aid0)
  unfolding reducing-sync-def using synced by auto
then have safe-recovery gs (src, CONFISCATE, aid0)
  using True by (simp add: safe-recovery-def)
then show ?thesis by blast
next
case False
— No terminal holding: the first disagreeing chain is non-terminal, and a non-
CONFISCATE action is enabled there.
have s1-nonterm: s1 ≠ CONFISCATED
  using False g1 unfolding terminal-present-def by blast
obtain act ns where act-nc: act ≠ CONFISCATE
  and atrans: reg-transition s1 act = Some ns
  using non-terminal-non-confiscate-step[OF s1-nonterm] by blast
have sync c1 act aid0 gs
  = Some (release-lock
    (update-all-chains gsl aid0 ns (connected-chains gs aid0)) aid0)
  unfolding sync-def using g1 atrans acq by (simp add: Let-def)
then obtain gs' where synced: sync c1 act aid0 gs = Some gs' by blast
have inconsistency-pairs gs' < inconsistency-pairs gs
  using sync-reduces-inconsistency[OF fin synced conn1 conn2 dis] .
then have reducing-sync gs (c1, act, aid0)
  unfolding reducing-sync-def using synced by auto
then have safe-recovery gs (c1, act, aid0)
  using False act-nc by (simp add: safe-recovery-def)
then show ?thesis by blast
qed
qed

```

Terminal completion is definitional: on a terminal-bearing asset a safe recovery can only be the completing confiscation.

corollary *recovery-terminal-completion:*

```

assumes safe-recovery gs (src, act, aid) and terminal-present gs aid
shows act = CONFISCATE
using assms by (simp add: safe-recovery-def)

```

The value a successful synchronization writes onto a connected chain is the output of the regulatory transition fired at the source. This exposes, for the safety argument below, the link between the broadcast value and the transition table that produced it.

lemma *sync-connected-value:*

```

assumes synced: sync src act aid0 gs = Some gs'
and conn: c ∈ connected-chains gs aid0
shows ∃ cur new-st. get-reg-state gs src aid0 = Some cur
  ∧ reg-transition cur act = Some new-st
  ∧ get-reg-state gs' c aid0 = Some new-st

```

proof —

```

from sync-components[OF synced] obtain cur new-st gs-locked where
cs: get-reg-state gs src aid0 = Some cur

```

```

and tr: reg-transition cur act = Some new-st
and lock: acquire-lock gs aid0 = Some gs-locked
and gs': gs' = release-lock
      (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0))
aid0
  by blast
  have lc: gs-chains gs-locked = gs-chains gs using acquire-lock-chains[OF lock] .
  from conn have asset-exists gs c aid0 by (simp add: connected-chains-def)
  then obtain ast where get-asset-state gs c aid0 = Some ast
    unfolding asset-exists-def by auto
  then have astl: get-asset-state gs-locked c aid0 = Some ast
    unfolding get-asset-state-def using lc by simp
  have get-reg-state
    (update-all-chains gs-locked aid0 new-st (connected-chains gs aid0)) c aid0
    = Some new-st
    using update-all-chains-reg-state[OF conn astl] .
  then have get-reg-state gs' c aid0 = Some new-st
    unfolding gs' by (simp add: release-lock-reg-state)
  with cs tr show ?thesis by blast
qed

```

The single-step safety payload of the terminal-faithfulness equivalence: a safe-recovery synchronization never makes the terminal state appear on an asset that did not already carry it. If the synchronized value were *CONFISCATED*, then by *to-confiscated-only-confiscate* the action was *CONFISCATE*, so by the equivalence the asset already carried the terminal state — a contradiction.

lemma *safe-recovery-sync-no-fresh-terminal*:

```

assumes sr: safe-recovery gs (src, act, aid0)
and synced: sync src act aid0 gs = Some gs'
and nt:  $\neg$  terminal-present gs aid
shows  $\neg$  terminal-present gs' aid

```

proof

```

assume terminal-present gs' aid
then obtain c where c: get-reg-state gs' c aid = Some CONFISCATED
  unfolding terminal-present-def by blast
show False
proof (cases aid = aid0)
  case False
    then have get-reg-state gs c aid = Some CONFISCATED
      using c sync-reg-other-asset[OF synced False] by simp
    then show False using nt unfolding terminal-present-def by blast
  next
    case True
      have c  $\in$  connected-chains gs' aid0
        using c True get-reg-state-connected by blast
      then have conn: c  $\in$  connected-chains gs aid0
        using sync-connected-chains-preserved[OF synced] by simp
      obtain cur new-st where

```

```

    tr: reg-transition cur act = Some new-st
    and val: get-reg-state gs' c aid0 = Some new-st
    using sync-connected-value[OF synced conn] by blast
    have new-st = CONFISCATED using c val True by simp
    then have act = CONFISCATE using to-confiscated-only-confiscate tr by
blast
    then have terminal-present gs aid0
    using sr by (simp add: safe-recovery-def)
    then show False using nt True by simp
qed
qed

```

Machine-checked regression witnesses for the two excluded behaviours. The two-chain builder places the given regulatory values on chains 0 and $\text{Suc } 0$ of asset 0 ; both witnesses evaluate by unfolding.

definition *mixed-pair-state* :: *reg-state* $\Rightarrow$ *reg-state* $\Rightarrow$ *global-state* **where**

```

mixed-pair-state v0 v1 =
  (| gs-chains = ( $\lambda c a$ . if  $a = 0 \wedge c = 0$ 
    then Some (| as-reg-state = v0 |)
    else if  $a = 0 \wedge c = \text{Suc } 0$ 
    then Some (| as-reg-state = v1 |)
    else None),
    gs-locks = ( $\lambda a$ . False) |)

```

lemma *mixed-pair-get-reg*:

```

get-reg-state (mixed-pair-state v0 v1) c a =
  (if  $a = 0 \wedge c = 0$  then Some v0 else if  $a = 0 \wedge c = \text{Suc } 0$  then Some v1 else
None)
by (simp add: mixed-pair-state-def get-reg-state-def get-asset-state-def)

```

Indiscriminate confiscation is excluded: on a plain *ACTIVE/FROZEN* disagreement with no terminal holding, no *CONFISCATE* propagation is a safe recovery, from any source chain.

lemma *blind-confiscate-excluded*:

```

 $\neg$  terminal-present (mixed-pair-state ACTIVE FROZEN) 0
 $\neg$  safe-recovery (mixed-pair-state ACTIVE FROZEN) (c, CONFISCATE, 0)
by (auto simp: terminal-present-def safe-recovery-def mixed-pair-get-reg split:
if-splits)

```

Confiscation erasure is excluded: on a *CONFISCATED/ACTIVE* disagreement the terminal state is present, so no non-*CONFISCATE* broadcast is a safe recovery — a recorded confiscation cannot be revived by overwriting it with a non-terminal value.

lemma *terminal-overwrite-excluded*:

```

terminal-present (mixed-pair-state CONFISCATED ACTIVE) 0
act  $\neq$  CONFISCATE  $\implies \neg$  safe-recovery (mixed-pair-state CONFISCATED
ACTIVE) (c, act, 0)
proof -

```

```

have get-reg-state (mixed-pair-state CONFISCATED ACTIVE) 0 0 = Some
CONFISCATED
by (simp add: mixed-pair-get-reg)
then show terminal-present (mixed-pair-state CONFISCATED ACTIVE) 0
unfolding terminal-present-def by blast
then show act ≠ CONFISCATE ⇒ ¬ safe-recovery (mixed-pair-state CON-
FISCATED ACTIVE) (c, act, 0)
by (simp add: safe-recovery-def)
qed

```

46 The Oraclizer as a Converging Composition

We package an abstract convergence specification as the data of a *converging-composition*: the carrier is the set of finite-domain, unlocked global states; the operations are synchronization triples; the invariant is cross-chain consistency; the progress measure is the inconsistency count; and the realization map uses Hilbert choice, in any inconsistent state, to pick a terminal-faithful safe recovery — a synchronization that reduces the measure and uses *CONFISCATE* exactly on terminal-bearing assets (existence guaranteed by *inconsistent-has-safe-recovery*). The event argument does not compute that recovery, so this is an existential fair-discharge model rather than an executable queue, priority, or BFT algorithm.

definition *oss-carrier* :: *global-state set* **where**
oss-carrier = {*gs*. *finite-domain gs* ∧ *no-locked-without-reason gs*}

definition *oss-step* ::
global-state ⇒ (*chain-id* × *reg-action* × *asset-id*) ⇒ *global-state option* **where**
oss-step gs p = (case *p* of (*src*, *act*, *aid*) ⇒ *sync src act aid gs*)

definition *oss-ops* :: (*chain-id* × *reg-action* × *asset-id*) *set* **where**
oss-ops = *UNIV*

definition *oss-guard* ::
global-state ⇒ (*chain-id* × *reg-action* × *asset-id*) ⇒ *bool* **where**
oss-guard gs p = (case *p* of (*src*, *act*, *aid*) ⇒ (∃ *gs'*. *sync src act aid gs* = Some *gs'*))

definition *oss-realize* ::
node-info ⇒ *global-state* ⇒ (*chain-id* × *reg-action* × *asset-id*) *option* **where**
oss-realize ev gs =
(if *consistent-state gs* then None else Some (*SOME p*. *safe-recovery gs p*))

The carrier is inhabited: every authenticated extraction with a finite revealed-chain set lies in it, so the convergence development below ranges over a non-empty carrier.

lemma *auth-state-in-oss-carrier*:
assumes *finite P*

shows $\text{auth-state } r \ P \in \text{oss-carrier}$
proof –
 have $\{(c, \text{aid}). \text{asset-exists } (\text{auth-state } r \ P) \ c \ \text{aid}\} = P \times \{0\}$
 by (auto simp: asset-exists-def get-asset-state-def auth-state-def split: if-splits)
 then have finite-domain (auth-state $r \ P$)
 unfolding finite-domain-def using assms by simp
 moreover have no-locked-without-reason (auth-state $r \ P$)
 by (simp add: no-locked-without-reason-def is-locked-def auth-state-def)
 ultimately show ?thesis by (simp add: oss-carrier-def)
qed

lemma *oss-step-closed*:
 assumes $gs \in \text{oss-carrier}$ and $\text{oss-step } gs \ p = \text{Some } gs'$
 shows $gs' \in \text{oss-carrier}$
proof –
 obtain $\text{src } \text{act } \text{aid}$ where $p: p = (\text{src}, \text{act}, \text{aid})$ by (cases p)
 with assms(2) have synced: $\text{sync } \text{src } \text{act } \text{aid } gs = \text{Some } gs'$ by (simp add: oss-step-def)
 from assms(1) have finite-domain gs and no-locked-without-reason gs
 by (simp-all add: oss-carrier-def)
 then have finite-domain gs' and no-locked-without-reason gs'
 using sync-preserves-finite-domain[OF - synced] sync-preserves-unlocked[OF - synced] by auto
 then show ?thesis by (simp add: oss-carrier-def)
qed

lemma *oss-guarded-preservation*:
 assumes $\text{carr}: gs \in \text{oss-carrier}$ and $\text{cons}: \text{consistent-state } gs$
 and $\text{synced}: \text{oss-step } gs \ p = \text{Some } gs'$
 shows $\text{consistent-state } gs'$
proof –
 obtain $\text{src } \text{act } \text{aid}$ where $p: p = (\text{src}, \text{act}, \text{aid})$ by (cases p)
 with synced have $s: \text{sync } \text{src } \text{act } \text{aid } gs = \text{Some } gs'$ by (simp add: oss-step-def)
 from carr have $nl: \text{no-locked-without-reason } gs$
 by (simp add: oss-carrier-def)
 have $\text{valid}: \text{valid-state } gs$ using cons nl by (simp add: valid-state-def)
 from sync-components[OF s] obtain $\text{current-st } \text{new-st } gs\text{-locked}$ where
 $\text{cur}: \text{get-reg-state } gs \ \text{src } \text{aid} = \text{Some } \text{current-st}$
 and $\text{tr}: \text{reg-transition } \text{current-st } \text{act} = \text{Some } \text{new-st}$ by blast
 show ?thesis
 using sync-preserves-consistent-state[OF valid cur tr s] .
qed

lemma *oss-realize-discharges*:
 assumes $\text{carr}: gs \in \text{oss-carrier}$ and $r: \text{oss-realize } ev \ gs = \text{Some } \text{opn}$
 shows $\text{opn} \in \text{oss-ops} \wedge \text{oss-guard } gs \ \text{opn}$
proof –
 from r have $\text{ninc}: \neg \text{consistent-state } gs$ and $\text{opn}: \text{opn} = (\text{SOME } p. \text{safe-recovery } gs \ p)$

```

  by (auto simp: oss-realize-def split: if-splits)
  from carr have fd: finite-domain gs and nl: no-locked-without-reason gs
  by (simp-all add: oss-carrier-def)
  have  $\exists p. \text{safe-recovery } gs \ p$  using inconsistent-has-safe-recovery[OF fd nl ninc]
  .
  then have safe-recovery gs opn using opn by (metis someI-ex)
  then have reducing-sync gs opn by (rule safe-recovery-reducing)
  then obtain src act aid gs' where  $opn = (src, act, aid)$  and  $sync \ src \ act \ aid$ 
   $gs = \text{Some } gs'$ 
  unfolding reducing-sync-def by (auto split: prod.splits)
  then have oss-guard gs opn by (auto simp: oss-guard-def)
  then show ?thesis by (simp add: oss-ops-def)
qed

```

lemma *oss-discharge-progresses*:

```

  assumes carr:  $gs \in \text{oss-carrier}$  and ninc:  $\neg \text{consistent-state } gs$ 
  shows  $\exists opn \ gs'. \text{oss-realize } ev \ gs = \text{Some } opn \wedge \text{oss-step } gs \ opn = \text{Some } gs'$ 
   $\wedge \text{inconsistency-pairs } gs' < \text{inconsistency-pairs } gs$ 

```

proof –

```

  from carr have fd: finite-domain gs and nl: no-locked-without-reason gs
  by (simp-all add: oss-carrier-def)
  have ex:  $\exists p. \text{safe-recovery } gs \ p$  using inconsistent-has-safe-recovery[OF fd nl ninc] .
  define opn where  $opn = (\text{SOME } p. \text{safe-recovery } gs \ p)$ 
  have sr: safe-recovery gs opn unfolding opn-def using ex by (rule someI-ex)
  have rs: reducing-sync gs opn using sr by (rule safe-recovery-reducing)
  have realize:  $\text{oss-realize } ev \ gs = \text{Some } opn$ 
  using ninc unfolding oss-realize-def opn-def by simp
  from rs obtain src act aid gs' where
   $p: opn = (src, act, aid)$  and  $s: sync \ src \ act \ aid \ gs = \text{Some } gs'$ 
  and dec:  $\text{inconsistency-pairs } gs' < \text{inconsistency-pairs } gs$ 
  unfolding reducing-sync-def by (auto split: prod.splits)
  have oss-step gs opn =  $\text{Some } gs'$  using p s by (simp add: oss-step-def)
  with realize dec show ?thesis by blast
qed

```

47 Guarded Bounded Convergence for the Oracizer

Inside the D-quencer liveness context (a fair-leader assumption over a Byzantine-threshold cardinality bound), the abstract data above forms a *converging-composition*: an honest-tagged event within every fairness window gates an existential measure-reducing synchronization. The fairness assumption is exactly the *fair-leader* assumption of *dquencer-liveness*; window positivity is derived from bounded occurrence inside the composition locale (*window-positive*), so no positivity obligation remains here.

The fairness assumption of *dquencer-liveness* is satisfiable in ev-

ery D-quencer system: the BFT threshold yields an honest node (*dquencer-system.honest-nonempty*), and the constant schedule on that node — a schedule drawn from the system’s own roster — meets every fairness window. This grounds the assume-guarantee abstraction: the system’s own threshold supplies an in-roster honest-node witness for a constant schedule, so the deterministic hypothesis is satisfiable. This is not a derivation from VRF probabilities or a model of adversarial consensus execution.

context *dquencer-system*
begin

lemma *fair-schedule-exists*:

$\exists \text{ sched} :: \text{nat} \Rightarrow \text{node-info}.$
 $\text{range sched} \subseteq \text{nodes} \wedge$
 $(\forall \text{ epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$
 $\text{ni-behavior (sched } e) = \text{Honest})$

proof —

obtain *h* **where** *hin*: $h \in \text{honest-nodes nodes}$ **using** *honest-nonempty* **by** *blast*
then have *hb*: $\text{ni-behavior } h = \text{Honest}$ **and** *hn*: $h \in \text{nodes}$
by (*simp-all add: honest-nodes-def*)
define *sched* :: $\text{nat} \Rightarrow \text{node-info}$ **where** *sched* = $(\lambda-. h)$
have $\text{range sched} \subseteq \text{nodes}$ **using** *hn* **by** (*auto simp: sched-def*)
moreover have $\forall \text{ epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound}$
 $\wedge \text{ni-behavior (sched } e) = \text{Honest}$
using *fairness-positive hb* **by** (*auto simp: sched-def*)
ultimately show *?thesis* **by** *blast*

qed

What the Byzantine threshold contributes here. The threshold guarantees an honest node *within the roster* (*honest-nonempty*, via *honest-majority*), and the constant schedule on that node witnesses the fair-leader hypothesis with a schedule drawn from the roster — so for *every* D-quencer system *dquencer-liveness* is inhabitable by the system’s own nodes (*liveness_inhabitable*, a satisfiability witness for the locale, terminal; the in-roster conjunct is what the threshold buys). The scope of this role is deliberately narrow: the headline bounded-convergence theorem below consumes the fair-leader assumption directly and does not route through this witness, being driven instead by the cross-chain inconsistency measure, and the honest *majority* beyond non-emptiness is not consumed by the inhabitability result.

theorem *liveness-inhabitable*:

$\exists (ls :: \text{nat} \Rightarrow \text{node-info}) (pc :: \text{nat} \Rightarrow \text{nat}).$
 $\text{range } ls \subseteq \text{nodes} \wedge \text{dquencer-liveness nodes } f\text{-max fairness-bound } ls \text{ } pc$

proof —

obtain *sched* :: $\text{nat} \Rightarrow \text{node-info}$ **where** *rng*: $\text{range sched} \subseteq \text{nodes}$
and *fair*: $\forall \text{ epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + \text{fairness-bound} \wedge$
 $\text{ni-behavior (sched } e) = \text{Honest}$

```

    using fair-schedule-exists by blast
    have dquencer-liveness nodes f-max fairness-bound sched ( $\lambda$ -. 0)
    by unfold-locales (use rng fair in auto)
    then show ?thesis using rng by blast
qed

end

context dquencer-liveness
begin

interpretation oss: converging-composition
  oss-carrier oss-step oss-ops consistent-state oss-guard
  leader-schedule  $\lambda n$ . ni-behavior  $n = \text{Honest}$  fairness-bound oss-realize inconsistency-pairs
proof unfold-locales
  show  $\bigwedge s \text{ opn } s'. s \in \text{oss-carrier} \implies \text{opn} \in \text{oss-ops} \implies \text{oss-step } s \text{ opn} = \text{Some } s'$ 
     $\implies s' \in \text{oss-carrier}$ 
    using oss-step-closed by blast
next
  show  $\bigwedge s \text{ opn } s'. s \in \text{oss-carrier} \implies \text{opn} \in \text{oss-ops} \implies \text{consistent-state } s$ 
     $\implies \text{oss-guard } s \text{ opn} \implies \text{oss-step } s \text{ opn} = \text{Some } s' \implies \text{consistent-state } s'$ 
    using oss-guarded-preservation by blast
next
  show  $\forall t. \exists t'. t \leq t' \wedge t' < t + \text{fairness-bound} \wedge \text{ni-behavior } (\text{leader-schedule } t') = \text{Honest}$ 
    using fair-leader by blast
next
  show  $\bigwedge s \text{ ev opn. } s \in \text{oss-carrier} \implies \text{ni-behavior } \text{ev} = \text{Honest} \implies \text{oss-realize } \text{ev}$ 
     $s = \text{Some } \text{opn}$ 
     $\implies \text{opn} \in \text{oss-ops} \wedge \text{oss-guard } s \text{ opn}$ 
    using oss-realize-discharges by blast
next
  show  $\bigwedge s \text{ ev. } s \in \text{oss-carrier} \implies \neg \text{consistent-state } s \implies \text{ni-behavior } \text{ev} = \text{Honest}$ 
     $\implies \exists \text{ opn } s'. \text{oss-realize } \text{ev } s = \text{Some } \text{opn} \wedge \text{oss-step } s \text{ opn} = \text{Some } s'$ 
     $\wedge \text{inconsistency-pairs } s' < \text{inconsistency-pairs } s$ 
    using oss-discharge-progresses by blast
qed

```

The headline of this entry's convergence layer. From an *arbitrary* finite-domain, unlocked global state — in particular with *no* assumption that the cross-chain consistency invariant holds initially — the model has an existential run that reaches, within *inconsistency-pairs* $gs_0 * \text{fairness-bound}$ evolution steps, a state that is valid (consistent and unlocked). The selected safe recovery is supplied non-constructively by *SOME*; this theorem is not an executable recovery procedure. Contrast *conditional-safety-preservation*, which assumes *valid-state* gs : the convergence layer frees the safety guarantee of its initial-validity hypothesis, at the price of the fair-leader context.

theorem *oracizer-guarded-bounded-convergence*:

assumes *fin*: *finite-domain* gs_0

and *unlocked*: *no-locked-without-reason* gs_0

shows $\exists t \leq \text{oss.convergence-bound } gs_0. \exists gs_t.$

$\text{oss.evolved-to } gs_0 \ t \ gs_t \wedge \text{valid-state } gs_t$

proof –

have $carr: gs_0 \in \text{oss-carrier}$ **using** *fin* *unlocked* **by** (*simp add: oss-carrier-def*)

obtain $t \ gs_t$ **where** $tb: t \leq \text{oss.convergence-bound } gs_0$

and *ev*: *oss.evolved-to* $gs_0 \ t \ gs_t$ **and** *cons*: *consistent-state* gs_t

using *oss.bounded-convergence-from-arbitrary*[*OF carr*] **by** *blast*

— The reached state remains in the carrier, hence is unlocked, hence valid.

have $gs_t = \text{oss.run-from } 0 \ gs_0 \ t$

using *ev* **by** (*simp add: oss.evolved-to-def oss.run-def*)

then have $gs_t \in \text{oss-carrier}$ **using** *oss.run-from-carrier*[*OF carr*] **by** *simp*

then have *no-locked-without-reason* gs_t **by** (*simp add: oss-carrier-def*)

then have *valid-state* gs_t **using** *cons* **by** (*simp add: valid-state-def*)

with *tb ev* **show** *?thesis* **by** *blast*

qed

Terminal faithfulness along whole recovery runs. Every evolution step either stutters or applies a realized safe recovery, and a safe recovery never confiscates an asset that carried no terminal holding (*safe-recovery-sync-no-fresh-terminal*); by induction the guarantee extends to the entire trajectory: an asset that starts with no recorded confiscation never acquires one, no matter how the run is scheduled. Together with *recovery-terminal-completion* — on a terminal-bearing asset a safe recovery can only complete the confiscation — this pins the recovery layer, on the terminal axis, to the two behaviours the regulatory semantics admits for *CONFISCATED*: indiscriminate confiscation and confiscation erasure are both excluded. On the non-terminal axis the broadcast value is constrained by the source chain’s transition table, not by the target chains’.

lemma *oss-evolve-step-no-fresh-terminal*:

assumes *carr*: $gs \in \text{oss-carrier}$

and *nt*: $\neg \text{terminal-present } gs \ \text{aid}$

shows $\neg \text{terminal-present } (\text{oss.evolve-step } t \ gs) \ \text{aid}$

proof (*cases ni-behavior (leader-schedule t) = Honest*)

case *False*

then show *?thesis* **using** *nt* **by** (*simp add: oss.evolve-step-def*)

next

case *True*

show *?thesis*

proof (*cases oss-realize (leader-schedule t) gs*)

case *None*

with *True* **show** *?thesis* **using** *nt* **by** (*simp add: oss.evolve-step-def*)

next

case (*Some opn*)

note $r = \text{this}$

show *?thesis*

```

proof (cases oss-step gs opn)
  case None
  with True r show ?thesis using nt by (simp add: oss.evolve-step-def)
next
  case (Some gs')
  note st = this
  from r have ninc:  $\neg$  consistent-state gs
    and opn-eq: opn = (SOME p. safe-recovery gs p)
    by (auto simp: oss-realize-def split: if-splits)
  from carr have fd: finite-domain gs and nl: no-locked-without-reason gs
    by (simp-all add: oss-carrier-def)
  have  $\exists p$ . safe-recovery gs p using inconsistent-has-safe-recovery[OF fd nl
ninc] .
  then have sr: safe-recovery gs opn using opn-eq by (metis someI-ex)
  obtain src act aid0 where p: opn = (src, act, aid0) by (cases opn)
  have synced: sync src act aid0 gs = Some gs' using st p by (simp add:
oss-step-def)
  have  $\neg$  terminal-present gs' aid
    using safe-recovery-sync-no-fresh-terminal[OF sr[unfolded p] synced nt] .
  with True r st show ?thesis by (simp add: oss.evolve-step-def)
  qed
qed
qed

lemma oss-run-from-no-fresh-terminal:
  assumes carr:  $gs_0 \in$  oss-carrier
  and nt:  $\neg$  terminal-present  $gs_0$  aid
  shows  $\neg$  terminal-present (oss.run-from k  $gs_0$  n) aid
proof (induction n)
  case 0
  show ?case using nt by simp
next
  case (Suc n)
  have c: oss.run-from k  $gs_0$  n  $\in$  oss-carrier using oss.run-from-carrier[OF carr]
  .
  show ?case using oss-evolve-step-no-fresh-terminal[OF c Suc.IH] by simp
qed

theorem recovery-no-fresh-terminal:
  assumes carr:  $gs_0 \in$  oss-carrier
  and nt:  $\neg$  terminal-present  $gs_0$  aid
  and ev: oss.evokes-to  $gs_0$  t  $gs_t$ 
  shows  $\neg$  terminal-present  $gs_t$  aid
proof -
  have  $gs_t =$  oss.run-from 0  $gs_0$  t
    using ev by (simp add: oss.evokes-to-def oss.run-def)
  then show ?thesis using oss-run-from-no-fresh-terminal[OF carr nt] by simp
qed

```

end

48 Global Model Witnesses for the Liveness Hosts

The liveness interpretations above live inside *dquencer-liveness*, so they establish the consistency of their conclusions only relative to the host locale's assumptions. The interpretations below discharge those assumptions globally, with no ambient hypotheses: a single-honest-node system with zero Byzantine budget and a unit fairness window, scheduled constantly on its one node, with no pending requests. Every assumption of *dquencer-system* and *dquencer-liveness* holds outright (the BFT threshold reads $1 \geq 3 \cdot 0 + 1$, the Byzantine census is empty, and the constant schedule meets every unit fairness window), so the assumption sets of the liveness hosts are consistent absolutely, not merely relative to a hypothetical deployment.

definition *singleton-node* :: *node-info* **where**

singleton-node = $\langle \text{ni-id} = 0, \text{ni-behavior} = \text{Honest} \rangle$

lemma *singleton-fair-leader*:

$\forall \text{epoch}. \exists e. \text{epoch} \leq e \wedge e < \text{epoch} + (1::\text{nat})$
 $\wedge \text{ni-behavior } ((\lambda-. \text{singleton-node}) e) = \text{Honest}$

proof

fix *epoch* :: *nat*

have $\text{epoch} \leq \text{epoch} \wedge \text{epoch} < \text{epoch} + 1$

$\wedge \text{ni-behavior } ((\lambda-. \text{singleton-node}) \text{epoch}) = \text{Honest}$

by (*simp add: singleton-node-def*)

then show $\exists e. \text{epoch} \leq e \wedge e < \text{epoch} + 1$

$\wedge \text{ni-behavior } ((\lambda-. \text{singleton-node}) e) = \text{Honest}$

by *blast*

qed

interpretation *singleton-dq: dquencer-liveness*

$\{ \text{singleton-node} \} 0 1 0 0 \lambda-. \text{singleton-node} \lambda-. 0$

by *unfold-locales*

(*auto simp: byzantine-nodes-def singleton-node-def intro: singleton-fair-leader*)

The priority sublocale receives the analogous witness: the one-message set over the same singleton system. The well-formedness bounds hold with both maxima at 0, and priority-key distinctness on a one-element set is immediate.

definition *singleton-msg* :: *dq-message* **where**

singleton-msg =

$\langle \text{msg-action} = \text{FREEZE}, \text{msg-timestamp} = 0,$
 $\text{dqm-authority-level} = \text{National}, \text{dqm-source-node} = 0 \rangle$

interpretation *singleton-dq-priority: dquencer-priority-concrete*

$\{ \text{singleton-node} \} 0 1 0 0 \{ \text{singleton-msg} \}$

by *unfold-locales*

(*auto simp: byzantine-nodes-def singleton-node-def singleton-msg-def*
intro: singleton-fair-leader)

A non-degenerate Byzantine witness. The singleton system above carries a zero Byzantine budget, so it meets the threshold vacuously ($1 \geq 3 \cdot 0 + 1$). The four-node system below exercises the threshold at its tight bound with a *real* fault: one Byzantine node among four ($4 \geq 3 \cdot 1 + 1$ and $\text{card } (\text{byzantine-nodes}) = 1 \leq 1$), an honest super-majority of three, scheduled constantly on one honest node. It discharges every assumption of *dquencer-liveness* with $f\text{-max} = 1$, so the Byzantine threshold is exercised at its tight bound by a real Byzantine-tagged node (a cardinality witness), not only the degenerate zero-fault case.

definition *quorum-nodes* :: *node-info set* **where**

quorum-nodes =
 $\{ \langle \text{ni-id} = 0, \text{ni-behavior} = \text{Byzantine} \rangle,$
 $\langle \text{ni-id} = 1, \text{ni-behavior} = \text{Honest} \rangle,$
 $\langle \text{ni-id} = 2, \text{ni-behavior} = \text{Honest} \rangle,$
 $\langle \text{ni-id} = 3, \text{ni-behavior} = \text{Honest} \rangle \}$

definition *quorum-leader* :: *nat* $\Rightarrow$ *node-info* **where**

quorum-leader = ($\lambda\text{-} \langle \text{ni-id} = 1, \text{ni-behavior} = \text{Honest} \rangle$)

lemma *quorum-finite*: *finite quorum-nodes*

by (*simp add: quorum-nodes-def*)

lemma *quorum-nonempty*: *quorum-nodes* $\neq \{\}$

by (*simp add: quorum-nodes-def*)

lemma *quorum-card*: *card quorum-nodes* = 4

by (*simp add: quorum-nodes-def*)

lemma *quorum-byzantine-eq*:

byzantine-nodes quorum-nodes = $\{ \langle \text{ni-id} = 0, \text{ni-behavior} = \text{Byzantine} \rangle \}$
by (*auto simp: quorum-nodes-def byzantine-nodes-def*)

lemma *quorum-byz-card*: *card (byzantine-nodes quorum-nodes)* = 1

by (*simp add: quorum-byzantine-eq*)

lemma *quorum-leader-in-nodes*: *range quorum-leader* $\subseteq$ *quorum-nodes*

by (*auto simp: quorum-leader-def quorum-nodes-def*)

interpretation *bft-quorum*: *dquencer-liveness*

quorum-nodes 1 1 0 0 *quorum-leader* $\lambda e. 1 - e$

proof *unfold-locales*

show *finite quorum-nodes* **by** (*rule quorum-finite*)

next

show $3 * (1::\text{nat}) + 1 \leq \text{card } \text{quorum-nodes}$

using *quorum-card* **by** *simp*


```

next
  show card (byzantine-nodes quorum-nodes) ≤ (1::nat)
  using quorum-byz-card by simp
next
  show 0 < (1::nat) by simp
next
  show range quorum-leader ⊆ quorum-nodes ∧
    (∀ epoch. ∃ e. epoch ≤ e ∧ e < epoch + (1::nat) ∧
      ni-behavior (quorum-leader e) = Honest)

  proof
    show range quorum-leader ⊆ quorum-nodes
    by (rule quorum-leader-in-nodes)
    show ∀ epoch. ∃ e. epoch ≤ e ∧ e < epoch + (1::nat) ∧
      ni-behavior (quorum-leader e) = Honest
    by (auto simp: quorum-leader-def)
  qed
next
  fix e
  assume ni-behavior (quorum-leader e) = Honest 0 < 1 - e
  then show 1 - Suc e < 1 - e by auto
next
  fix e
  show 1 - Suc e ≤ 1 - e by auto
qed

```

Deterministic selection is exercised on a genuine conflict: two well-formed messages from different authorities over the four-node quorum system. The international confiscation outranks the regional unfreeze in the lexicographic priority order, and selection picks it deterministically — the conflict-resolution reading of Property 2 on a non-degenerate candidate set.

definition *conflict-msg-intl* :: dq-message **where**

```

conflict-msg-intl =
  ⟨ msg-action = CONFISCATE, msg-timestamp = 0,
    dqm-authority-level = International, dqm-source-node = 0 ⟩

```

definition *conflict-msg-regional* :: dq-message **where**

```

conflict-msg-regional =
  ⟨ msg-action = UNFREEZE, msg-timestamp = 0,
    dqm-authority-level = Regional, dqm-source-node = 0 ⟩

```

interpretation *conflict-dq-priority*: dqencer-priority-concrete
 quorum-nodes 1 1 0 0 {conflict-msg-intl, conflict-msg-regional}

by unfold-locales

```

(auto simp: quorum-card quorum-byz-card quorum-finite quorum-nonempty
  conflict-msg-intl-def conflict-msg-regional-def
  make-priority-key-def)

```

lemma *conflict-selection-deterministic*:

```

dq-priority.select-highest conflict-dq-priority.msg-priority-keys
= Some (make-priority-key 0 0 conflict-msg-intl)
proof –
  have keys: conflict-dq-priority.msg-priority-keys
    = {make-priority-key 0 0 conflict-msg-intl,
       make-priority-key 0 0 conflict-msg-regional}
  by (simp add: conflict-dq-priority.msg-priority-keys-def)
  have ki: make-priority-key 0 0 conflict-msg-intl = (3, 0, 7, 0)
  by (simp add: conflict-msg-intl-def make-priority-key-def)
  have kr: make-priority-key 0 0 conflict-msg-regional = (1, 0, 2, 0)
  by (simp add: conflict-msg-regional-def make-priority-key-def)
  show ?thesis
  unfolding dq-priority.select-highest-def keys ki kr
  by (auto intro!: the-equality simp: less-eq-prod-def)
qed

end

```

```

theory Hierarchy
imports Functor-Laws
begin

```

49 Sync Degree Stratification

A degree- k broadcast drives every chain $c \leq k$ that currently holds the asset aid to the regulatory value v , leaving every other chain and every other asset untouched. It is the state-level effect of synchronizing the asset across the chains coupled at degree k .

definition *broadcast-le* ::
 $\text{nat} \Rightarrow \text{asset-id} \Rightarrow \text{reg-state} \Rightarrow \text{global-state} \Rightarrow \text{global-state}$ **where**
broadcast-le k aid v gs =
 $gs \downarrow gs\text{-chains} :=$
 $(\lambda c. \text{if } c \leq k$
 $\text{then } (\lambda aid'. \text{if } aid' = aid$
 $\text{then } \text{map-option } (\lambda ast. \text{ast} \downarrow \text{as-reg-state} := v) \downarrow$
 $(gs\text{-chains } gs \ c \ aid)$
 $\text{else } gs\text{-chains } gs \ c \ aid')$
 $\text{else } gs\text{-chains } gs \ c) \downarrow)$

lemma *broadcast-le-locks* [simp]:
 $gs\text{-locks } (broadcast\text{-le } k \ aid \ v \ gs) = gs\text{-locks } gs$
by (simp add: broadcast-le-def)

lemma *broadcast-le-exists* [simp]:
 $\text{asset-exists } (broadcast\text{-le } k \ aid \ v \ gs) \ c \ aid' = \text{asset-exists } gs \ c \ aid'$
by (auto simp: broadcast-le-def asset-exists-def get-asset-state-def
split: option.splits)

lemma *broadcast-le-reg-other-asset:*

$aid' \neq aid \implies$
 $get-reg-state \ (broadcast-le \ k \ aid \ v \ gs) \ c \ aid' = get-reg-state \ gs \ c \ aid'$
by (*simp add: broadcast-le-def get-reg-state-def get-asset-state-def*)

lemma *broadcast-le-reg-out:*

$\neg c \leq k \implies$
 $get-reg-state \ (broadcast-le \ k \ aid \ v \ gs) \ c \ aid' = get-reg-state \ gs \ c \ aid'$
by (*simp add: broadcast-le-def get-reg-state-def get-asset-state-def*)

lemma *broadcast-le-reg-in:*

$c \leq k \implies$
 $get-reg-state \ (broadcast-le \ k \ aid \ v \ gs) \ c \ aid =$
 $(case \ get-reg-state \ gs \ c \ aid \ of \ None \Rightarrow \ None \mid \ Some \ - \Rightarrow \ Some \ v)$
by (*simp add: broadcast-le-def get-reg-state-def get-asset-state-def*
 $map-option-case \ split: \ option.splits$)

Pointwise reading of the chain map after a broadcast: a clean rewrite that avoids unfolding the raw record update inside nested terms.

lemma *broadcast-le-chains:*

$gs-chains \ (broadcast-le \ k \ aid \ v \ gs) \ c =$
 $(if \ c \leq k$
 $then \ (\lambda aid'. \ if \ aid' = aid$
 $then \ map-option \ (\lambda ast. \ ast(\ as-reg-state := v \)) \ (gs-chains \ gs \ c \ aid)$
 $else \ gs-chains \ gs \ c \ aid')$
 $else \ gs-chains \ gs \ c)$
by (*simp add: broadcast-le-def*)

Degree demotion: *degree-forget j* drops the holdings of chain *j*, the top chain coupled one degree up. It keeps locks and every other chain.

definition *degree-forget :: nat $\Rightarrow$ global-state $\Rightarrow$ global-state* **where**

$degree-forget \ j \ gs = gs(\ gs-chains := (gs-chains \ gs)(j := (\lambda aid. \ None)) \)$

lemma *degree-forget-locks [simp]:*

$gs-locks \ (degree-forget \ j \ gs) = gs-locks \ gs$
by (*simp add: degree-forget-def*)

lemma *degree-forget-reg-eq:*

$c \neq j \implies get-reg-state \ (degree-forget \ j \ gs) \ c \ aid = get-reg-state \ gs \ c \ aid$
by (*simp add: degree-forget-def get-reg-state-def get-asset-state-def*)

lemma *degree-forget-exists:*

$asset-exists \ (degree-forget \ j \ gs) \ c \ aid =$
 $(c \neq j \wedge asset-exists \ gs \ c \ aid)$
by (*simp add: degree-forget-def asset-exists-def get-asset-state-def*)

lemma *degree-forget-chains:*

$gs-chains \ (degree-forget \ j \ gs) \ c =$

(if $c = j$ then $(\lambda aid. None)$ else $gs\text{-}chains\ gs\ c$)
by (*simp add: degree-forget-def*)

Demotion forgets, never invents: $degree\text{-}forget\ j\ gs$ is a blinding refinement of gs in the sense of the glue layer of *Cross-Domain-State-Preservation.Functor-Laws*. Every chain still known after demotion carries exactly the regulatory state it had before. This is the state-level reading of the ADS blinding order bo (*state-refines*): degree demotion lands inside the partial-view preorder on which the authenticated layer is built, so the hierarchy sits on top of the functor laws rather than beside them.

lemma *degree-forget-refines: state-refines (degree-forget j gs) gs*
unfolding *state-refines-def*
proof (*intro allI impI*)
fix *aid c*
assume $c \in connected\text{-}chains\ (degree\text{-}forget\ j\ gs)\ aid$
then have *asset-exists (degree-forget j gs) c aid*
by (*simp add: connected-chains-def*)
then have *cne: $c \neq j$ and ce: asset-exists gs c aid*
by (*simp-all add: degree-forget-exists*)
from ce have $c \in connected\text{-}chains\ gs\ aid$ **by** (*simp add: connected-chains-def*)
moreover have *get-reg-state (degree-forget j gs) c aid = get-reg-state gs c aid*
using cne by (*rule degree-forget-reg-eq*)
ultimately show
 $c \in connected\text{-}chains\ gs\ aid \wedge$
 $get\text{-}reg\text{-}state\ (degree\text{-}forget\ j\ gs)\ c\ aid = get\text{-}reg\text{-}state\ gs\ c\ aid$
by blast
qed

The degree- k transition step. An operation (act, aid) reads the hub chain's regulatory state for aid , fires the regulatory transition, and broadcasts the result across the chains coupled at degree k . It is undefined exactly when the hub does not hold the asset or the regulatory action is not enabled there.

definition *deg-step ::*
 $nat \Rightarrow (reg\text{-}action \times asset\text{-}id) \Rightarrow global\text{-}state \Rightarrow global\text{-}state\ option$ **where**
 $deg\text{-}step\ k\ opn\ gs =$
 $(case\ opn\ of\ (act, aid) \Rightarrow$
 $(case\ get\text{-}reg\text{-}state\ gs\ 0\ aid\ of$
 $None \Rightarrow None$
 $| Some\ s \Rightarrow (case\ reg\text{-}transition\ s\ act\ of$
 $None \Rightarrow None$
 $| Some\ s' \Rightarrow Some\ (broadcast\text{-}le\ k\ aid\ s'\ gs))))$

The carrier of the degree- k functor: states supported on chains $0..k$, i.e. only chains within the degree's coupling reach hold assets.

definition *deg-carrier :: nat $\Rightarrow$ global-state set where*
 $deg\text{-}carrier\ k = \{gs. \forall c\ aid. asset\text{-}exists\ gs\ c\ aid \longrightarrow c \leq k\}$

The carrier is inhabited — a non-vacuity witness for the degree functors: every authenticated extraction supported on chains $0..k$ lies in it.

lemma *deg-carrier-inhabited*:

assumes $P \subseteq \{0..k\}$

shows $\text{auth-state } r \ P \in \text{deg-carrier } k$

using *assms*

by (*auto simp: deg-carrier-def asset-exists-def get-asset-state-def auth-state-def split: if-splits*)

50 Degree Functors and Their Natural Transformations

We package each degree as the data of a transition functor: a set of states and an action-indexed family of partial transitions. Reading the one-object action category into *Set*, a natural transformation between two such functors is a single state-component map whose naturality squares commute. The conditions below mirror, on *global-state*, the well-definedness and naturality conditions of *state-preservation* (with the action map the identity, since the regulatory action vocabulary is shared across degrees); this is the same commuting square that defines the morphisms of the functor in *Cross-Domain-State-Preservation.Functor-Laws*.

record *gobj* =

obj-states :: *global-state set*

obj-step :: (*reg-action* $\times$ *asset-id*) $\Rightarrow$ *global-state* $\Rightarrow$ *global-state option*

definition $F :: \text{nat} \Rightarrow \text{gobj}$ **where**

$F\ k = \langle \text{obj-states} = \text{deg-carrier } k, \text{obj-step} = \text{deg-step } k \rangle$

definition *natural-transformation* ::

$\text{gobj} \Rightarrow \text{gobj} \Rightarrow (\text{global-state} \Rightarrow \text{global-state}) \Rightarrow \text{bool}$ **where**

natural-transformation *Fhi Flo* $\eta \longleftrightarrow$

$(\forall \text{gs}. \text{gs} \in \text{obj-states } Fhi \longrightarrow \eta \text{gs} \in \text{obj-states } Flo)$

$\wedge (\forall \text{opn gs gs}'. \text{gs} \in \text{obj-states } Fhi \longrightarrow \text{obj-step } Fhi \text{ opn gs} = \text{Some gs}'$
 $\longrightarrow \text{obj-step } Flo \text{ opn } (\eta \text{gs}) = \text{Some } (\eta \text{gs}'))$

$\wedge (\forall \text{opn gs}. \text{gs} \in \text{obj-states } Fhi \longrightarrow \text{obj-step } Fhi \text{ opn gs} = \text{None}$
 $\longrightarrow \text{obj-step } Flo \text{ opn } (\eta \text{gs}) = \text{None})$

Vertical composition of natural transformations is a natural transformation. This is the *global-state*-level analogue of *preservation-compose*: the naturality square of the lower transformation is stacked on that of the upper one, exactly as the second morphism's square is stacked on the first there. It is the structural fact that makes the whole degree ladder cohere.

lemma *nt-vertical-compose*:

assumes *hi*: *natural-transformation* *A B* η

and *lo*: *natural-transformation* *B C* ϑ

shows *natural-transformation* *A C* $(\vartheta \circ \eta)$

```

proof –
  from hi have
    hi1:  $\bigwedge gs. gs \in \text{obj-states } A \implies \eta \text{ } gs \in \text{obj-states } B$ 
    and hi2:  $\bigwedge \text{opn } gs \text{ } gs'. gs \in \text{obj-states } A \implies \text{obj-step } A \text{ } \text{opn } gs = \text{Some } gs'$ 
       $\implies \text{obj-step } B \text{ } \text{opn } (\eta \text{ } gs) = \text{Some } (\eta \text{ } gs')$ 
    and hi3:  $\bigwedge \text{opn } gs. gs \in \text{obj-states } A \implies \text{obj-step } A \text{ } \text{opn } gs = \text{None}$ 
       $\implies \text{obj-step } B \text{ } \text{opn } (\eta \text{ } gs) = \text{None}$ 
    unfolding natural-transformation-def by blast+
  from lo have
    lo1:  $\bigwedge gs. gs \in \text{obj-states } B \implies \vartheta \text{ } gs \in \text{obj-states } C$ 
    and lo2:  $\bigwedge \text{opn } gs \text{ } gs'. gs \in \text{obj-states } B \implies \text{obj-step } B \text{ } \text{opn } gs = \text{Some } gs'$ 
       $\implies \text{obj-step } C \text{ } \text{opn } (\vartheta \text{ } gs) = \text{Some } (\vartheta \text{ } gs')$ 
    and lo3:  $\bigwedge \text{opn } gs. gs \in \text{obj-states } B \implies \text{obj-step } B \text{ } \text{opn } gs = \text{None}$ 
       $\implies \text{obj-step } C \text{ } \text{opn } (\vartheta \text{ } gs) = \text{None}$ 
    unfolding natural-transformation-def by blast+
  show ?thesis
    unfolding natural-transformation-def
  proof (intro conjI allI impI)
    fix gs assume  $gs \in \text{obj-states } A$ 
    then show  $(\vartheta \circ \eta) \text{ } gs \in \text{obj-states } C$ 
      using hi1 lo1 by (simp add: comp-def)
  next
    fix opn gs gs'
    assume a:  $gs \in \text{obj-states } A$  and s:  $\text{obj-step } A \text{ } \text{opn } gs = \text{Some } gs'$ 
    have  $\text{obj-step } C \text{ } \text{opn } (\vartheta (\eta \text{ } gs)) = \text{Some } (\vartheta (\eta \text{ } gs'))$ 
      using lo2 [OF hi1 [OF a] hi2 [OF a s]] .
    then show  $\text{obj-step } C \text{ } \text{opn } ((\vartheta \circ \eta) \text{ } gs) = \text{Some } ((\vartheta \circ \eta) \text{ } gs')$ 
      by (simp add: comp-def)
  next
    fix opn gs
    assume a:  $gs \in \text{obj-states } A$  and s:  $\text{obj-step } A \text{ } \text{opn } gs = \text{None}$ 
    have  $\text{obj-step } C \text{ } \text{opn } (\vartheta (\eta \text{ } gs)) = \text{None}$ 
      using lo3 [OF hi1 [OF a] hi3 [OF a s]] .
    then show  $\text{obj-step } C \text{ } \text{opn } ((\vartheta \circ \eta) \text{ } gs) = \text{None}$ 
      by (simp add: comp-def)
  qed
qed

```

The key commutation: broadcasting at degree k after forgetting chain $\text{Suc } k$ equals forgetting chain $\text{Suc } k$ after broadcasting at degree $\text{Suc } k$. This is the heart of the naturality square: on chains $c \leq k$ both sides broadcast, on chain $\text{Suc } k$ both sides clear, and on higher chains both sides leave gs alone.

lemma *broadcast-forget-commute*:

$$\begin{aligned}
 & \text{broadcast-le } k \text{ aid } v \text{ (degree-forget (Suc } k) \text{ } gs) \\
 &= \text{degree-forget (Suc } k) \text{ (broadcast-le (Suc } k) \text{ aid } v \text{ } gs)
 \end{aligned}$$

proof (*rule global-state.equality, goal-cases chains locks more*)

case *chains*

show *?case*

```

proof (intro ext)
  fix c aid'
  consider (le)  $c \leq k$  | (top)  $c = \text{Suc } k$  | (gt)  $\text{Suc } k < c$  by linarith
  then show gs-chains (broadcast-le k aid v (degree-forget (Suc k) gs)) c aid'
    = gs-chains (degree-forget (Suc k) (broadcast-le (Suc k) aid v gs)) c
aid'
  proof cases
    case le
      then have  $c \neq \text{Suc } k$  and  $c \leq \text{Suc } k$  by linarith+
      with le show ?thesis
      by (simp add: broadcast-le-chains degree-forget-chains)
    next
      case top
      then have  $\neg c \leq k$  by linarith
      with top show ?thesis
      by (simp add: broadcast-le-chains degree-forget-chains)
    next
      case gt
      then have  $\neg c \leq k$  and  $\neg c \leq \text{Suc } k$  and  $c \neq \text{Suc } k$  by linarith+
      then show ?thesis
      by (simp add: broadcast-le-chains degree-forget-chains)
  qed
qed
next
  case locks then show ?case by simp
next
  case more then show ?case by simp
qed

```

The degree demotion *degree-forget* (*Suc k*) is a natural transformation F (*Suc k*) $\Rightarrow F$ *k*: it maps degree-*Suc k* states to degree-*k* states, and its naturality square commutes (forget-after-step equals step-after-forget, in both the enabled and the disabled case). Because the hub chain *0* is never the dropped chain *Suc k*, demotion leaves the transition's enabling condition — the hub's regulatory state — untouched.

theorem *degree-natural-transformation:*

natural-transformation (F (*Suc k*)) (F *k*) (*degree-forget* (*Suc k*))

unfolding *natural-transformation-def*

proof (intro conjI allI impI)

fix gs **assume** gsin: $gs \in \text{obj-states } (F \text{ (Suc } k))$

show *degree-forget* (*Suc k*) $gs \in \text{obj-states } (F \text{ } k)$

using gsin **by** (fastforce simp: *F-def* *deg-carrier-def* *degree-forget-exists*)

next

fix opn gs gs'

assume $gs \in \text{obj-states } (F \text{ (Suc } k))$

and *step*: *obj-step* ($F \text{ (Suc } k)$) opn $gs = \text{Some } gs'$

obtain act aid **where** opn: $\text{opn} = (\text{act}, \text{aid})$ **by** (cases opn)

from *step* opn **have** *deg-step* (*Suc k*) (act, aid) $gs = \text{Some } gs'$

by (simp add: *F-def*)

```

then obtain  $s \ s'$  where
   $hub$ :  $get\_reg\_state \ gs \ 0 \ aid = Some \ s$ 
  and  $tr$ :  $reg\_transition \ s \ act = Some \ s'$ 
  and  $gs'$ :  $gs' = broadcast\_le \ (Suc \ k) \ aid \ s' \ gs$ 
  by ( $auto \ simp$ :  $deg\_step\_def \ split$ :  $option.splits$ )
have  $hub'$ :  $get\_reg\_state \ (degree\_forget \ (Suc \ k) \ gs) \ 0 \ aid = Some \ s$ 
  using  $hub$  by ( $simp \ add$ :  $degree\_forget\_reg\_eq$ )
have  $obj\_step \ (F \ k) \ opn \ (degree\_forget \ (Suc \ k) \ gs)$ 
  =  $Some \ (broadcast\_le \ k \ aid \ s' \ (degree\_forget \ (Suc \ k) \ gs))$ 
  using  $hub' \ tr$  by ( $simp \ add$ :  $F\_def \ deg\_step\_def \ opn$ )
also have  $broadcast\_le \ k \ aid \ s' \ (degree\_forget \ (Suc \ k) \ gs)$ 
  =  $degree\_forget \ (Suc \ k) \ (broadcast\_le \ (Suc \ k) \ aid \ s' \ gs)$ 
  by ( $rule \ broadcast\_forget\_commute$ )
also have  $\dots = degree\_forget \ (Suc \ k) \ gs'$  using  $gs'$  by  $simp$ 
finally show  $obj\_step \ (F \ k) \ opn \ (degree\_forget \ (Suc \ k) \ gs)$ 
  =  $Some \ (degree\_forget \ (Suc \ k) \ gs')$  .

next
fix  $opn \ gs$ 
assume  $gs \in obj\_states \ (F \ (Suc \ k))$ 
  and  $step$ :  $obj\_step \ (F \ (Suc \ k)) \ opn \ gs = None$ 
obtain  $act \ aid$  where  $opn$ :  $opn = (act, aid)$  by ( $cases \ opn$ )
from  $step \ opn$  have  $none$ :  $deg\_step \ (Suc \ k) \ (act, aid) \ gs = None$ 
  by ( $simp \ add$ :  $F\_def$ )
have  $key$ :  $get\_reg\_state \ (degree\_forget \ (Suc \ k) \ gs) \ 0 \ aid = get\_reg\_state \ gs \ 0 \ aid$ 
  by ( $simp \ add$ :  $degree\_forget\_reg\_eq$ )
show  $obj\_step \ (F \ k) \ opn \ (degree\_forget \ (Suc \ k) \ gs) = None$ 
proof ( $cases \ get\_reg\_state \ gs \ 0 \ aid$ )
  case  $None$ 
  then show  $?thesis$  using  $key$  by ( $simp \ add$ :  $F\_def \ deg\_step\_def \ opn$ )
next
case ( $Some \ s$ )
with  $none$  have  $reg\_transition \ s \ act = None$ 
  by ( $simp \ add$ :  $deg\_step\_def \ split$ :  $option.splits$ )
with  $key \ Some$  show  $?thesis$  by ( $simp \ add$ :  $F\_def \ deg\_step\_def \ opn$ )
qed
qed

```

Natural transformations of degree functors compose. The two-step demotion $F \ (k+2) \Rightarrow F \ k$ is a natural transformation, obtained by vertically composing $F \ (k+2) \Rightarrow F \ (k+1)$ with $F \ (k+1) \Rightarrow F \ k$. This is what lifts the degree ladder from a point-to-point collection of forgetful maps to a structurally coherent tower: the whole chain $S_3 \Rightarrow S_2 \Rightarrow S_1 \Rightarrow S_0$ is natural at once. The degree functors share one action category as common source, so vertical composition is the only composition that arises for this tower; horizontal composition and whiskering do not typecheck here and are not claimed. This is a layer that, to our knowledge, `ADS_Functor` does not develop: it closes functors under composition but does not build a hierarchy of functors with natural transformations between them.


```

theorem nt-compose:
  natural-transformation (F (Suc (Suc k))) (F k)
    (degree-forget (Suc k) ∘ degree-forget (Suc (Suc k)))
proof (rule nt-vertical-compose)
  show natural-transformation (F (Suc (Suc k))) (F (Suc k)) (degree-forget (Suc
    (Suc k)))
    using degree-natural-transformation[of Suc k] .
next
  show natural-transformation (F (Suc k)) (F k) (degree-forget (Suc k))
    using degree-natural-transformation[of k] .
qed

```

51 Functoriality of the Degree Transition Functor on Action Words

The natural-transformation results above operate on single regulatory operations. To make F a functor in the genuine sense — and the degree demotion a natural transformation on the whole one-object action category, not only on its generators — we must extend the transition step to action *words* and prove it preserves the category structure (identity word and word concatenation). The action category is the free monoid of regulatory action sequences; its non-degenerate content is supplied by the regulatory transition relation, which composes the sequence into a single composite state.

```

fun reg-run :: reg-state ⇒ reg-action list ⇒ reg-state option where
  reg-run s [] = Some s
| reg-run s (a # w) =
  (case reg-transition s a of None ⇒ None | Some s' ⇒ reg-run s' w)

```

Regulatory composition: running a concatenated action word is running the first part and then, from the resulting state, the second. This is the monoid structure the degree functor must transport.

```

lemma reg-run-append:
  reg-run s (v @ w) = (case reg-run s v of None ⇒ None | Some s' ⇒ reg-run s'
    w)
  by (induction v arbitrary: s) (auto split: option.splits)

```

```

fun deg-run ::
  nat ⇒ asset-id ⇒ reg-action list ⇒ global-state ⇒ global-state option where
  deg-run k aid [] gs = Some gs
| deg-run k aid (a # w) gs =
  (case deg-step k (a, aid) gs of None ⇒ None | Some gs' ⇒ deg-run k aid w gs')

```

Broadcast overwrite: a second degree- k broadcast of the same asset obliterates the first. This is not a free-monoid triviality — it holds because the broadcast set is the chains that *already* hold the asset (which the first broadcast does not change) so the second writes over exactly the same chains. It is the structural reason a word of synchronizations collapses to a single one.

```

lemma broadcast-le-overwrite:
  broadcast-le k aid v2 (broadcast-le k aid v1 gs) = broadcast-le k aid v2 gs
proof (rule global-state.equality, goal-cases chains locks more)
  case chains
  show ?case
  proof (intro ext)
    fix c aid'
    show gs-chains (broadcast-le k aid v2 (broadcast-le k aid v1 gs)) c aid'
      = gs-chains (broadcast-le k aid v2 gs) c aid'
    proof (cases c ≤ k)
      case True
      show ?thesis
        by (cases aid' = aid; cases gs-chains gs c aid)
          (simp-all add: broadcast-le-chains True)
      next
      case False
      then show ?thesis by (simp add: broadcast-le-chains)
    qed
  qed
next
  case locks then show ?case by (simp add: broadcast-le-locks)
next
  case more then show ?case by simp
qed

```

Hub tracking: because the hub chain 0 is always within the coupling reach ($0 \leq k$), a broadcast that finds the asset at the hub leaves the hub carrying exactly the broadcast value, so the next step in a word reads the updated state.

```

lemma broadcast-le-hub:
  assumes get-reg-state gs 0 aid = Some s0
  shows get-reg-state (broadcast-le k aid v gs) 0 aid = Some v
  using assms broadcast-le-reg-in[of 0 k aid v gs] by simp

```

The collapse, on an already-broadcast state. Running an action word at degree k from a state of the form $\text{broadcast-le } k \text{ aid } v \text{ gs}$ reduces to the regulatory composite $\text{reg-run } v \text{ w}$ broadcast once. The induction is where the degree/broadcast structure does real work: each successive step reads the hub value left by the previous broadcast (broadcast-le-hub) and overwrites it ($\text{broadcast-le-overwrite}$).

```

lemma deg-run-on-broadcast:
  assumes asset-exists gs 0 aid
  shows deg-run k aid w (broadcast-le k aid v gs)
    = (case reg-run v w of None ⇒ None
      | Some sn ⇒ Some (broadcast-le k aid sn gs))
  using assms
proof (induction w arbitrary: v)
  case Nil

```

```

  show ?case by simp
next
case (Cons a w)
have hub: get-reg-state (broadcast-le k aid v gs) 0 aid = Some v
proof -
  from Cons.premis obtain ast where get-asset-state gs 0 aid = Some ast
  by (auto simp: asset-exists-def)
  then have get-reg-state gs 0 aid = Some (as-reg-state ast)
  by (simp add: get-reg-state-def)
  then show ?thesis by (rule broadcast-le-hub)
qed
show ?case
proof (cases reg-transition v a)
  case None
  then show ?thesis using hub by (simp add: deg-step-def)
next
case (Some v')
have step: deg-step k (a, aid) (broadcast-le k aid v gs)
  = Some (broadcast-le k aid v' gs)
  using hub Some by (simp add: deg-step-def broadcast-le-overwrite)
show ?thesis using Cons.IH[OF Cons.premis, of v'] Some by (simp add: step)
qed
qed

```

The general collapse: a non-empty word of degree- k synchronizations on an asset held at the hub equals broadcasting the composite regulatory state once.

lemma *deg-run-collapse*:

```

  assumes get-reg-state gs 0 aid = Some s0
  shows deg-run k aid (a # w) gs
    = (case reg-run s0 (a # w) of None  $\Rightarrow$  None
      | Some sn  $\Rightarrow$  Some (broadcast-le k aid sn gs))
proof -
  have ex: asset-exists gs 0 aid
  using assms by (auto simp: asset-exists-def get-reg-state-def get-asset-state-def
    split: option.splits)
  show ?thesis
proof (cases reg-transition s0 a)
  case None
  then show ?thesis using assms by (simp add: deg-step-def)
next
case (Some s1)
have deg-step k (a, aid) gs = Some (broadcast-le k aid s1 gs)
  using assms Some by (simp add: deg-step-def)
then show ?thesis
  using deg-run-on-broadcast[OF ex, of k w s1] Some by simp
qed
qed

```

The degree- k step keeps a state inside the degree- k carrier: a broadcast

rewrites regulatory values but never changes which chains hold the asset, so the support bound $c \leq k$ is preserved.

lemma *deg-step-carrier*:

assumes $gs \in \text{deg-carrier } k$ **and** $\text{deg-step } k \ (a, \text{aid}) \ gs = \text{Some } gs'$
shows $gs' \in \text{deg-carrier } k$

proof –

from *assms(2)* **obtain** $s \ s'$ **where**

$\text{get-reg-state } gs \ 0 \ \text{aid} = \text{Some } s$ **and** $\text{reg-transition } s \ a = \text{Some } s'$

and $gs': gs' = \text{broadcast-le } k \ \text{aid } s' \ gs$

by (*auto simp: deg-step-def split: option.splits*)

have $\bigwedge c \ \text{aid}''. \text{asset-exists } gs' \ c \ \text{aid}'' = \text{asset-exists } gs \ c \ \text{aid}''$

unfolding gs' **by** (*simp add: broadcast-le-exists*)

with *assms(1)* **show** *?thesis* **by** (*auto simp: deg-carrier-def*)

qed

Functor laws on action words. The identity word acts as the identity, and a concatenation of words is the Kleisli composition of the two segment maps: *deg-run* is a functor from the free monoid of regulatory action sequences into the Kleisli category of the option monad on *global-state*. These two laws are the categorical skeleton; the semantic content that the functor is non-degenerate — that its action on a word is the regulatory composite broadcast once, not an arbitrary fold — is *deg-run-collapse*.

lemma *deg-run-Nil*: $\text{deg-run } k \ \text{aid } [] = \text{Some}$

by (*simp add: fun-eq-iff*)

lemma *deg-run-append*:

$\text{deg-run } k \ \text{aid } (v @ w) \ gs = \text{Option.bind } (\text{deg-run } k \ \text{aid } v \ gs) \ (\text{deg-run } k \ \text{aid } w)$

by (*induction v arbitrary: gs*) (*auto split: option.splits*)

Non-degeneracy witness. A genuine two-action regulatory word — *FREEZE* then *CONFISCATE* — composes to *CONFISCATED* through the intermediate *FROZEN*, and at the degree level its second synchronization overwrites the first: the word $[\text{FREEZE}, \text{CONFISCATE}]$ broadcasts *CONFISCATED* while its prefix $[\text{FREEZE}]$ broadcasts *FROZEN*. The two differ, so the composition is not idempotent padding; the functor transports real regulatory content.

lemma *deg-run-nondegenerate*:

assumes $\text{get-reg-state } gs \ 0 \ 0 = \text{Some } \text{ACTIVE}$

shows $\text{deg-run } k \ 0 \ [\text{FREEZE}, \text{CONFISCATE}] \ gs = \text{Some } (\text{broadcast-le } k \ 0 \ \text{CONFISCATED } gs)$

and $\text{deg-run } k \ 0 \ [\text{FREEZE}] \ gs = \text{Some } (\text{broadcast-le } k \ 0 \ \text{FROZEN } gs)$

and $\text{broadcast-le } k \ 0 \ \text{CONFISCATED } gs \neq \text{broadcast-le } k \ 0 \ \text{FROZEN } gs$

proof –

have $f1: \text{reg-transition } \text{ACTIVE } \text{FREEZE} = \text{Some } \text{FROZEN}$ **by** *simp*

have $f2: \text{reg-transition } \text{FROZEN } \text{CONFISCATE} = \text{Some } \text{CONFISCATED}$ **by** *simp*

have $rc: \text{deg-run } \text{ACTIVE } [\text{FREEZE}, \text{CONFISCATE}] = \text{Some } \text{CONFISCATED}$
using $f1 \ f2$ **by** *simp*

```

have rf: reg-run ACTIVE [FREEZE] = Some FROZEN using f1 by simp
show deg-run k 0 [FREEZE, CONFISCATE] gs = Some (broadcast-le k 0 CONFISCATED gs)
using deg-run-collapse[OF assms, of k FREEZE [CONFISCATE]] rc by simp
show deg-run k 0 [FREEZE] gs = Some (broadcast-le k 0 FROZEN gs)
using deg-run-collapse[OF assms, of k FREEZE []] rf by simp
have c1: get-reg-state (broadcast-le k 0 CONFISCATED gs) 0 0 = Some CONFISCATED
using assms by (simp add: broadcast-le-reg-in)
have c2: get-reg-state (broadcast-le k 0 FROZEN gs) 0 0 = Some FROZEN
using assms by (simp add: broadcast-le-reg-in)
show broadcast-le k 0 CONFISCATED gs ≠ broadcast-le k 0 FROZEN gs
proof
  assume broadcast-le k 0 CONFISCATED gs = broadcast-le k 0 FROZEN gs
  then have get-reg-state (broadcast-le k 0 CONFISCATED gs) 0 0
    = get-reg-state (broadcast-le k 0 FROZEN gs) 0 0 by simp
  with c1 c2 show False by simp
qed
qed

```

The single-step naturality square of *degree-natural-transformation*, re-exposed at the level of *deg-step* for use inside the word induction below.

lemma *deg-step-forget-some*:

```

assumes gs ∈ deg-carrier (Suc k) and deg-step (Suc k) opn gs = Some gs'
shows deg-step k opn (degree-forget (Suc k) gs) = Some (degree-forget (Suc k) gs')
proof –
  obtain a b where opn: opn = (a, b) by (cases opn)
  show ?thesis
    using degree-natural-transformation[of k] assms
    unfolding opn natural-transformation-def F-def by auto
qed

```

lemma *deg-step-forget-none*:

```

assumes gs ∈ deg-carrier (Suc k) and deg-step (Suc k) opn gs = None
shows deg-step k opn (degree-forget (Suc k) gs) = None
proof –
  obtain a b where opn: opn = (a, b) by (cases opn)
  show ?thesis
    using degree-natural-transformation[of k] assms
    unfolding opn natural-transformation-def F-def by auto
qed

```

Naturality on the whole action category. The single-step square above lifts, by induction on the action word, to the statement that degree demotion commutes with running an *entire* word: forgetting the top chain and then running at degree k equals running at degree $\text{Suc } k$ and then forgetting. Together with the functor laws *deg-run-Nil* and *deg-run-append*, this promotes *degree-natural-transformation* from a square on generators to a genuine nat-

ural transformation between the degree functors $F (Suc\ k)$ and $F\ k$ on the free action category — the functor-tower structure the entry claims.

theorem *degree-forget-natural-run:*

assumes $gs \in \text{deg-carrier}\ (Suc\ k)$

shows $\text{deg-run}\ k\ aid\ w\ (\text{degree-forget}\ (Suc\ k)\ gs)$

$= \text{map-option}\ (\text{degree-forget}\ (Suc\ k))\ (\text{deg-run}\ (Suc\ k)\ aid\ w\ gs)$

using *assms*

proof (*induction w arbitrary: gs*)

case *Nil*

show *?case* **by** *simp*

next

case (*Cons a w*)

show *?case*

proof (*cases deg-step (Suc k) (a, aid) gs*)

case *None*

then have $\text{deg-step}\ k\ (a, aid)\ (\text{degree-forget}\ (Suc\ k)\ gs) = \text{None}$

using *deg-step-forget-none[OF Cons.prem] by simp*

then show *?thesis* **using** *None* **by** *simp*

next

case (*Some gs'*)

have $\text{deg-step}\ k\ (a, aid)\ (\text{degree-forget}\ (Suc\ k)\ gs) = \text{Some}\ (\text{degree-forget}\ (Suc\ k)\ gs')$

using *deg-step-forget-some[OF Cons.prem Some]* .

moreover have $gs' \in \text{deg-carrier}\ (Suc\ k)$

using *deg-step-carrier[OF Cons.prem Some]* .

ultimately show *?thesis* **using** *Cons.IH Some* **by** *simp*

qed

qed

52 Degree Classes and Hierarchy Monotonicity

Each asset carries a required coupling strength, its *degree*, ranging over the four classes $S_0..S_3$. An asset belongs to the degree- k class when its required strength is at least k ; since a requirement of at least k' entails a requirement of at least any $k \leq k'$, the classes are nested (*reduction-containment*). A system, in turn, has a capability degree; it is the comparison of the asset's required degree with the system's capability degree that decides over- and under-provisioning.

The assignment of degrees to assets is operational data, not structure: the whole degree-class layer below is therefore parametric in an arbitrary assignment $\text{asset-degree} :: \text{asset-id} \Rightarrow \text{nat}$, fixed in an anonymous context. Every theorem of the layer holds for every assignment; a concrete example assignment witnesses non-vacuity at the end of the section.

Processing an asset at system degree d reconciles it across the chains the system couples (those with index $\leq d$): it broadcasts the hub chain's regulatory value over that reach. Reconciling to the hub value is the state-

level action of a degree- d synchronization. The processing machinery is independent of the degree assignment, so it lives outside the parametric context.

definition *process-at-degree* :: *nat* $\Rightarrow$ *global-state* $\Rightarrow$ *asset-id* $\Rightarrow$ *global-state* **where**
process-at-degree *d* *gs* *aid* =
 (case *get-reg-state* *gs* 0 *aid* of
 None $\Rightarrow$ *gs*
 | Some *v* $\Rightarrow$ *broadcast-le* *d* *aid* *v* *gs*)

On a consistent state every chain holding the asset already agrees with the hub, so broadcasting the hub value over any reach leaves the state consistent.

lemma *broadcast-le-consistent*:

assumes *cons*: *consistent-state* *gs* **and** *hub*: *get-reg-state* *gs* 0 *aid* = *Some* *v*

shows *consistent-state* (*broadcast-le* *d* *aid* *v* *gs*)

unfolding *consistent-state-def*

proof (*intro allI impI*)

fix *c1* *c2* *aid''* *s1* *s2*

assume *g1*: *get-reg-state* (*broadcast-le* *d* *aid* *v* *gs*) *c1* *aid''* = *Some* *s1*

and *g2*: *get-reg-state* (*broadcast-le* *d* *aid* *v* *gs*) *c2* *aid''* = *Some* *s2*

have *val*: $\bigwedge c$ *s*. *get-reg-state* (*broadcast-le* *d* *aid* *v* *gs*) *c* *aid''* = *Some* *s* $\implies$ *aid''* = *aid* $\implies$ *s* = *v*

proof -

fix *c* *s*

assume *gc*: *get-reg-state* (*broadcast-le* *d* *aid* *v* *gs*) *c* *aid''* = *Some* *s* **and** *eqaid*: *aid''* = *aid*

show *s* = *v*

proof (*cases* *c* $\leq$ *d*)

case *True*

with *gc* *eqaid* **have** (case *get-reg-state* *gs* *c* *aid* of *None* $\Rightarrow$ *None* | *Some* - $\Rightarrow$ *Some* *v*) = *Some* *s*

by (*simp* *add*: *broadcast-le-reg-in*)

then show *s* = *v* **by** (*auto* *split*: *option.splits*)

next

case *False*

with *gc* *eqaid* **have** *get-reg-state* *gs* *c* *aid* = *Some* *s* **by** (*simp* *add*: *broadcast-le-reg-out*)

with *cons* *hub* **show** *s* = *v* **unfolding** *consistent-state-def* **by** *blast*

qed

qed

show *s1* = *s2*

proof (*cases* *aid''* = *aid*)

case *True*

from *val*[*OF* *g1* *True*] *val*[*OF* *g2* *True*] **show** *?thesis* **by** *simp*

next

case *False*

then have *get-reg-state* *gs* *c1* *aid''* = *Some* *s1* **and** *get-reg-state* *gs* *c2* *aid''* = *Some* *s2*

using *g1* *g2* **by** (*simp-all* *add*: *broadcast-le-reg-other-asset*)

```

    then show ?thesis using cons unfolding consistent-state-def by blast
qed
qed

```

Validity preservation under processing holds with no reference to degrees at all: on a consistent state the broadcast value agrees with every chain it overwrites, and the broadcast touches no lock. Isolating this as its own lemma records that the degree hypothesis of the over-provisioning headline below is not load-bearing for validity — the regime-sensitive content, where over- and under-provisioning genuinely diverge, is the pair *over-provisioning-guarantees / no-downward-safety*.

```

lemma processing-preserves-validity:
  assumes valid-state gs
  shows valid-state (process-at-degree d gs aid)
proof (cases get-reg-state gs 0 aid)
  case None
  then show ?thesis using assms by (simp add: process-at-degree-def)
next
  case (Some v)
  from assms have c: consistent-state gs and nl: no-locked-without-reason gs
  by (simp-all add: valid-state-def)
  have consistent-state (broadcast-le d aid v gs)
  using broadcast-le-consistent[OF c Some] .
  moreover have no-locked-without-reason (broadcast-le d aid v gs)
  using nl by (simp add: no-locked-without-reason-def is-locked-def)
  ultimately show ?thesis using Some by (simp add: process-at-degree-def
valid-state-def)
qed

```

The timestamp order `lamport_hb` underlying the degree boundary is likewise independent of the degree assignment; it is a plain strict order on integer timestamps, not a distributed causal order.

```

definition lamport-hb :: timestamp  $\Rightarrow$  timestamp  $\Rightarrow$  bool where
  lamport-hb t1 t2  $\longleftrightarrow$  t1 < t2

```

The degree-class layer, parametric in the degree assignment. Everything proved inside this context holds for an *arbitrary* assignment *asset-degree*; on export each constant and theorem generalises over the assignment.

```

context
  fixes asset-degree :: asset-id  $\Rightarrow$  nat
begin

```

```

definition degree-capable :: nat  $\Rightarrow$  asset-id  $\Rightarrow$  bool where
  degree-capable k aid  $\longleftrightarrow$  k  $\leq$  asset-degree aid

```

Reduction containment: the degree classes form a descending chain, so membership in a higher class entails membership in every lower one. This is the partial-order closure of the hierarchy.

theorem *reduction-containment*:

$k \leq k' \implies \text{degree-capable } k' \text{ aid} \longrightarrow \text{degree-capable } k \text{ aid}$

unfolding *degree-capable-def* **by** *simp*

Validity preservation in the over-provisioning regime (auxiliary). A system preserves global validity when it processes an asset. This is a *degree-free* fact: its proof is exactly *processing-preserves-validity*, and it carries no provisioning hypothesis. It is therefore recorded as an auxiliary lemma aliasing that fact, *not* the headline monotonicity result. The genuinely degree-sensitive, one-directional monotonicity is carried by *over-provisioning-guarantees* together with *no-downward-safety* below, where the provisioning hypothesis is load-bearing.

lemma *hierarchy-monotonicity*:

assumes *valid-state gs*

shows *valid-state (process-at-degree system-degree gs aid)*

using *assms* **by** (*rule processing-preserves-validity*)

The coupling guarantee a degree- d system actually delivers: after processing, all of the asset's *required* chains — those within its degree class, $\text{index} \leq \text{asset-degree aid}$ — that hold the asset agree on its regulatory state.

definition *guarantees-preservation* :: $\text{nat} \Rightarrow \text{global-state} \Rightarrow \text{asset-id} \Rightarrow \text{bool}$ **where**
guarantees-preservation d gs aid $\longleftrightarrow$

$(\forall c1\ c2. c1 \leq \text{asset-degree aid} \longrightarrow c2 \leq \text{asset-degree aid}$
 $\longrightarrow \text{get-reg-state (process-at-degree d gs aid) } c1 \text{ aid} \neq \text{None}$
 $\longrightarrow \text{get-reg-state (process-at-degree d gs aid) } c2 \text{ aid} \neq \text{None}$
 $\longrightarrow \text{get-reg-state (process-at-degree d gs aid) } c1 \text{ aid}$
 $= \text{get-reg-state (process-at-degree d gs aid) } c2 \text{ aid})$

Under over-provisioning the guarantee is met: every required chain lies within the system's coupling reach ($\leq \text{asset-degree aid} \leq d$), so all are driven to the single hub value.

theorem *over-provisioning-guarantees*:

assumes *deg: asset-degree aid $\leq d$ and val: valid-state gs*

shows *guarantees-preservation d gs aid*

proof (*cases get-reg-state gs 0 aid*)

case *None*

then have *proc: process-at-degree d gs aid = gs* **by** (*simp add: process-at-degree-def*)

show *?thesis*

unfolding *guarantees-preservation-def proc*

proof (*intro allI impI*)

fix *c1 c2*

assume *c1 $\leq$ asset-degree aid and c2 $\leq$ asset-degree aid*

and *get-reg-state gs c1 aid $\neq$ None and get-reg-state gs c2 aid $\neq$ None*

then obtain *s1 s2* **where**

s1: get-reg-state gs c1 aid = Some s1 and s2: get-reg-state gs c2 aid = Some

s2

by *auto*

```

    have s1 = s2 using val s1 s2 unfolding valid-state-def consistent-state-def
  by blast
    with s1 s2 show get-reg-state gs c1 aid = get-reg-state gs c2 aid by simp
  qed
next
  case (Some v)
  then have proc: process-at-degree d gs aid = broadcast-le d aid v gs
    by (simp add: process-at-degree-def)
  show ?thesis
    unfolding guarantees-preservation-def
  proof (intro allI impI)
    fix c1 c2
    assume c1d: c1 ≤ asset-degree aid and c2d: c2 ≤ asset-degree aid
      and d1: get-reg-state (process-at-degree d gs aid) c1 aid ≠ None
      and d2: get-reg-state (process-at-degree d gs aid) c2 aid ≠ None
    from c1d deg have l1: c1 ≤ d by simp
    from c2d deg have l2: c2 ≤ d by simp
    have get-reg-state (process-at-degree d gs aid) c1 aid = Some v
      using l1 d1 unfolding proc by (auto simp: broadcast-le-reg-in split: option.splits)
    moreover have get-reg-state (process-at-degree d gs aid) c2 aid = Some v
      using l2 d2 unfolding proc by (auto simp: broadcast-le-reg-in split: option.splits)
    ultimately show get-reg-state (process-at-degree d gs aid) c1 aid
      = get-reg-state (process-at-degree d gs aid) c2 aid by simp
  qed
qed

```

The degree hypothesis is load-bearing in its own right: from an *arbitrary* hub-defined state — with no validity assumption, so the chains may well disagree beforehand — over-provisioning alone drives every required chain to the hub value. Without it, *no-downward-safety* below exhibits the failure on exactly such a disagreeing state, so the contrast pair is matched: both directions speak about arbitrary states.

theorem *over-provisioning-reconciles*:

```

  assumes deg: asset-degree aid ≤ d
    and hub: get-reg-state gs 0 aid = Some v
  shows guarantees-preservation d gs aid

```

proof —

```

  have proc: process-at-degree d gs aid = broadcast-le d aid v gs
    using hub by (simp add: process-at-degree-def)
  show ?thesis
    unfolding guarantees-preservation-def
  proof (intro allI impI)
    fix c1 c2
    assume c1d: c1 ≤ asset-degree aid and c2d: c2 ≤ asset-degree aid
      and d1: get-reg-state (process-at-degree d gs aid) c1 aid ≠ None
      and d2: get-reg-state (process-at-degree d gs aid) c2 aid ≠ None
    from c1d deg have l1: c1 ≤ d by simp

```

```

from  $c2d$  deg have  $l2: c2 \leq d$  by simp
have get-reg-state (process-at-degree  $d$   $gs$   $aid$ )  $c1$   $aid = \text{Some } v$ 
  using  $l1$   $d1$  unfolding proc by (auto simp: broadcast-le-reg-in split: option.splits)
moreover have get-reg-state (process-at-degree  $d$   $gs$   $aid$ )  $c2$   $aid = \text{Some } v$ 
  using  $l2$   $d2$  unfolding proc by (auto simp: broadcast-le-reg-in split: option.splits)
ultimately show get-reg-state (process-at-degree  $d$   $gs$   $aid$ )  $c1$   $aid$ 
   $= \text{get-reg-state} (\text{process-at-degree } d \text{ } gs \text{ } aid) \text{ } c2 \text{ } aid$  by simp

qed
qed

```

No downward safety. Under-provisioning carries no preservation guarantee: if the asset's required degree exceeds the system's capability, then some state defeats every preservation claim. The witness places the hub at *ACTIVE* and a required chain just beyond the system's reach (index *Suc system-degree*) at *FROZEN*; processing reconciles only the chains within reach, leaving the out-of-reach required chain in disagreement. This is the formal counterpart of over-provisioning safety: the monotonicity is genuinely one-directional.

theorem *no-downward-safety:*

assumes *asset-degree* $aid > \text{system-degree}$

shows $\neg (\forall gs. \text{guarantees-preservation } \text{system-degree } gs \text{ } aid)$

proof –

define *ast0* **where** $ast0 = \langle \langle \text{as-reg-state} = \text{ACTIVE} \rangle \rangle$

define *ast1* **where** $ast1 = \langle \langle \text{as-reg-state} = \text{FROZEN} \rangle \rangle$

define *gs0* **where** $gs0 =$

$\langle \langle \text{gs-chains} = (\lambda c. \text{if } c = 0 \text{ then } (\lambda a. \text{if } a = aid \text{ then } \text{Some } ast0 \text{ else } \text{None})$

$\text{else if } c = \text{Suc system-degree} \text{ then } (\lambda a. \text{if } a = aid \text{ then } \text{Some}$

$ast1 \text{ else } \text{None})$

$\text{else } (\lambda a. \text{None})) \rangle \rangle,$

$gs\text{-locks} = (\lambda a. \text{False}) \rangle \rangle$

have $c0: \text{get-reg-state } gs0 \text{ } 0 \text{ } aid = \text{Some } \text{ACTIVE}$

by (*simp add: gs0-def get-reg-state-def get-asset-state-def ast0-def*)

have $c1: \text{get-reg-state } gs0 \text{ } (\text{Suc system-degree}) \text{ } aid = \text{Some } \text{FROZEN}$

by (*simp add: gs0-def get-reg-state-def get-asset-state-def ast1-def*)

have *proc: process-at-degree system-degree gs0 aid = broadcast-le system-degree aid ACTIVE gs0*

using $c0$ **by** (*simp add: process-at-degree-def*)

have $p0: \text{get-reg-state} (\text{process-at-degree system-degree } gs0 \text{ } aid) \text{ } 0 \text{ } aid = \text{Some } \text{ACTIVE}$

using *proc* $c0$ **by** (*simp add: broadcast-le-reg-in*)

have $p1: \text{get-reg-state} (\text{process-at-degree system-degree } gs0 \text{ } aid) \text{ } (\text{Suc system-degree}) \text{ } aid = \text{Some } \text{FROZEN}$

using *proc* $c1$ **by** (*simp add: broadcast-le-reg-out*)

have $le0: (0::nat) \leq \text{asset-degree } aid$ **by** *simp*

have $le1: \text{Suc system-degree} \leq \text{asset-degree } aid$ **using** *assms* **by** *simp*

have $\neg \text{guarantees-preservation system-degree } gs0 \text{ } aid$

proof

```

assume guarantees-preservation system-degree gs0 aid
then have get-reg-state (process-at-degree system-degree gs0 aid) 0 aid
           = get-reg-state (process-at-degree system-degree gs0 aid) (Suc sys-
tem-degree) aid
unfolding guarantees-preservation-def using le0 le1 p0 p1 by blast
with p0 p1 show False by simp
qed
then show ?thesis by blast
qed

```

The timestamp-order degree boundary. The classes S_1 and S_2 are separated by a degree-2 threshold. The legacy identifiers *requires-causal* and *causal-consistent-at* label that threshold, but the formal relation is only the strict order on integer timestamps, not a distributed happened-before relation. We show the boundary is well-defined on $(asset, system)$ pairs: the requirement is a single-valued function of the asset's degree, it is met whenever the system over-provisions the asset, and the underlying happened-before relation is a strict partial order (irreflexive and asymmetric; in fact linear, as timestamps are linearly ordered), so the timestamp-order boundary is well-formed.

The degree index corresponds to the product's synchronization-degree hierarchy $S_0..S_3$: $F\ k$ formalizes the *coupling breadth* of degree k — the containment and monotonicity structure of the ladder — while the per-degree synchronization semantics itself (static, unidirectional observation, bidirectional coupling, atomic state binding) is abstracted. The timestamp-order boundary sits at $k = 2$, the S1/S2 transition below.

definition *requires-causal* :: *asset-id* $\Rightarrow$ *bool* **where**
requires-causal aid $\longleftrightarrow 2 \leq asset-degree\ aid$

definition *causal-consistent-at* :: *asset-id* $\Rightarrow$ *nat* $\Rightarrow$ *bool* **where**
causal-consistent-at aid d $\longleftrightarrow (requires-causal\ aid \longrightarrow 2 \leq d)$

theorem *boundary-well-defined*:

```

(causal-consistent-at aid d  $\longleftrightarrow (2 \leq asset-degree\ aid \longrightarrow 2 \leq d)$ )
 $\wedge$  (asset-degree aid  $\leq d \longrightarrow$  causal-consistent-at aid d)
 $\wedge$  ( $\forall t1\ t2. lamport-hb\ t1\ t2 \longrightarrow \neg lamport-hb\ t2\ t1$ )
 $\wedge$  ( $\forall t. \neg lamport-hb\ t\ t$ )
 $\wedge$  ( $\forall t1\ t2\ t3. lamport-hb\ t1\ t2 \longrightarrow lamport-hb\ t2\ t3 \longrightarrow lamport-hb\ t1\ t3$ )
by (auto simp: causal-consistent-at-def requires-causal-def lamport-hb-def)

```

end

Non-vacuity of the parametric layer: a concrete example assignment, cycling the four degree classes over asset identifiers. The over-provisioning guarantee instantiates to it directly.

definition *example-degree* :: *asset-id* $\Rightarrow$ *nat* **where**
example-degree aid = *aid mod 4*

corollary *example-degree-over-provisioning:*

example-degree $aid \leq d \implies \text{valid-state } gs$

$\implies \text{guarantees-preservation } \text{example-degree } d \text{ } gs \text{ } aid$

by (*rule over-provisioning-guarantees*)

Static promotion safety. A promotion re-assigns an asset’s required degree; statically — between synchronizations — the change is just a different assignment parameter, so as long as the promoted degree stays within the system’s capability, the over-provisioning guarantee transfers verbatim to the new assignment. In-flight promotion (a re-assignment crossing a live synchronization) is out of scope for this entry; subsequent work on retry-queue semantics covers it.

corollary *static-promotion-safety:*

assumes *asset-degree'* $aid \leq \text{system-degree}$ **and** *valid-state* gs

shows *guarantees-preservation* *asset-degree'* *system-degree* $gs \text{ } aid$

using *assms* **by** (*rule over-provisioning-guarantees*)

end

theory *External-Instance*

imports *Proof-Automation*

begin

53 A TCP-Inspired Endpoint State Machine

A toy single-endpoint lifecycle, restricted to six states and five packet events and using names inspired by RFC 793. It is not an RFC 793 subset or a trace-conformant TCP model: in particular, *FIN*–*WAIT* followed by *ACK* goes directly to *CLOSED*, deliberately collapsing the peer-FIN and TIME-WAIT stages. The purpose is only to instantiate the generic locales outside the regulatory vocabulary. A *RST* aborts any modeled connection in flight.

Two modelling notes. First, *CLOSED* is terminal: a re-connection is a new connection identity (a new tracker entry), not a transition of the old one. Second, a *RST* in *listen* is ignored by RFC 793; the partial transition function returns *None* there, as for every other state/event pair the subset does not admit.

datatype *tcp-state* =

LISTEN | *SYN-SENT* | *SYN-RCVD* | *ESTABLISHED* | *FIN-WAIT* | *CLOSED*

datatype *tcp-event* = *EvSyn* | *EvSynAck* | *EvAck* | *EvFin* | *EvRst*

fun *tcp-transition* :: *tcp-state* $\Rightarrow$ *tcp-event* $\Rightarrow$ *tcp-state option* **where**

tcp-transition *LISTEN* *EvSyn* = *Some SYN-RCVD*

| *tcp-transition* *SYN-SENT* *EvSynAck* = *Some ESTABLISHED*

```

| tcp-transition SYN-RCVD  EvAck  = Some ESTABLISHED
| tcp-transition ESTABLISHED EvFin  = Some FIN-WAIT
| tcp-transition FIN-WAIT  EvAck  = Some CLOSED
| tcp-transition SYN-SENT  EvRst  = Some CLOSED
| tcp-transition SYN-RCVD  EvRst  = Some CLOSED
| tcp-transition ESTABLISHED EvRst  = Some CLOSED
| tcp-transition FIN-WAIT  EvRst  = Some CLOSED
| tcp-transition -        -      = None

```

definition *tcp-states* :: *tcp-state set* **where**

tcp-states = {*LISTEN*, *SYN-SENT*, *SYN-RCVD*, *ESTABLISHED*, *FIN-WAIT*, *CLOSED*}

definition *tcp-events* :: *tcp-event set* **where**

tcp-events = {*EvSyn*, *EvSynAck*, *EvAck*, *EvFin*, *EvRst*}

definition *tcp-terminal* :: *tcp-state set* **where**

tcp-terminal = {*CLOSED*}

lemma *tcp-states-UNIV*: *tcp-states* = *UNIV*

unfolding *tcp-states-def* **by** (*auto*, *case-tac x*, *auto*)

lemma *finite-tcp-states*: *finite tcp-states*

unfolding *tcp-states-def* **by** *simp*

lemma *finite-tcp-state-UNIV*: *finite (UNIV :: tcp-state set)*

using *finite-tcp-states* **by** (*simp add: tcp-states-UNIV*)

lemma *closed-terminal*: *tcp-transition CLOSED e = None*

by (*cases e*) *simp-all*

54 The Connection-Tracker Representation

The tracker entry is a structured record: a phase tag together with the peer-endpoint metadata the tracker keeps for a connection in flight. The well-formedness invariant ties the auxiliary field to the tag — the peer is recorded exactly while a connection attempt or connection exists — mirroring the layer-crossing pattern of the regulatory development. The peer identifier value itself is a representative placeholder, exactly as in the DAML permission record; what the invariant tracks is its presence. The tracker’s state space is the image of the representation map, and its transition function is the lift of the endpoint transition through that map, guarded by the image domain.

datatype *ct-phase* =

CtNew | *CtSynSent* | *CtSynRecv* | *CtEstablished* | *CtFinWait* | *CtClosed*

record *ct-entry* =

ct-state :: *ct-phase*
ct-peer :: *nat option*

definition *default-peer* :: *nat* **where**
default-peer = 0

fun *tcp-to-ct* :: *tcp-state* $\Rightarrow$ *ct-entry* **where**
tcp-to-ct *LISTEN* = ($\lfloor$ *ct-state* = *CtNew*, *ct-peer* = *None* $\rfloor$)
 $\mid$ *tcp-to-ct* *SYN-SENT* = ($\lfloor$ *ct-state* = *CtSynSent*, *ct-peer* = *Some default-peer*
 $\rfloor$)
 $\mid$ *tcp-to-ct* *SYN-RCVD* = ($\lfloor$ *ct-state* = *CtSynRecv*, *ct-peer* = *Some default-peer*
 $\rfloor$)
 $\mid$ *tcp-to-ct* *ESTABLISHED* = ($\lfloor$ *ct-state* = *CtEstablished*, *ct-peer* = *Some default-peer*
 $\rfloor$)
 $\mid$ *tcp-to-ct* *FIN-WAIT* = ($\lfloor$ *ct-state* = *CtFinWait*, *ct-peer* = *Some default-peer*
 $\rfloor$)
 $\mid$ *tcp-to-ct* *CLOSED* = ($\lfloor$ *ct-state* = *CtClosed*, *ct-peer* = *None* $\rfloor$)

fun *ct-to-tcp* :: *ct-entry* $\Rightarrow$ *tcp-state* **where**
ct-to-tcp *p* = (case *ct-state p* of
CtNew $\Rightarrow$ *LISTEN*
 $\mid$ *CtSynSent* $\Rightarrow$ *SYN-SENT*
 $\mid$ *CtSynRecv* $\Rightarrow$ *SYN-RCVD*
 $\mid$ *CtEstablished* $\Rightarrow$ *ESTABLISHED*
 $\mid$ *CtFinWait* $\Rightarrow$ *FIN-WAIT*
 $\mid$ *CtClosed* $\Rightarrow$ *CLOSED*)

definition *valid-ct-entry* :: *ct-entry* $\Rightarrow$ *bool* **where**
valid-ct-entry *p* $\longleftrightarrow$ (*ct-state p* $\in$ {*CtNew*, *CtClosed*}) = (*ct-peer p* = *None*)

lemma *tcp-to-ct-valid*: *valid-ct-entry* (*tcp-to-ct s*)
by (cases *s*) (auto simp: *valid-ct-entry-def*)

lemma *ct-to-tcp-to-ct-id*: *ct-to-tcp* (*tcp-to-ct s*) = *s*
by (cases *s*) *simp-all*

definition *ct-states* :: *ct-entry set* **where**
ct-states = range *tcp-to-ct*

definition *ct-terminal* :: *ct-entry set* **where**
ct-terminal = {*tcp-to-ct CLOSED*}

55 The Tracked Event Subset and the Tracker Transition

The abstract tracker observes the modeled handshake and teardown events but excludes *RST* by design. This is an action-scoping choice, not a claim about a concrete operating-system conntrack implementation. The source

action set of the preservation morphism is therefore the strict subset of tracked events, and the tracker has its own event alphabet, mapped one-to-one from that subset.

datatype *ct-event* = *CT-SYN* | *CT-SYNACK* | *CT-ACK* | *CT-FIN*

definition *tracked-events* :: *tcp-event* set **where**

tracked-events = {*EvSyn*, *EvSynAck*, *EvAck*, *EvFin*}

definition *ct-events* :: *ct-event* set **where**

ct-events = {*CT-SYN*, *CT-SYNACK*, *CT-ACK*, *CT-FIN*}

fun *tcp-to-ct-event* :: *tcp-event* $\Rightarrow$ *ct-event* **where**

tcp-to-ct-event EvSyn = *CT-SYN*
| *tcp-to-ct-event EvSynAck* = *CT-SYNACK*
| *tcp-to-ct-event EvAck* = *CT-ACK*
| *tcp-to-ct-event EvFin* = *CT-FIN*
| *tcp-to-ct-event EvRst* = *CT-SYN*

— The last clause is unused by the morphism, since *actions_s* is the tracked subset, which excludes *EvRst*; it only makes the function total, exactly as in the escalation action map.

fun *ct-event-to-tcp* :: *ct-event* $\Rightarrow$ *tcp-event* **where**

ct-event-to-tcp CT-SYN = *EvSyn*
| *ct-event-to-tcp CT-SYNACK* = *EvSynAck*
| *ct-event-to-tcp CT-ACK* = *EvAck*
| *ct-event-to-tcp CT-FIN* = *EvFin*

definition *ct-transition* :: *ct-entry* $\Rightarrow$ *ct-event* $\Rightarrow$ *ct-entry* option **where**

ct-transition p e =
(if *p* $\in$ *ct-states*
then map-option *tcp-to-ct* (*tcp-transition* (*ct-to-tcp p*) (*ct-event-to-tcp e*))
else None)

lemma *ct-event-roundtrip*:

e $\in$ *tracked-events* $\implies$ *ct-event-to-tcp* (*tcp-to-ct-event e*) = *e*
by (*auto simp: tracked-events-def*)

56 Discharging the Instance with the Generic Methods

The helper lemmas mirror, point for point, those of the regulatory development: finiteness in the simplifier's normal form, terminal absorption, closure, domain, the naturality of the representation map on the tracked subset, and the terminal/well-definedness facts of the map. They are declared into the discharge collections, after which the machine and preservation instances close by one method invocation each.

lemma *ct-states-finite: finite ct-states*
by (*auto simp: ct-states-def finite-tcp-state-UNIV intro: finite-imageI*)

lemma *ct-terminal-subset: ct-terminal $\subseteq$ ct-states*
unfolding *ct-terminal-def ct-states-def* **by** *blast*

lemma *ct-closed-absorbing:*
 $\llbracket p \in \text{ct-terminal}; e \in \text{ct-events} \rrbracket \implies \text{ct-transition } p \ e = \text{None}$
by (*auto simp: ct-terminal-def ct-transition-def ct-to-tcp-to-ct-id closed-terminal*)

lemma *ct-transition-closed:*
 $\llbracket p \in \text{ct-states}; e \in \text{ct-events}; \text{ct-transition } p \ e = \text{Some } p' \rrbracket \implies p' \in \text{ct-states}$
unfolding *ct-transition-def ct-states-def*
by (*auto split: if_splits option.splits*)

lemma *ct-transition-outside-states:*
 $p \notin \text{ct-states} \implies \text{ct-transition } p \ e = \text{None}$
by (*simp add: ct-transition-def*)

lemma *tcp-to-ct-in-states: tcp-to-ct s $\in$ ct-states*
by (*simp add: ct-states-def*)

lemma *tcp-to-ct-terminal:*
 $s \in \text{tcp-terminal} \implies \text{tcp-to-ct } s \in \text{ct-terminal}$
by (*auto simp: tcp-terminal-def ct-terminal-def*)

lemma *tcp-to-ct-naturality-some:*
 $\llbracket s \in \text{tcp-states}; e \in \text{tracked-events}; \text{tcp-transition } s \ e = \text{Some } s' \rrbracket$
 $\implies \text{ct-transition } (\text{tcp-to-ct } s) \ (\text{tcp-to-ct-event } e) = \text{Some } (\text{tcp-to-ct } s')$
by (*simp add: ct-transition-def tcp-to-ct-in-states ct-to-tcp-to-ct-id*
ct-event-roundtrip
del: ct-to-tcp.simps tcp-to-ct.simps)

lemma *tcp-to-ct-naturality-none:*
 $\llbracket s \in \text{tcp-states}; e \in \text{tracked-events}; \text{tcp-transition } s \ e = \text{None} \rrbracket$
 $\implies \text{ct-transition } (\text{tcp-to-ct } s) \ (\text{tcp-to-ct-event } e) = \text{None}$
by (*simp add: ct-transition-def tcp-to-ct-in-states ct-to-tcp-to-ct-id*
ct-event-roundtrip
del: ct-to-tcp.simps tcp-to-ct.simps)

lemmas [*discharge-simps*] =
tcp-states-UNIV finite-tcp-state-UNIV
tcp-events-def tracked-events-def ct-events-def
tcp-terminal-def closed-terminal
ct-states-finite ct-terminal-subset
ct-closed-absorbing ct-transition-outside-states
tcp-to-ct-naturality-some tcp-to-ct-naturality-none
tcp-to-ct-terminal tcp-to-ct-in-states

lemmas [*discharge-intros*] =
ct-transition-closed

lemmas [*discharge-dels*] =
tcp-to-ct.simps ct-to-tcp.simps

The two state machines and the preservation morphism, each discharged by the corresponding generic method. No obligation is weakened: the statements are the full locale predicates.

theorem *tcp-state-machine*:
state-machine tcp-states tcp-events tcp-transition tcp-terminal
by *discharge-state-machine*

theorem *conntrack-state-machine*:
state-machine ct-states ct-events ct-transition ct-terminal
by *discharge-state-machine*

theorem *tcp-conntrack-preservation*:
state-preservation tcp-states tracked-events tcp-transition tcp-terminal
ct-states ct-events ct-transition ct-terminal
tcp-to-ct tcp-to-ct-event
by *discharge-preservation*

Registered form of the morphism, giving access to the locale's sequential payload: every tracked packet sequence accepted by the endpoint is mirrored, packet for packet, by the tracker, ending in the corresponding tracker entry.

interpretation *tcp-ct*:
state-preservation tcp-states tracked-events tcp-transition tcp-terminal
ct-states ct-events ct-transition ct-terminal
tcp-to-ct tcp-to-ct-event
by (*rule tcp-conntrack-preservation*)

corollary *tracked-sequence-mirrored*:
assumes *s* ∈ *tcp-states*
and $\forall e \in \text{set } es. e \in \text{tracked-events}$
and *tcp-ct.source.apply-actions* *s es* = *Some s'*
shows *tcp-ct.target.apply-actions* (*tcp-to-ct s*) (*map tcp-to-ct-event es*)
= *Some (tcp-to-ct s')*
using *assms* **by** (*rule tcp-ct.sequential-preservation*)

Non-vacuity witnesses: the three-way handshake is accepted by the endpoint machine, and its tracked image steps the tracker entry from the fresh entry to the established entry.

lemma *three-way-handshake-endpoint*:
tcp-transition LISTEN EvSyn = Some SYN-RCVD
tcp-transition SYN-RCVD EvAck = Some ESTABLISHED
by *simp-all*

lemma *three-way-handshake-tracked*:

```

    ct-transition (tcp-to-ct LISTEN) CT-SYN = Some (tcp-to-ct SYN-RCVD)
    ct-transition (tcp-to-ct SYN-RCVD) CT-ACK = Some (tcp-to-ct ESTAB-
LISHED)
  by (simp-all add: ct-transition-def tcp-to-ct-in-states ct-to-tcp-to-ct-id
    del: ct-to-tcp.simps tcp-to-ct.simps)

end

```

```

theory Canton-Bridge
imports
  Functor-Laws
  ADS-Functor.ADS-Construction
  ADS-Functor.Canton-Transaction-Tree
begin

```

57 Auxiliary list predicate for synchronization sequences

A blinding path: each element of the list is a blinding of its successor. As blindings narrow what is revealed, the last element is the most-revealed view of the sequence and every earlier element is a partial view of it.

definition *blinding-path* :: $('d \Rightarrow 'd \Rightarrow \text{bool}) \Rightarrow 'd \text{ list} \Rightarrow \text{bool}$ **where**
blinding-path bl xs $\longleftrightarrow (\forall i. \text{Suc } i < \text{length } xs \longrightarrow \text{bl } (xs ! i) (xs ! \text{Suc } i))$

58 Composite functor laws: cross-domain preservation after the ADS Merkle functor

Inside *authenticated-state* the ADS layer supplies a thin category: objects are authenticated values, and the unique morphism $a \rightarrow b$ is the blinding relation $bo\ a\ b$, a preorder (reflexive and transitive by the Merkle interface). The cross-domain layer supplies another thin category: objects are global states and the morphism $s \rightarrow s'$ is *state-refines* $s\ s'$, also a preorder. Because *extract-map* is partial, its extractable values and endpoint-extractable blinding morphisms form the relevant sub-preorder. On that domain the extraction map is the composite functor: it sends a value to the state it extracts to, and a blinding morphism to a refinement morphism. The next four results are the category axioms of this composite functor together with the associativity of the lifted merge (join) algebra.

```

context authenticated-state
begin

```

Identity is preserved: the identity (blinding) morphism on a , which exists by reflexivity of bo , is sent to the identity (refinement) morphism on the extracted state.

lemma *cdsp-ads-id*:

assumes *extract-map a = Some sa*
shows *bo a a ∧ valid-state sa ∧ state-refines sa sa*
using *mk.reflp extract-preserves-validity[OF assms] state-refines-refl*
by (*simp add: reflp-def*)

The functor's action on a single morphism: a blinding $a \rightarrow b$ is sent to a refinement $\text{extract } a \rightarrow \text{extract } b$, with both extracted states valid. This is *blinded-view-preserves-validity* with the validity hypothesis on b discharged internally from *extract-preserves-validity*, so the morphism map is total on the downward blinding closure of extractable endpoints.

lemma *cdsp-ads-morphism*:

assumes *bo: bo a b and eb: extract-map b = Some sb*
shows $\exists sa. \text{extract-map } a = \text{Some } sa \wedge \text{valid-state } sa \wedge \text{valid-state } sb \wedge \text{state-refines } sa \ sb$
proof –
have *hab: h a = h b using bo-hash-eq[OF bo]* .
obtain *sa where ea: extract-map a = Some sa and sr: state-refines sa sb*
using *extract-under-blinding[OF hab bo eb]* **by** *blast*
have *valid-state sa using extract-preserves-validity[OF ea]* .
moreover **have** *valid-state sb using extract-preserves-validity[OF eb]* .
ultimately **show** *?thesis using ea sr by blast*

qed

Composition is preserved (closed). Two composable blinding morphisms $a \rightarrow b$ and $b \rightarrow c$ compose in the source category to $a \rightarrow c$ (transitivity of *bo*), and the functor sends the composite to the composite of the images $\text{extract } a \rightarrow \text{extract } b \rightarrow \text{extract } c$ (transitivity of *state-refines*). This is the cross-domain, authenticated analogue of *preservation-compose*.

theorem *cdsp-ads-compose*:

assumes *ab: bo a b and bc: bo b c and ec: extract-map c = Some sc*
shows *bo a c*
 $\wedge (\exists sa \ sb. \text{extract-map } a = \text{Some } sa \wedge \text{extract-map } b = \text{Some } sb$
 $\wedge \text{valid-state } sa \wedge \text{valid-state } sb \wedge \text{valid-state } sc$
 $\wedge \text{state-refines } sa \ sb \wedge \text{state-refines } sb \ sc \wedge \text{state-refines } sa \ sc)$

proof –

have *boac: bo a c using mk.transp ab bc by (metis transpD)*
obtain *sb where eb: extract-map b = Some sb*
and *srbc: state-refines sb sc and vsb: valid-state sb and vsc: valid-state sc*
using *cdsp-ads-morphism[OF bc ec]* **by** *blast*
obtain *sa where ea: extract-map a = Some sa*
and *srab: state-refines sa sb and vsa: valid-state sa*
using *cdsp-ads-morphism[OF ab eb]* **by** *blast*
have *srac: state-refines sa sc using state-refines-trans[OF srab srbc]* .
show *?thesis using boac ea eb vsa vsb vsc srab srbc srac by blast*

qed

Associativity. Three composable blinding morphisms yield the same composite $a \rightarrow d$ regardless of the bracketing; in this thin category the

composite morphism is the relation $bo\ a\ d$ (and its image $state-refines\ sa\ sd$), which is bracketing-independent because the relations are propositional. The substantive associativity of the lifted merge appears in the merge-associativity result below.

theorem *cdsp-ads-assoc*:

assumes $ab: bo\ a\ b$ **and** $bc: bo\ b\ c$ **and** $cd: bo\ c\ d$
and $ed: extract-map\ d = Some\ sd$
shows $bo\ a\ d \wedge (\exists\ sa. extract-map\ a = Some\ sa \wedge valid-state\ sa \wedge state-refines\ sa\ sd)$
proof –
have $bo\ b\ d$ **using** $mk.transp\ bc\ cd$ **by** (*metis transpD*)
hence $boad: bo\ a\ d$ **using** $mk.transp\ ab$ **by** (*metis transpD*)
obtain sa **where** $extract-map\ a = Some\ sa \wedge valid-state\ sa \wedge state-refines\ sa\ sd$
using *cdsp-ads-morphism*[*OF boad ed*] **by** *blast*
thus *?thesis* **using** $boad$ **by** *blast*
qed

The functor’s action on the merge (join) product: a merge of two views is sent to a join of the two extracted states, with the joined state valid and refining each input. This lifts *extract-respects-merging* to a theorem carrying validity, and is exactly *authenticated-preservation-soundness* read as the product map of the composite functor.

lemma *cdsp-ads-merge*:

assumes $mab: m\ a\ b = Some\ ab$
and $ea: extract-map\ a = Some\ sa$ **and** $eb: extract-map\ b = Some\ sb$
shows $\exists\ sab. extract-map\ ab = Some\ sab \wedge valid-state\ sab$
 $\wedge state-join\ sa\ sb\ sab$
 $\wedge state-refines\ sa\ sab \wedge state-refines\ sb\ sab$
using *authenticated-preservation-soundness*[*OF mab ea eb*]
by *blast*

Associativity of the lifted merge/join algebra. The ADS merge is associative (Merkle-interface axiom *mk.assoc*), so the three-way combination of authenticated views is independent of the bracketing: both $(a \cdot b) \cdot c$ and $a \cdot (b \cdot c)$ land on the same merged value, hence (the extraction being a function) on the same joined state. This is the substantive “associativity preserved” of the composite functor. Commutativity and idempotence of the lifted combination descend directly from the interface laws (*mk.commute*, *mk.idem*) with the extraction a function; associativity is the one law that needs the rearrangement below, so it is the law recorded as a theorem.

theorem *cdsp-ads-merge-assoc*:

assumes $mab: m\ a\ b = Some\ ab$ **and** $mabc: m\ ab\ c = Some\ abc$
shows $\exists\ bc. m\ b\ c = Some\ bc \wedge m\ a\ bc = Some\ abc$
proof –
have $m\ b\ c \gg m\ a = m\ a\ b \gg m\ c$ **by** (*rule mk.assoc[symmetric]*)
also have $m\ a\ b \gg m\ c = m\ c\ ab$ **using** mab **by** *simp*
also have $m\ c\ ab = m\ ab\ c$ **by** (*rule mk.commute*)

also have $\dots = \text{Some } abc$ using *mabc* by *simp*
 finally have $m\ b\ c \gg m\ a = \text{Some } abc$.
 thus *?thesis* by (cases *m b c*) *auto*
 qed
 end

59 Authenticity along blinding paths and merge folds

Four authenticity properties are shown to hold not only across a single blinding or merge step but along a blinding path or merge fold:

1. **validity** — every view in the sequence extracts to a valid state, along a blinding path (theorem *sequence-authenticity-preservation*) and along a merge fold (theorem *sequence-merge-soundness*);
2. **need-to-know (refinement)** — every view is a partial view refining the most-revealed endpoint (theorem *sequence-authenticity-preservation*, extended to arbitrary blinding-step reachability by corollary *reachable-view-authenticity*); dually, every contributor to a merge fold refines the combined view (theorem *sequence-merge-soundness*);
3. **hash soundness** — each path or fold shares one authenticating root, in both the blinding direction (theorem *blinding-path-hash-soundness*) and the merge direction (theorem *merge-seq-hash*);
4. **inclusion integrity** — theorem *sequence-inclusion-integrity*.

Scope qualifier (read carefully). The inclusion result is *state-level*: it is about *revealed holdings* — a holding exhibited by any view in the sequence is genuinely present in the endpoint with the same regulatory state, and no view fabricates a chain the endpoint does not authenticate. This is *not* a claim about the *concrete Merkle inclusion path* (the witness list authenticating a leaf against a root hash); that construction lives in *ADS-Constructor.Inclusion-Proof-Construction* and belongs to the concrete recursive-tree layer, not to this abstract layer.

context *authenticated-state*
 begin

59.1 Need-to-know blinding along a sequence

Along a blinding path, an earlier (more blinded) element blinds any later element: the blinding relation is monotone along the path by transitivity.

lemma *blinding-path-mono*:

```

assumes path: blinding-path bo xs and ij:  $i \leq j$  and j:  $j < \text{length } xs$ 
shows bo (xs ! i) (xs ! j)
proof –
  from ij obtain d where jd:  $j = i + d$  using le-Suc-ex by blast
  from j jd show ?thesis unfolding jd
  proof (induction d arbitrary: j)
    case 0
    show ?case using mk.reflp by (simp add: reflp-def)
  next
    case (Suc d)
    have lt:  $i + d < \text{length } xs$  using Suc.prem by simp
    have ih: bo (xs ! i) (xs ! (i + d)) using Suc.IH lt by simp
    have step: bo (xs ! (i + d)) (xs ! Suc (i + d))
      using path Suc.prem unfolding blinding-path-def by simp
    show ?case using mk.transp ih step by (metis transpD add-Suc-right)
  qed
qed

```

Path-level need-to-know preservation. Given an authenticated-view list presented as a blinding path whose most-revealed endpoint (the last element) extracts to a valid cross-domain state, *every* intermediate view extracts to a valid state that refines the endpoint. Authenticity (validity together with the refinement guarantee of a partial view) is preserved along the whole path, not merely across one blinding step. This generalises *blinded-view-preserves-validity*.

theorem *sequence-authenticity-preservation*:

```

assumes path: blinding-path bo xs and ne:  $xs \neq []$ 
and elast: extract-map (last xs) = Some s-last
and i:  $i < \text{length } xs$ 
shows  $\exists s_i. \text{extract-map } (xs ! i) = \text{Some } s_i \wedge \text{valid-state } s_i \wedge \text{state-refines } s_i \text{ s-last}$ 
proof –
  have le:  $i \leq \text{length } xs - 1$  using i by simp
  have lt:  $\text{length } xs - 1 < \text{length } xs$  using ne by (cases xs) auto
  have bo (xs ! i) (xs ! (length xs - 1))
    using blinding-path-mono[OF path le lt] .
  also have  $xs ! (\text{length } xs - 1) = \text{last } xs$  using ne by (simp add: last-conv-nth)
  finally have bo (xs ! i) (last xs) .
  from cdsp-ads-morphism[OF this elast] show ?thesis by blast
qed

```

Hash soundness along the sequence. Every view in a blinding path commits to the same root hash as the most-revealed endpoint; a blinded view cannot present a different root. This is the sequence-level form of the ADS guarantee *bo-hash-eq*: along the blinding path there is a single authenticating root, so any inclusion claim made anywhere along the sequence is checked against the same commitment.

theorem *blinding-path-hash-soundness*:

```

assumes path: blinding-path bo xs and ne:  $xs \neq []$  and i:  $i < \text{length } xs$ 

```

```

shows  $h (xs ! i) = h (last\ xs)$ 
proof –
  have  $le: i \leq length\ xs - 1$  using  $i$  by  $simp$ 
  have  $lt: length\ xs - 1 < length\ xs$  using  $ne$  by  $(cases\ xs)\ auto$ 
  have  $bo\ (xs ! i)\ (xs ! (length\ xs - 1))$  using  $blinding-path-mono[OF\ path\ le\ lt]$  .
  moreover have  $xs ! (length\ xs - 1) = last\ xs$  using  $ne$  by  $(simp\ add: last-conv-nth)$ 
  ultimately show  $?thesis$  using  $bo-hash-eq$  by  $simp$ 
qed

```

Inclusion-proof integrity along the sequence. Every holding revealed by any view in the sequence is genuinely included in the most-revealed endpoint with the same regulatory state, and the revealing chain is connected in the endpoint. No view along the sequence can therefore exhibit a holding that the endpoint does not authenticate. Together with *blinding-path-hash-soundness* (one root for the whole sequence) this is the state-level reading of inclusion-proof integrity preserved across the authenticated-view path.

At this abstract layer inclusion reduces to the refinement guarantee (*sequence-authenticity-preservation*) read through *state-refines-def*: a partial view's revealed leaves are exactly the leaves it shares with the endpoint. The concrete Merkle inclusion-path machinery lives one level down in *ADS-Functor.Inclusion-Proof-Construction*.

theorem *sequence-inclusion-integrity*:

```

assumes  $path: blinding-path\ bo\ xs$  and  $ne: xs \neq []$ 
  and  $elast: extract-map\ (last\ xs) = Some\ s-last$ 
  and  $i: i < length\ xs$  and  $ei: extract-map\ (xs ! i) = Some\ s_i$ 
  and  $rev: get-reg-state\ s_i\ c\ aid = Some\ r$ 
shows  $get-reg-state\ s-last\ c\ aid = Some\ r \wedge c \in connected-chains\ s-last\ aid$ 
proof –
  obtain  $s'$  where  $es': extract-map\ (xs ! i) = Some\ s'$  and  $sr: state-refines\ s'\ s-last$ 
    using  $sequence-authenticity-preservation[OF\ path\ ne\ elast\ i]$  by  $blast$ 
  from  $es'\ ei$  have  $eq: s' = s_i$  by  $simp$ 
  have  $conn: c \in connected-chains\ s_i\ aid$ 
    using  $rev$  unfolding  $connected-chains-def\ asset-exists-def$ 
     $get-reg-state-def\ get-asset-state-def$  by  $(auto\ split: option.splits)$ 
  have  $c \in connected-chains\ s-last\ aid$ 
     $\wedge\ get-reg-state\ s_i\ c\ aid = get-reg-state\ s-last\ c\ aid$ 
    using  $sr\ eq\ conn$  unfolding  $state-refines-def$  by  $metis$ 
  thus  $?thesis$  using  $rev$  by  $simp$ 
qed

```

The same guarantee phrased over the reflexive-transitive closure: any view reachable from b by any number of blinding steps extracts to a valid state refining b . (The closure collapses because bo is already a preorder, which is precisely why arbitrary-length sequences add no new obligation.)

corollary *reachable-view-authenticity*:


```

assumes reach:  $bo^{**} a b$  and eb: extract-map  $b = \text{Some } sb$ 
shows  $\exists sa. \text{extract-map } a = \text{Some } sa \wedge \text{valid-state } sa \wedge \text{state-refines } sa sb$ 
proof –
  from reach have  $bo a b$ 
  proof (induction rule: rtranclp-induct)
    case base
    show ?case using mk.reflp by (simp add: reflp-def)
  next
    case (step  $y z$ )
    show ?case using mk.transp step.IH step.hyps(2) by (metis transpD)
  qed
from cdsp-ads-morphism[OF this eb] show ?thesis by blast
qed

```

59.2 Re-revealing merge along a sequence

The n-ary right fold of the (locale) merge along a non-empty sequence of authenticated views. It is defined with the locale merge m fixed, so its induction principle keeps m in place (a theory-level fold parameterised by the merge would generalise it away under induction).

```

fun merge-seq :: 'd list  $\Rightarrow$  'd option where
  merge-seq [] = None
| merge-seq [ $x$ ] = Some  $x$ 
| merge-seq ( $x \# y \# ys$ ) =
  (case merge-seq ( $y \# ys$ ) of None  $\Rightarrow$  None | Some  $z \Rightarrow m x z$ )

```

A single merge preserves the common hash, since each input blinds to the merge and blinding respects hashes.

```

lemma merge-preserves-hash:
  assumes  $m a b = \text{Some } ab$ 
  shows  $h ab = h a$ 
proof –
  have  $bo a ab$  using mk.join assms by blast
  thus ?thesis using bo-hash-eq by simp
qed

```

Sequence-level hash preservation for the merge fold. Combining a whole hash-compatible sequence of partial views yields a value with that same common hash: re-revealing along the sequence never changes the authenticating root. This is the merge-direction companion of *blinding-path-hash-soundness* (which covers the blinding direction), so along either form of synchronization sequence the root hash is preserved.

```

theorem merge-seq-hash:
  assumes merge-seq  $xs = \text{Some } r$  and  $xs \neq []$ 
  and  $\forall x \in \text{set } xs. h x = h (hd xs)$ 
  shows  $h r = h (hd xs)$ 
  using assms
proof (induction xs rule: merge-seq.induct)

```

```

case ( $\exists x y ys$ )
obtain  $z$  where  $z$ :  $\text{merge-seq } (y \# ys) = \text{Some } z$  and  $mzx$ :  $m x z = \text{Some } r$ 
using  $\mathcal{I}.\text{prems}(1)$  by ( $\text{auto split: option.splits}$ )
have  $h r = h x$  using  $\text{merge-preserves-hash}[OF mzx]$  .
thus  $?case$  by  $\text{simp}$ 
qed  $\text{simp-all}$ 

```

Merge-fold soundness. A non-empty list of partial authenticated views over the same committed object (all sharing a hash) folds into a single combined view that extracts to a valid cross-domain state refining every contributor. This is the n -ary generalisation of *authenticated-preservation-soundness*: re-revealing along the entire fold preserves validity and reconstructs a consistent joint view.

theorem *sequence-merge-soundness*:

```

 $\llbracket xs \neq []; \forall x \in \text{set } xs. \exists s. \text{extract-map } x = \text{Some } s;$ 
 $\quad \forall x \in \text{set } xs. h x = h (hd xs) \rrbracket$ 
 $\implies \exists ab s. \text{merge-seq } xs = \text{Some } ab \wedge \text{extract-map } ab = \text{Some } s \wedge \text{valid-state } s$ 
 $\quad \wedge (\forall x \in \text{set } xs. \exists sx. \text{extract-map } x = \text{Some } sx \wedge \text{state-refines } sx s)$ 

```

proof (*induction xs rule: merge-seq.induct*)

case 1

then show $?case$ **by** simp

next

case ($2 x$)

obtain sx **where** ex : $\text{extract-map } x = \text{Some } sx$ **using** $\mathcal{I}.\text{prems}$ **by** auto

have $\text{valid-state } sx$ **using** $\text{extract-preserves-validity}[OF ex]$.

thus $?case$ **using** ex $\text{state-refines-refl}$ **by** auto

next

case ($\exists x y ys$)

have $ne\text{-rest}$: $(y \# ys) \neq []$ **by** simp

have hx : $\bigwedge w. w \in \text{set } (y \# ys) \implies h w = h x$

using $\mathcal{I}.\text{prems}(3)$ **by** auto

have $hrest$: $\forall w \in \text{set } (y \# ys). h w = h (hd (y \# ys))$ **using** hx **by** auto

have $erest$: $\forall w \in \text{set } (y \# ys). \exists s. \text{extract-map } w = \text{Some } s$

using $\mathcal{I}.\text{prems}(2)$ **by** auto

obtain $z sz$ **where** mz : $\text{merge-seq } (y \# ys) = \text{Some } z$

and ez : $\text{extract-map } z = \text{Some } sz$ **and** vz : $\text{valid-state } sz$

and $refz$: $\forall w \in \text{set } (y \# ys). \exists sw. \text{extract-map } w = \text{Some } sw \wedge \text{state-refines}$

$sw sz$

using $\mathcal{I}.\text{IH}[OF ne\text{-rest } erest hrest]$ **by** blast

obtain sx **where** ex : $\text{extract-map } x = \text{Some } sx$ **using** $\mathcal{I}.\text{prems}(2)$ **by** auto

have hz : $h z = h x$

proof –

have $h z = h (hd (y \# ys))$ **using** $\text{merge-seq-hash}[OF mz ne\text{-rest } hrest]$.

thus $?thesis$ **using** hx **by** simp

qed

have $\exists ab. m x z = \text{Some } ab$

using hz $\text{mk.merge-respects-hashes}[of x z]$ **by** simp

then obtain xz **where** mzx : $m x z = \text{Some } xz$ **by** blast

have $mseq$: $\text{merge-seq } (x \# y \# ys) = \text{Some } xz$ **using** $mz mzx$ **by** simp

```

obtain szx where exz: extract-map xz = Some szx and vxz: valid-state szx
and rx: state-refines sx szx and rz: state-refines sz szx
using authenticated-preservation-soundness[OF mzx ex ez] by blast
have  $\forall w \in \text{set } (x \# y \# ys). \exists sw. \text{extract-map } w = \text{Some } sw \wedge \text{state-refines } sw$ 
szx
proof
  fix w assume  $w \in \text{set } (x \# y \# ys)$ 
  then consider  $w = x \mid w \in \text{set } (y \# ys)$  by auto
  then show  $\exists sw. \text{extract-map } w = \text{Some } sw \wedge \text{state-refines } sw \text{ } szx$ 
  proof cases
    case 1
    then show ?thesis using ex rx by blast
  next
    case 2
    then obtain sw where extract-map w = Some sw and state-refines sw sz
    using refz by blast
    then show ?thesis using state-refines-trans[OF - rz] by blast
  qed
qed
then show ?case using mseq exz vxz by blast
qed
end

```

60 Instantiation on the ADS blidable-position functor

We instantiate *authenticated-state* on a concrete ADS construction: the blidable-position functor of *ADS-Functor.ADS-Construction* applied to the Oraclizer Merkle interface *merkle-interface-auth*. The carrier is $(\text{reg-state} \times \text{chain-id set}, \text{reg-state})$ *blidable_m*: an authenticated datum is either *Unblinded* (r, P) (the fully revealed consensus state *r* with revealed-chain set *P*) or *Blinded* *x* (a hash-only commitment that reveals nothing). The blidable hash, blinding order and merge are inherited from the ADS construction; the Merkle-interface obligation is discharged by *merkle-blidable* from *merkle-interface-auth*, so no new interface proof is needed.

The empty extracted view does not depend on the (unreachable) consensus label, so all hash-only commitments extract to the same state.

lemma *auth-state-empty-eq*: *auth-state* *r* {} = *auth-state* *r'* {}
by (*simp add: auth-state-def*)

Extraction: a revealed datum yields its cross-domain state; a hash-only commitment yields the empty (no-chains) view, which is valid and refines every state. The commitment must still extract to a state – a blinding of a value must itself be extractable – so it maps to the empty view rather than

to *None*.

fun *bl-extract* :: (*reg-state* $\times$ *chain-id set*, *reg-state*) *blindable_m* $\Rightarrow$ *global-state option* **where**

bl-extract (*Unblinded* *a*) = *Some* (*auth-state* (*fst* *a*) (*snd* *a*))
| *bl-extract* (*Blinded* *-*) = *Some* (*auth-state* *ACTIVE* *{}*)

interpretation *oss-blindable*:

authenticated-state
hash-blindable *auth-hash*
blinding-of-blindable *auth-hash* *auth-bo*
merge-blindable *auth-hash* *auth-merge*
bl-extract

proof (*rule* *authenticated-state.intro*)

show *merkle-interface* (*hash-blindable* *auth-hash*)
(*blinding-of-blindable* *auth-hash* *auth-bo*) (*merge-blindable* *auth-hash* *auth-merge*)

by (*rule* *merkle-blindable*[*OF* *merkle-interface-auth*])

next

show *authenticated-state-axioms* (*hash-blindable* *auth-hash*)
(*blinding-of-blindable* *auth-hash* *auth-bo*) (*merge-blindable* *auth-hash* *auth-merge*) *bl-extract*

proof

— *extract* respects merging

fix *a b ab sa sb*

assume *Hab*: *hash-blindable* *auth-hash* *a* = *hash-blindable* *auth-hash* *b*

and *Mab*: *merge-blindable* *auth-hash* *auth-merge* *a b* = *Some* *ab*

and *ea*: *bl-extract* *a* = *Some* *sa* **and** *eb*: *bl-extract* *b* = *Some* *sb*

show $\exists$ *sab*. *bl-extract* *ab* = *Some* *sab* $\wedge$ *state-join* *sa sb sab*

proof (*cases* *a*)

case (*Unblinded* *p*) **note** *A* = *this*

show *?thesis*

proof (*cases* *b*)

case (*Unblinded* *q*) **note** *B* = *this*

have *fp*: *fst* *p* = *fst* *q* **using** *Hab* **by** (*simp* *add*: *A B* *auth-hash-def*)

have *ab-eq*: *ab* = *Unblinded* (*fst* *p*, *snd* *p* $\cup$ *snd* *q*)

using *Mab* *fp* **by** (*simp* *add*: *A B* *auth-merge-def*)

have *sa-eq*: *sa* = *auth-state* (*fst* *p*) (*snd* *p*) **using** *ea* **by** (*simp* *add*: *A*)

have *sb-eq*: *sb* = *auth-state* (*fst* *p*) (*snd* *q*) **using** *eb* *fp* **by** (*simp* *add*: *B*)

have *bl-extract* *ab* = *Some* (*auth-state* (*fst* *p*) (*snd* *p* $\cup$ *snd* *q*)) **by** (*simp*

add: *ab-eq*)

moreover **have** *state-join* *sa sb* (*auth-state* (*fst* *p*) (*snd* *p* $\cup$ *snd* *q*))

unfolding *sa-eq* *sb-eq* **by** (*rule* *state-join-auth*)

ultimately **show** *?thesis* **by** *blast*

next

case (*Blinded* *y*) **note** *B* = *this*

have *ab-eq*: *ab* = *Unblinded* *p* **using** *Mab* *Hab* **by** (*simp* *add*: *A B*

auth-hash-def)

have *sa-eq*: *sa* = *auth-state* (*fst* *p*) (*snd* *p*) **using** *ea* **by** (*simp* *add*: *A*)

have *sb-eq*: *sb* = *auth-state* *ACTIVE* *{}* **using** *eb* **by** (*simp* *add*: *B*)

```

    have bl-extract ab = Some (auth-state (fst p) (snd p)) by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state (fst p) (snd p))
      unfolding sa-eq sb-eq
      using state-join-auth[of fst p snd p {}] auth-state-empty-eq[of ACTIVE fst
p]
      by simp
    ultimately show ?thesis by blast
  qed
next
case (Blinded x) note A = this
show ?thesis
proof (cases b)
  case (Unblinded q) note B = this
    have ab-eq: ab = Unblinded q using Mab Hab by (simp add: A B
auth-hash-def)
    have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
    have sb-eq: sb = auth-state (fst q) (snd q) using eb by (simp add: B)
    have bl-extract ab = Some (auth-state (fst q) (snd q)) by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state (fst q) (snd q))
      unfolding sa-eq sb-eq
      using state-join-auth[of fst q {} snd q] auth-state-empty-eq[of ACTIVE fst
q]
      by simp
    ultimately show ?thesis by blast
  next
case (Blinded y) note B = this
    have ab-eq: ab = Blinded y using Mab Hab by (simp add: A B)
    have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
    have sb-eq: sb = auth-state ACTIVE {} using eb by (simp add: B)
    have bl-extract ab = Some (auth-state ACTIVE {}) by (simp add: ab-eq)
    moreover have state-join sa sb (auth-state ACTIVE {})
      unfolding sa-eq sb-eq using state-join-auth[of ACTIVE {} {}] by simp
    ultimately show ?thesis by blast
  qed
qed
next
— extract under blinding
fix a b sb
assume Hab: hash-blindable auth-hash a = hash-blindable auth-hash b
  and BOab: blinding-of-blindable auth-hash auth-bo a b
  and eb: bl-extract b = Some sb
show  $\exists sa. bl-extract a = Some sa \wedge state-refines sa sb$ 
proof (cases b)
  case (Unblinded q) note B = this
  show ?thesis
  proof (cases a)
    case (Unblinded p) note A = this
    have bo: auth-bo p q using BOab by (simp add: A B)
    have rfst: fst p = fst q and sub: snd p  $\subseteq$  snd q

```

```

    using bo by (simp-all add: auth-bo-def)
  have sr: state-refines (auth-state (fst p) (snd p)) (auth-state (fst p) (snd q))
    using sub by (rule state-refines-auth)
  have bl-extract a = Some (auth-state (fst p) (snd p)) by (simp add: A)
  moreover have state-refines (auth-state (fst p) (snd p)) sb
    using sr rfst eb by (simp add: B)
  ultimately show ?thesis by blast
next
case (Blinded x) note A = this
have sr: state-refines (auth-state (fst q) {}) (auth-state (fst q) (snd q))
  using empty-subsetI by (rule state-refines-auth)
have bl-extract a = Some (auth-state (fst q) {})
  by (simp add: A auth-state-empty-eq[of ACTIVE fst q])
moreover have state-refines (auth-state (fst q) {}) sb
  using sr eb by (simp add: B)
ultimately show ?thesis by blast
qed
next
case (Blinded y) note B = this
— Only a blinded value can blind to a blinded value, and then they are equal.
have a = Blinded y using BOab by (cases a) (simp-all add: B)
then have bl-extract a = Some (auth-state ACTIVE {}) by simp
moreover have sb = auth-state ACTIVE {} using eb by (simp add: B)
ultimately show ?thesis using state-refines-refl by auto
qed
next
— extract preserves validity
fix a s
assume bl-extract a = Some s
then show valid-state s
  by (cases a) (auto simp: auth-state-valid)
qed
qed

```

Non-degeneracy of the blindable instance: revealed two-chain views merge to their union, and a hash-only commitment merges back to the revealed view.

lemma *oss-blindable-nontrivial*:

```

merge-blindable auth-hash auth-merge (Unblinded (ACTIVE, {0})) (Unblinded
(ACTIVE, {Suc 0}))
= Some (Unblinded (ACTIVE, {0, Suc 0}))
bl-extract (Unblinded (ACTIVE, {0, Suc 0})) = Some (auth-state ACTIVE {0,
Suc 0})
merge-blindable auth-hash auth-merge (Blinded (Content ACTIVE)) (Unblinded
(ACTIVE, {0}))
= Some (Unblinded (ACTIVE, {0}))
by (simp-all add: auth-merge-def auth-hash-def insert-commute)

```

Regression witness exhibiting a non-vacuous instance for the sequence-level results above. The generalised predicate *blinding-path*

is satisfied by a genuinely increasing need-to-know sequence at this instance: a hash-only commitment, then a one-chain revealed view, then a two-chain revealed view, each blinding its successor. The most-revealed endpoint extracts to a non-degenerate two-chain state. This pins *authenticated-state.sequence-authenticity-preservation*, *authenticated-state.blinding-path-hash-soundness* and *authenticated-state.sequence-inclusion-integrity* to a non-vacuous instance.

lemma *blinding-path-witness*:

blinding-path (*blinding-of-blindable* *auth-hash* *auth-bo*)
 [Blinded (*Content* *ACTIVE*),
 Unblinded (*ACTIVE*, {0::nat}),
 Unblinded (*ACTIVE*, {0::nat, Suc 0})]
unfolding *blinding-path-def*
by (*auto simp: nth-Cons' auth-hash-def auth-bo-def*)

lemma *blinding-path-witness-endpoint*:

bl-extract (Unblinded (*ACTIVE*, {0::nat, Suc 0})) = *Some* (*auth-state* *ACTIVE* {0, Suc 0})
connected-chains (*auth-state* *ACTIVE* {0::nat, Suc 0}) (0::nat) = {0, Suc 0}
by (*simp-all add: auth-state-connected*)

61 Instantiation on the top-level Canton transaction commitment

Theory *ADS-Functor.Canton-Transaction-Tree* formalises the production Canton transaction tree as an authenticated data structure, but over abstract content: *view-data*, *view-metadata*, *common-metadata* and *participant-metadata* are introduced by **typeddecl** and carry no regulatory meaning, so a cross-domain state cannot be read out of *transaction_m* without an inadmissible axiom relating those opaque types to the regulatory model. We therefore model the same public structure with concrete content. A Canton transaction is *Transaction_m* applied to one top-level blindable position over a payload of metadata and a view list; we mirror exactly this one-constructor-over-a-blindable shape, taking the payload to be the cross-domain regulatory datum — the consensus regulatory state (in the role of the common metadata) together with the set of chains whose views attest it (the per-participant view list, abstracted to its attesting-chain set). Hash, blinding and merge are the ADS blindable-position operations on the payload: precisely the top-level authenticated operation Canton performs, since *Transaction_m* wraps a single *blindable_m*. The Merkle interface and the three coherence obligations are transported from the blindable instance *oss-blindable* through the constructor bijection, so no new axiom and no residual proof obligation arise. The one open item is model fidelity: whether this payload faithfully abstracts Canton’s metadata-and-view-list datatype.

That is a sanity check of the model against the Canton specification, not a proof gap.

datatype *reg-transaction* =

RegTransaction (*the-reg-tx*: (*reg-state* × *chain-id set*, *reg-state*) *blindable_m*)

definition *rtx-hash* :: *reg-transaction* ⇒ *reg-state blindable_h* **where**

rtx-hash *t* = *hash-blindable auth-hash* (*the-reg-tx* *t*)

definition *rtx-bo* :: *reg-transaction* ⇒ *reg-transaction* ⇒ *bool* **where**

rtx-bo *s t* = *blinding-of-blindable auth-hash auth-bo* (*the-reg-tx* *s*) (*the-reg-tx* *t*)

definition *rtx-merge* :: *reg-transaction* ⇒ *reg-transaction* ⇒ *reg-transaction option* **where**

rtx-merge *s t* =
map-option RegTransaction
(merge-blindable auth-hash auth-merge (*the-reg-tx* *s*) (*the-reg-tx* *t*))

definition *rtx-extract* :: *reg-transaction* ⇒ *global-state option* **where**

rtx-extract *t* = *bl-extract* (*the-reg-tx* *t*)

The single constructor is a bijection, so the Merkle interface of the payload blindable transports to the transaction wrapper.

lemma *merkle-reg-transaction*: *merkle-interface rtx-hash rtx-bo rtx-merge*

proof –

have *bind-simp*: *rtx-merge* *a b* ≫≫ *rtx-merge* *c*

= *map-option RegTransaction*

(merge-blindable auth-hash auth-merge (*the-reg-tx* *a*) (*the-reg-tx* *b*)

≫≫ *merge-blindable auth-hash auth-merge* (*the-reg-tx* *c*)) **for** *a b c*

by (*cases merge-blindable auth-hash auth-merge* (*the-reg-tx* *a*) (*the-reg-tx* *b*))
(simp-all add: rtx-merge-def)

have *mrh*: (*rtx-hash* *a* = *rtx-hash* *b*) = (∃ *ab*. *rtx-merge* *a b* = *Some ab*) **for** *a b*

by (*cases merge-blindable auth-hash auth-merge* (*the-reg-tx* *a*) (*the-reg-tx* *b*))

(auto simp: rtx-hash-def rtx-merge-def oss-blindable.mk.merge-respects-hashes)

have *midem*: *rtx-merge* *a a* = *Some a* **for** *a*

by (*simp add: rtx-merge-def oss-blindable.mk.idem*)

have *mcomm*: *rtx-merge* *a b* = *rtx-merge* *b a* **for** *a b*

by (*simp add: rtx-merge-def oss-blindable.mk.commute*)

have *massoc*: *rtx-merge* *a b* ≫≫ *rtx-merge* *c* = *rtx-merge* *b c* ≫≫ *rtx-merge* *a* **for** *a b c*

using *bind-simp*[*of a b c*] *bind-simp*[*of b c a*]

oss-blindable.mk.assoc[*of the-reg-tx a the-reg-tx b the-reg-tx c*] **by** *simp*

have *mbod*: *rtx-bo* *a b* = (*rtx-merge* *a b* = *Some b*) **for** *a b*

by (*cases merge-blindable auth-hash auth-merge* (*the-reg-tx* *a*) (*the-reg-tx* *b*))

(auto simp: rtx-bo-def rtx-merge-def oss-blindable.mk.bo-def reg-transaction.expand)

show *?thesis*

by (*rule merkle-interface.intro*[*OF mrh midem mcomm massoc mbod*])

qed

Extraction reads the revealed cross-domain datum out of the transac-

tion's top-level commitment; the three coherence obligations transport from the blindable instance *oss-blindable*.

interpretation *canton-authenticated*:

authenticated-state rtx-hash rtx-bo rtx-merge rtx-extract

proof (*rule authenticated-state.intro*)

show *merkle-interface rtx-hash rtx-bo rtx-merge* **by** (*rule merkle-reg-transaction*)

next

show *authenticated-state-axioms rtx-hash rtx-bo rtx-merge rtx-extract*

proof

— extract respects merging, transported through the constructor

fix *a b ab sa sb*

assume *H: rtx-hash a = rtx-hash b* **and** *M: rtx-merge a b = Some ab*

and *ea: rtx-extract a = Some sa* **and** *eb: rtx-extract b = Some sb*

have *h': hash-blindable auth-hash (the-reg-tx a) = hash-blindable auth-hash (the-reg-tx b)*

using *H* **by** (*simp add: rtx-hash-def*)

have *m': merge-blindable auth-hash auth-merge (the-reg-tx a) (the-reg-tx b) = Some (the-reg-tx ab)*

using *M* **by** (*auto simp: rtx-merge-def split: option.splits*)

have *ea': bl-extract (the-reg-tx a) = Some sa* **using** *ea* **by** (*simp add: rtx-extract-def*)

have *eb': bl-extract (the-reg-tx b) = Some sb* **using** *eb* **by** (*simp add: rtx-extract-def*)

obtain *sab* **where** *bl-extract (the-reg-tx ab) = Some sab* *state-join sa sb sab*

using *oss-blindable.extract-respects-merging[OF h' m' ea' eb']* **by** *blast*

then show $\exists sab. rtx-extract ab = Some sab \wedge state-join sa sb sab$

by (*auto simp: rtx-extract-def*)

next

— extract under blinding, transported through the constructor

fix *a b sb*

assume *H: rtx-hash a = rtx-hash b* **and** *BO: rtx-bo a b*

and *eb: rtx-extract b = Some sb*

have *h': hash-blindable auth-hash (the-reg-tx a) = hash-blindable auth-hash (the-reg-tx b)*

using *H* **by** (*simp add: rtx-hash-def*)

have *bo': blinding-of-blindable auth-hash auth-bo (the-reg-tx a) (the-reg-tx b)*

using *BO* **by** (*simp add: rtx-bo-def*)

have *eb': bl-extract (the-reg-tx b) = Some sb* **using** *eb* **by** (*simp add: rtx-extract-def*)

obtain *sa* **where** *bl-extract (the-reg-tx a) = Some sa* *state-refines sa sb*

using *oss-blindable.extract-under-blinding[OF h' bo' eb']* **by** *blast*

then show $\exists sa. rtx-extract a = Some sa \wedge state-refines sa sb$

by (*auto simp: rtx-extract-def*)

next

— extract preserves validity

fix *a s*

assume *rtx-extract a = Some s*

then have *bl-extract (the-reg-tx a) = Some s* **by** (*simp add: rtx-extract-def*)

then show *valid-state s* **using** *oss-blindable.extract-preserves-validity* **by** *blast*

qed
qed

Non-degeneracy: a revealed two-chain transaction joins with a hash-only commitment to the same chains, re-revealing the cross-domain view.

lemma *canton-authenticated-nontrivial*:

```

    rtx-merge (RegTransaction (Unblinded (ACTIVE, {0})))
      (RegTransaction (Unblinded (ACTIVE, {Suc 0})))
    = Some (RegTransaction (Unblinded (ACTIVE, {0, Suc 0})))
    rtx-extract (RegTransaction (Unblinded (ACTIVE, {0, Suc 0})))
    = Some (auth-state ACTIVE {0, Suc 0})
    rtx-merge (RegTransaction (Blinded (Content ACTIVE)))
      (RegTransaction (Unblinded (ACTIVE, {0})))
    = Some (RegTransaction (Unblinded (ACTIVE, {0})))
    by (simp-all add: rtx-merge-def rtx-extract-def auth-merge-def auth-hash-def in-
      sert-commute)

```

62 Recursive instantiation on the Canton transaction tree

The coarse instance *canton-authenticated* above models a Canton transaction as one top-level blindable over an abstract payload. Here we keep the public *recursive* shape: a view is the rose tree $rose-tree_m$ of *ADS-Functor.ADS-Construction* — the very datatype the public Canton formalisation uses to build $view_m$ ($view_m \cong view_content\ rose-tree_m$, with $merge_view = merge_tree \dots$). We instantiate that content-agnostic machine with concrete regulatory content: each node carries the *chain-id* it attests, and the children are its subviews. No new datatype, no new Merkle proof: the interface is inherited from *merkle-tree*.

type-synonym *reg-view* = (*chain-id*, *chain-id*) $rose-tree_m$

abbreviation *rv-hash* :: (*reg-view*, *chain-id* $rose-tree_h$) *hash* **where**

rv-hash $\equiv hash_tree\ (hash_discrete :: chain-id \Rightarrow chain-id)$

abbreviation *rv-bo* :: *reg-view* *blinding-of* **where**

rv-bo $\equiv blinding_of_tree\ hash_discrete\ (blinding_of_discrete :: chain-id\ blinding_of)$

abbreviation *rv-merge* :: *reg-view* *merge* **where**

rv-merge $\equiv merge_tree\ hash_discrete\ (merge_discrete :: chain-id\ merge)$

lemma *merkle-reg-view*: *merkle-interface* *rv-hash* *rv-bo* *rv-merge*

by (*rule* *merkle-tree*[*OF merkle-discrete*])

Extraction collects the attested chains from every revealed node, recursing into subviews; a blinded node contributes nothing. By the Merkle property a subview is reachable only through its (revealed) parent. Lemma *reg-chains-blinding* below makes this monotone: blinding a node can only shrink the attested-chain set.

```

fun reg-chains :: reg-view  $\Rightarrow$  chain-id set where
  reg-chains (Treem (Unblinded (c, kids))) = insert c ( $\bigcup_{x \in \text{set kids. reg-chains } x}$ )
  | reg-chains (Treem (Blinded -)) = {}

```

62.1 Extraction commutes with blinding and merge

A pointwise containment along a list lifts to the union of the collected chains.

lemma *UN-reg-chains-mono*:

```

assumes list-all2 ( $\lambda x y. \text{reg-chains } x \subseteq \text{reg-chains } y$ ) xs ys
shows ( $\bigcup_{x \in \text{set xs. reg-chains } x} \subseteq \bigcup_{y \in \text{set ys. reg-chains } y}$ )
using assms by (induction xs ys rule: list-all2-induct) auto

```

Positional characterisation of the ADS list merge (used for the children lists of a rose-tree node). The length alignment is *derived*, not assumed: the lemmas *merge-list-NC* / *merge-list-CN* below reject lists of different length, exactly because the list hash (*map*) commits to the length.

context begin

interpretation list-R1 .

lemma *merge-list-Cons*:

```

merge-list m (x # xs) (y # ys) =
  (case m x y of None  $\Rightarrow$  None | Some z  $\Rightarrow$  map-option ((#) z) (merge-list m xs
ys))
by (simp add: merge-list-def merge-F-def merge-R1.simps
      list-to-list-R1.simps list-R1-to-list-simps merge-sum.simps
merge-prod.simps
      merge-discrete-def option.map-comp o-def split: option.splits)

```

lemma *merge-list-NN* [simp]: merge-list m [] [] = Some []

```

by (simp add: merge-list-def merge-F-def merge-R1.simps
      list-to-list-R1.simps list-R1-to-list-simps merge-sum.simps
merge-discrete-def)

```

lemma *merge-list-CN* [simp]: merge-list m (x # xs) [] = None

```

by (simp add: merge-list-def merge-F-def merge-R1.simps list-to-list-R1.simps
merge-sum.simps)

```

lemma *merge-list-NC* [simp]: merge-list m [] (y # ys) = None

```

by (simp add: merge-list-def merge-F-def merge-R1.simps list-to-list-R1.simps
merge-sum.simps)

```

end

The collected chains of a list-merge are the union of the contributors' chains, given that the elementwise merge has that property.

lemma *reg-chains-merge-list*:

```

assumes merge-list m k1 k2 = Some k
and  $\bigwedge s t u. s \in \text{set } k1 \implies m s t = \text{Some } u \implies \text{reg-chains } u = \text{reg-chains } s \cup \text{reg-chains } t$ 

```

```

shows ( $\bigcup_{s \in \text{set } k}. \text{reg-chains } s$ ) = ( $\bigcup_{s \in \text{set } k1}. \text{reg-chains } s$ )  $\cup$  ( $\bigcup_{t \in \text{set } k2}. \text{reg-chains } t$ )
using assms
proof (induction k1 arbitrary: k2 k)
  case Nil
  then show ?case by (cases k2) auto
next
  case (Cons x xs)
  show ?case
  proof (cases k2)
    case Nil then show ?thesis using Cons.prem1 by simp
  next
    case (Cons y ys)
    from Cons.prem1 obtain z k' where z: m x y = Some z
      and k': merge-list m xs ys = Some k' and keq: k = z # k'
      by (auto simp: Cons merge-list-Cons split: option.splits)
    have hz: reg-chains z = reg-chains x  $\cup$  reg-chains y
      using Cons.prem2[of x y z] z by simp
    have ih: ( $\bigcup_{s \in \text{set } k'}. \text{reg-chains } s$ ) = ( $\bigcup_{s \in \text{set } xs}. \text{reg-chains } s$ )  $\cup$  ( $\bigcup_{t \in \text{set } ys}. \text{reg-chains } t$ )
      using k' by (intro Cons.IH) (use Cons.prem2) in auto
    show ?thesis using keq hz ih by (simp add: Cons) blast
  qed
qed

```

Blinding shrinks the revealed chains, recursively through subviews: a blinded view's chains are contained in those of the view it blinds.

```

lemma reg-chains-blinding:
  rv-bo a b  $\implies$  reg-chains a  $\subseteq$  reg-chains b
proof (induction a arbitrary: b rule: rose-treem.induct)
  case (Treem x b)
  obtain t2 where bt2: b = Treem t2 by (cases b)
  note rt = Treem.prems[unfolded bt2 blinding-of-tree-simps]
  show ?case
  proof (cases x)
    case (Blinded h) then show ?thesis by simp
  next
    case (Unblinded p)
    obtain c kids1 where p: p = (c, kids1) by (cases p)
    from rt obtain kids2 where t2eq: t2 = Unblinded (c, kids2) and
      la: list-all2 rv-bo kids1 kids2
      by (auto simp: Unblinded p rel-prod-inject)
    have la2: list-all2 ( $\lambda s t. \text{reg-chains } s \subseteq \text{reg-chains } t$ ) kids1 kids2
      proof (rule list-all2-all-nthI)
        show length kids1 = length kids2 using la list-all2-lengthD by blast
        fix n assume n: n < length kids1
        have m1: (c, kids1)  $\in$  set1-blindablem x by (simp add: Unblinded p)
        have sm: kids1  $\in$  snds (c, kids1) by (simp add: prod-set-simps)
        have memn: kids1 ! n  $\in$  set kids1 using n by simp
      qed
    qed
  qed

```

```

    have rvn: rv-bo (kids1 ! n) (kids2 ! n) using la n list-all2-nthD by blast
    show reg-chains (kids1 ! n)  $\subseteq$  reg-chains (kids2 ! n)
      using Treem.IH[OF m1 sm memn rvn] .
  qed
  from UN-reg-chains-mono[OF la2] show ?thesis
    by (auto simp: Unblinded p bt2 t2eq)
  qed
  qed

  Merging re-reveals exactly the union of the two contributors' chains,
  recursively through subviews.

  lemma reg-chains-merge:
    rv-merge a b = Some ab  $\implies$  reg-chains ab = reg-chains a  $\cup$  reg-chains b
  proof (induction a arbitrary: b ab rule: rose-treem.induct)
    case (Treem x b ab)
    obtain y where bY: b = Treem y by (cases b)
    obtain z where abz: ab = Treem z by (cases ab)
    from Treem.prems[unfolded bY abz merge-tree.simps]
    have mz: merge-rt-Fm hash-discrete merge-discrete (hash-tree hash-discrete)
  rv-merge x y = Some z
    by (auto split: option.splits)
    note mb = mz[unfolded merge-rt-Fm-def]
    show ?case
  proof (cases x)
    case (Blinded hx)
    from mb have z = y  $\vee$  ( $\exists q. y = \text{Unblinded } q \wedge z = \text{Unblinded } q$ )
      by (cases y) (auto simp: Blinded split: if-splits)
    then show ?thesis using bY abz Blinded by auto
  next
    case (Unblinded px)
    obtain c1 k1 where px: px = (c1, k1) by (cases px)
    show ?thesis
  proof (cases y)
    case (Blinded hy)
    from mb have z = Unblinded (c1, k1)
      by (auto simp: Unblinded px Blinded split: if-splits)
    then show ?thesis using bY abz Unblinded px Blinded by simp
  next
    case (Unblinded py)
    obtain c2 k2 where py: py = (c2, k2) by (cases py)
    from mb obtain k where c12: c1 = c2 and kk: merge-list rv-merge k1 k2
  = Some k
      and zk: z = Unblinded (c1, k)
      by (auto simp:  $\langle x = \text{Unblinded } px \rangle$  px Unblinded py merge-discrete-def
        split: option.splits if-splits)
    have IHc:  $\bigwedge s t u. s \in \text{set } k1 \implies \text{rv-merge } s t = \text{Some } u$ 
       $\implies \text{reg-chains } u = \text{reg-chains } s \cup \text{reg-chains } t$ 
  proof -
    fix s t u assume sset: s  $\in$  set k1 and st: rv-merge s t = Some u

```

```

    have m1: (c1, k1) ∈ set1-blindablem x by (simp add: ⟨x = Unblinded px⟩
px)
    have sm: k1 ∈ snds (c1, k1) by (simp add: prod-set-simps)
    show reg-chains u = reg-chains s ∪ reg-chains t
      using Treem.IH[OF m1 sm sset st] .
  qed
  have (⋃ s∈set k. reg-chains s) = (⋃ s∈set k1. reg-chains s) ∪ (⋃ t∈set k2.
reg-chains t)
    using reg-chains-merge-list[OF kk IHc] .
  then show ?thesis
    using bY abz zk c12 by (simp add: ⟨x = Unblinded px⟩ px Unblinded py)
  qed
qed
qed

```

Regression witnesses for the recursive structure. First, deep-nested chains are not dropped: extraction genuinely recurses into subviews. Second, list-merge length alignment is enforced (mismatched lengths are rejected), so it is derived from the hash rather than assumed. Third, a blinded node contributes nothing. The single-consensus invariant (load-bearing, requiring the extraction map) is exhibited with the interpretation below.

lemma *reg-chains-deep-nesting*:

```

reg-chains (Treem (Unblinded (0::chain-id, [Treem (Unblinded (Suc 0, []))]))) =
{0, Suc 0}
by simp

```

lemma *reg-chains-blinded-contributes-nothing*:

```

reg-chains (Treem (Blinded h)) = {}
by simp

```

lemma *merge-list-length-aligned*:

```

merge-list rv-merge [t] [] = None
merge-list rv-merge [] [t] = None
by simp-all

```

62.2 The recursive transaction-tree instance

A Canton transaction wraps its consensus metadata and its view list in one top-level blindable ($transaction_m = Transaction_m (blindable_m \dots)$). We mirror that exactly: the carrier is a top-level blindable over a payload $reg-state \times reg-view\ list$ — the consensus regulatory state together with the list of recursive view trees. Hash, blinding and merge come from the public product/list/blindable building blocks, so the Merkle interface is inherited (lemma *merkle-reg-tx* below).

Honesty (model-fidelity boundary, read carefully). The recursive view structure is now *faithful*: subviews nest as in Canton, and subview-level blinding is alive (the witnesses below). Two points remain modelling

choices, *not* a 1:1 match with Canton's datatype. First, leaf content is concrete *chain-id/reg-state*, whereas Canton's leaves are the opaque *view-data/view-metadata*: this is *recursive-faithful*, *content-abstracted*. Second, the consensus *reg-state* is modelled as a *bare, non-independently-blindable* field of the payload, while Canton's *common-metadata_m* is itself an independently blindable position. This is what structurally forces the single-consensus invariant (a view is reachable only by revealing the whole transaction, hence its consensus), which in turn gives validity with no assumption. The price is a scope limit: the case “a view is revealed while the consensus is blinded” is *out of this model's scope*. Lifting it would require an independently-blindable consensus and a fresh consistency argument; that is recorded as future work and as the residual fidelity check for the Canton authors, not closed here.

type-synonym *reg-transaction-tree* =

(*reg-state* × *reg-view list*, *reg-state* × *chain-id rose-tree_h list*) *blindable_m*

abbreviation *rtt-chash* :: (*reg-state* × *reg-view list*, *reg-state* × *chain-id rose-tree_h list*) *hash*

where *rtt-chash* ≡ *hash-prod hash-discrete (hash-list rv-hash)*

abbreviation *rtt-cbo* :: (*reg-state* × *reg-view list*) *blinding-of*

where *rtt-cbo* ≡ *blinding-of-prod blinding-of-discrete (blinding-of-list rv-bo)*

abbreviation *rtt-cmerge* :: (*reg-state* × *reg-view list*) *merge*

where *rtt-cmerge* ≡ *merge-prod merge-discrete (merge-list rv-merge)*

abbreviation *rtt-hash* :: (*reg-transaction-tree*, (*reg-state* × *chain-id rose-tree_h list*) *blindable_h*) *hash*

where *rtt-hash* ≡ *hash-blindable rtt-chash*

abbreviation *rtt-bo* :: *reg-transaction-tree* *blinding-of*

where *rtt-bo* ≡ *blinding-of-blindable rtt-chash rtt-cbo*

abbreviation *rtt-merge* :: *reg-transaction-tree* *merge*

where *rtt-merge* ≡ *merge-blindable rtt-chash rtt-cmerge*

lemma *merkle-reg-tx-content*: *merkle-interface rtt-chash rtt-cbo rtt-cmerge*

by (rule *merkle-product*[*OF merkle-discrete merkle-list*[*OF merkle-reg-view*]])

lemma *merkle-reg-tx*: *merkle-interface rtt-hash rtt-bo rtt-merge*

by (rule *merkle-blindable*[*OF merkle-reg-tx-content*])

The two list-level corollaries of the recursive lemmas above, lifted over the view list of a transaction.

lemma *reg-chains-views-blinding*:

assumes *list-all2 rv-bo va vb*

shows ($\bigcup v \in \text{set } va. \text{reg-chains } v$) $\subseteq$ ($\bigcup w \in \text{set } vb. \text{reg-chains } w$)

proof (rule *UN-reg-chains-mono*)

show *list-all2* ($\lambda x y. \text{reg-chains } x \subseteq \text{reg-chains } y$) *va vb*

using *assms* **by** (*auto elim!*: *list-all2-mono reg-chains-blinding*)

qed

```

lemma reg-chains-views-merge:
  assumes merge-list rv-merge va vb = Some v
  shows  $(\bigcup_{x \in \text{set } v}. \text{reg-chains } x) = (\bigcup_{x \in \text{set } va}. \text{reg-chains } x) \cup (\bigcup_{y \in \text{set } vb}. \text{reg-chains } y)$ 
proof (rule reg-chains-merge-list[OF assms])
  fix s t u assume rv-merge s t = Some u
  then show reg-chains u = reg-chains s  $\cup$  reg-chains t by (rule reg-chains-merge)
qed

```

Extraction: a revealed transaction yields the cross-domain state in which every attested chain (collected recursively from the revealed views) holds the asset in the consensus state; a fully blinded transaction yields the empty view.

```

fun rtt-extract :: reg-transaction-tree  $\Rightarrow$  global-state option where
  rtt-extract (Unblinded (r, tvs)) = Some (auth-state r  $(\bigcup_{v \in \text{set } tvs}. \text{reg-chains } v)$ )
| rtt-extract (Blinded -) = Some (auth-state ACTIVE  $\{\}$ )

```

interpretation *reg-tx-authenticated*:

```

  authenticated-state rtt-hash rtt-bo rtt-merge rtt-extract
proof (rule authenticated-state.intro)
  show merkle-interface rtt-hash rtt-bo rtt-merge by (rule merkle-reg-tx)
next
  note mlist = merkle-list[OF merkle-reg-view]
  show authenticated-state-axioms rtt-hash rtt-bo rtt-merge rtt-extract
proof
  — extract respects merging
  fix a b ab sa sb
  assume Hab: rtt-hash a = rtt-hash b and Mab: rtt-merge a b = Some ab
  and ea: rtt-extract a = Some sa and eb: rtt-extract b = Some sb
  show  $\exists \text{ sab}. \text{rtt-extract } ab = \text{Some } sab \wedge \text{state-join } sa \text{ } sb \text{ } sab$ 
proof (cases a)
  case (Unblinded pa) note A = this
  obtain ra va where pa: pa = (ra, va) by (cases pa)
  show ?thesis
proof (cases b)
  case (Unblinded pb) note B = this
  obtain rb vb where pb: pb = (rb, vb) by (cases pb)
  have hh: ra = rb  $\wedge$  map rv-hash va = map rv-hash vb
  using Hab by (simp add: A B pa pb)
  then have req: ra = rb and meq: hash-list rv-hash va = hash-list rv-hash vb
by simp-all
  from meq obtain v where mlv: merge-list rv-merge va vb = Some v
  using merkle-interface.merge-respects-hashes[OF mlist] by blast
  have ab-eq: ab = Unblinded (ra, v)
  using Mab req mlv by (simp add: A B pa pb merge-discrete-def)
  have sa-eq: sa = auth-state ra  $(\bigcup_{x \in \text{set } va}. \text{reg-chains } x)$  using ea by (simp add: A pa)
  have sb-eq: sb = auth-state ra  $(\bigcup_{y \in \text{set } vb}. \text{reg-chains } y)$  using eb req by (simp add: B pb)

```



```

have uv: ( $\bigcup x \in \text{set } v. \text{reg-chains } x$ )
      = ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ )  $\cup$  ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ )
  using reg-chains-views-merge[OF mlv] .
have rtt-extract ab = Some (auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ )
                              $\cup$  ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ )))
  by (simp add: ab-eq uv)
moreover have state-join sa sb (auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ )
                                      $\cup$  ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ )))
  unfolding sa-eq sb-eq by (rule state-join-auth)
ultimately show ?thesis by blast
next
case (Blinded hb) note B = this
have ab-eq: ab = Unblinded (ra, va) using Mab Hab by (simp add: A B pa)
have sa-eq: sa = auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ ) using ea by (simp
add: A pa)
have sb-eq: sb = auth-state ACTIVE {} using eb by (simp add: B)
have rtt-extract ab = Some (auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ ))
  by (simp add: ab-eq)
moreover have state-join sa sb (auth-state ra ( $\bigcup x \in \text{set } va. \text{reg-chains } x$ ))
  unfolding sa-eq sb-eq
  using state-join-auth[of ra  $\bigcup x \in \text{set } va. \text{reg-chains } x$  {}]
    auth-state-empty-eq[of ACTIVE ra] by simp
ultimately show ?thesis by blast
qed
next
case (Blinded ha) note A = this
show ?thesis
proof (cases b)
  case (Unblinded pb) note B = this
  obtain rb vb where pb: pb = (rb, vb) by (cases pb)
  have ab-eq: ab = Unblinded (rb, vb) using Mab Hab by (simp add: A B pb)
  have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
  have sb-eq: sb = auth-state rb ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ ) using eb by (simp
add: B pb)
  have rtt-extract ab = Some (auth-state rb ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ ))
    by (simp add: ab-eq)
  moreover have state-join sa sb (auth-state rb ( $\bigcup y \in \text{set } vb. \text{reg-chains } y$ ))
    unfolding sa-eq sb-eq
    using state-join-auth[of rb {}  $\bigcup y \in \text{set } vb. \text{reg-chains } y$ ]
      auth-state-empty-eq[of ACTIVE rb] by simp
  ultimately show ?thesis by blast
next
case (Blinded hb) note B = this
have ab-eq: ab = Blinded hb using Mab Hab by (simp add: A B)
have sa-eq: sa = auth-state ACTIVE {} using ea by (simp add: A)
have sb-eq: sb = auth-state ACTIVE {} using eb by (simp add: B)
have rtt-extract ab = Some (auth-state ACTIVE {}) by (simp add: ab-eq)
moreover have state-join sa sb (auth-state ACTIVE {})
  unfolding sa-eq sb-eq using state-join-auth[of ACTIVE {} {}] by simp

```

```

      ultimately show ?thesis by blast
    qed
  qed
next
  — extract under blinding
  fix a b sb
  assume Hab: rtt-hash a = rtt-hash b and BOab: rtt-bo a b
  and eb: rtt-extract b = Some sb
  show  $\exists sa. rtt-extract a = Some sa \wedge state-refines sa sb$ 
  proof (cases b)
    case (Unblinded pb) note B = this
    obtain rb vb where pb: pb = (rb, vb) by (cases pb)
    show ?thesis
    proof (cases a)
      case (Unblinded pa) note A = this
      obtain ra va where pa: pa = (ra, va) by (cases pa)
      have bo: rtt-cbo (ra, va) (rb, vb) using BOab by (simp add: A B pa pb)
      then have req: ra = rb and la: list-all2 rv-bo va vb
        by (simp-all add: rel-prod-inject)
      have sr:  $(\bigcup_{x \in set va} reg-chains x) \subseteq (\bigcup_{y \in set vb} reg-chains y)$ 
        using reg-chains-views-blinding[OF la] .
      have srf: state-refines (auth-state ra  $(\bigcup_{x \in set va} reg-chains x)$ )
        (auth-state ra  $(\bigcup_{y \in set vb} reg-chains y)$ )
        using sr by (rule state-refines-auth)
      have rtt-extract a = Some (auth-state ra  $(\bigcup_{x \in set va} reg-chains x)$ )
        by (simp add: A pa)
      moreover have state-refines (auth-state ra  $(\bigcup_{x \in set va} reg-chains x)$ ) sb
        using srf req eb by (simp add: B pb)
      ultimately show ?thesis by blast
    next
      case (Blinded ha) note A = this
      have sr: state-refines (auth-state rb  $\{\}$ ) (auth-state rb  $(\bigcup_{y \in set vb} reg-chains$ 
y))
        using empty-subsetI by (rule state-refines-auth)
      have rtt-extract a = Some (auth-state rb  $\{\}$ )
        by (simp add: A auth-state-empty-eq[of ACTIVE rb])
      moreover have state-refines (auth-state rb  $\{\}$ ) sb
        using sr eb by (simp add: B pb)
      ultimately show ?thesis by blast
    qed
  next
    case (Blinded hb) note B = this
    have a = Blinded hb using BOab by (cases a) (simp-all add: B)
    then have rtt-extract a = Some (auth-state ACTIVE  $\{\}$ ) by simp
    moreover have sb = auth-state ACTIVE  $\{\}$  using eb by (simp add: B)
    ultimately show ?thesis using state-refines-refl by auto
  qed
next
  — extract preserves validity: single consensus per transaction, hence consistent

```

```

fix  $a\ s$ 
assume  $\text{rtt-extract } a = \text{Some } s$ 
then show  $\text{valid-state } s$ 
  by ( $\text{cases } a$ ) ( $\text{auto simp: auth-state-valid}$ )
qed
qed

```

Multi-level (subview) non-degeneracy witness. A transaction with one view that has two subviews (chains 1 and 2 nested under chain 0). Revealing everything attests all three chains; blinding *only the chain-2 subview* — a subview-level, not top-level, blinding — drops exactly chain 2 , leaving chains 0 and 1 attested. Selective disclosure is alive below the root.

definition $\text{demo-subview1} :: \text{reg-view}$ **where**

$\text{demo-subview1} = \text{Tree}_m (\text{Unblinded} (\text{Suc } 0, []))$

definition $\text{demo-subview2} :: \text{reg-view}$ **where**

$\text{demo-subview2} = \text{Tree}_m (\text{Unblinded} (\text{Suc} (\text{Suc } 0), []))$

definition $\text{demo-view} :: \text{reg-view}$ **where**

$\text{demo-view} = \text{Tree}_m (\text{Unblinded} (0, [\text{demo-subview1}, \text{demo-subview2}]))$

definition $\text{demo-view-sv2blinded} :: \text{reg-view}$ **where**

$\text{demo-view-sv2blinded} = \text{Tree}_m (\text{Unblinded} (0, [\text{demo-subview1}, \text{Tree}_m (\text{Blinded} (\text{Garbage } 0))]))$

lemma $\text{demo-subview-disclosure}$:

$\text{reg-chains } \text{demo-view} = \{0, \text{Suc } 0, \text{Suc} (\text{Suc } 0)\}$

$\text{reg-chains } \text{demo-view-sv2blinded} = \{0, \text{Suc } 0\}$

by ($\text{auto simp: demo-view-def demo-view-sv2blinded-def demo-subview1-def demo-subview2-def}$)

lemma $\text{demo-multilevel-extract}$:

$\text{rtt-extract} (\text{Unblinded} (\text{ACTIVE}, [\text{demo-view}]))$

$= \text{Some} (\text{auth-state } \text{ACTIVE } \{0, \text{Suc } 0, \text{Suc} (\text{Suc } 0)\})$

$\text{rtt-extract} (\text{Unblinded} (\text{ACTIVE}, [\text{demo-view-sv2blinded}]))$

$= \text{Some} (\text{auth-state } \text{ACTIVE } \{0, \text{Suc } 0\})$

by ($\text{simp-all add: demo-subview-disclosure}$)

The single-consensus invariant is load-bearing. If a model allowed a chain to be attested in a regulatory state different from another chain's (which a per-node *reg-state* would permit), validity would collapse: the state below pins chain 0 to *ACTIVE* and chain $\text{Suc } 0$ to *FROZEN* on the same asset, and is *not* valid. The recursive extraction *rtt-extract* never produces such a state, because every attested chain shares the *one* consensus *reg-state* of its transaction — exactly the structural reason validity holds with no assumption.

definition $\text{rogue-inconsistent-state} :: \text{global-state}$ **where**

$\text{rogue-inconsistent-state} =$

$\langle \text{gs-chains} = (\lambda c\ a.$

$\text{if } a = 0 \wedge c = 0$

$\text{then Some } \langle \text{as-reg-state} = \text{ACTIVE} \rangle$

else if $a = 0 \wedge c = \text{Suc } 0$
 then $\text{Some } (\text{as-reg-state} = \text{FROZEN})$
 else None ,
 $\text{gs-locks} = (\lambda\text{-}. \text{False})$

lemma *single-consensus-load-bearing*:

$\neg \text{valid-state rogue-inconsistent-state}$

by (*auto simp: rogue-inconsistent-state-def valid-state-def consistent-state-def*
get-reg-state-def get-asset-state-def)

Non-degeneracy of the recursive transaction instance: a single-chain transaction extracts to a genuine one-chain cross-domain state, and a transaction with a non-empty view list genuinely merges at the transaction-tree level (the *extract-respects-merging* path is exercised directly, not only through *reg-chains-merge*).

lemma *reg-tx-authenticated-nontrivial*:

rtt-extract (*Unblinded* (*ACTIVE*, [*Tree_m* (*Unblinded* (*0*, []))]))
 $= \text{Some } (\text{auth-state } \text{ACTIVE } \{0\})$

rtt-merge (*Unblinded* (*ACTIVE*, [*Tree_m* (*Unblinded* (*0*, []))]))
 $(\text{Unblinded } (\text{ACTIVE}, [\text{Tree}_m (\text{Unblinded } (0, []))]))$

$= \text{Some } (\text{Unblinded } (\text{ACTIVE}, [\text{Tree}_m (\text{Unblinded } (0, []))]))$

by (*simp-all add: merge-discrete-def merge-list-Cons merge-tree.simps*
merge-rt-F_m-def)

63 Path-level Merkle inclusion for the recursive view tree

The sequence-level result *authenticated-state.sequence-inclusion-integrity* is stated at the level of revealed holdings. Here we sharpen it to the concrete Merkle *inclusion path*. The recursive view *reg-view* is exactly an instance of the generic rose-tree inclusion-proof machinery of *ADS-Functor.Inclusion-Proof-Construction*: we take the source-content embedding to be the identity on *chain-id* (*rv-embed*) and the content hash to be *id*. An inclusion proof is a zipper in which the target subview is revealed while every node on the path to the root and all off-path siblings are blinded (*blind-path*). Because the discrete content already satisfies the blinding-order locale (*blinding-of-on-discrete*), the two soundness results transfer with no new assumption. This is the same construction the public Canton formalisation applies to its own view tree; it operates on the view rose tree, so the consensus scope limit of the preceding section is untouched.

abbreviation *rv-embed* :: *chain-id* $\Rightarrow$ *chain-id* **where**

rv-embed $\equiv id$

abbreviation *zippers-reg-view* :: *chain-id zipper* $\Rightarrow$ (*chain-id*, *chain-id*) *zipper_m*
list **where**

zippers-reg-view $\equiv \text{zippers-rose-tree } \text{rv-embed } \text{hash-discrete}$

abbreviation $\text{blind-reg-path} :: \text{chain-id path} \Rightarrow (\text{chain-id}, \text{chain-id}) \text{ path}_m$ **where**
 $\text{blind-reg-path} \equiv \text{blind-path rv-embed hash-discrete}$

abbreviation $\text{embed-reg-path} :: \text{chain-id path} \Rightarrow (\text{chain-id}, \text{chain-id}) \text{ path}_m$ **where**
 $\text{embed-reg-path} \equiv \text{embed-path rv-embed}$

Inclusion soundness: every enumerated inclusion proof commits to the same authenticating root hash as the fully revealed view tree, so no inclusion proof can fabricate a tree the root does not authenticate.

theorem *reg-view-inclusion-same-hash*:

assumes $z \in \text{set } (\text{zippers-reg-view } (p, t))$
shows $\text{rv-hash } (\text{tree-of-zipper}_m z)$
 $\quad = \text{rv-hash } (\text{tree-of-zipper}_m (\text{embed-reg-path } p, \text{embed-source-tree rv-embed } t))$
using *assms* **by** $(\text{rule } \text{zippers-rose-tree-same-hash})$

Each inclusion proof is a blinding of the canonical inclusion-path tree, which carries the same root hash as the fully revealed tree: a partial, need-to-know view that refines the whole. The single content premise is discharged from the discrete building block, so no assumption is added.

theorem *reg-view-inclusion-blinding-of*:

assumes $z \in \text{set } (\text{zippers-reg-view } (p, t))$
shows $\text{rv-bo } (\text{tree-of-zipper}_m z)$
 $\quad (\text{tree-of-zipper}_m (\text{blind-reg-path } p, \text{embed-source-tree rv-embed } t))$
by $(\text{rule } \text{zippers-rose-tree-blinding-of}[\text{OF blinding-of-on-discrete assms}])$

The attested chains carried by an inclusion proof are contained in those of the fully revealed tree: a path-level reading of need-to-know, from *reg-chains-blinding*.

corollary *reg-view-inclusion-chains-sound*:

assumes $z \in \text{set } (\text{zippers-reg-view } (p, t))$
shows $\text{reg-chains } (\text{tree-of-zipper}_m z)$
 $\quad \subseteq \text{reg-chains } (\text{tree-of-zipper}_m (\text{blind-reg-path } p, \text{embed-source-tree rv-embed } t))$
using $\text{reg-chains-blinding}[\text{OF reg-view-inclusion-blinding-of}[\text{OF assms}]]$.

Non-degeneracy. Over a genuine two-level view tree the canonical inclusion endpoint reveals all three attested chains, and a selective inclusion proof in the enumeration exposes its target subview (chains 0 and $\text{Suc } 0$) while the unrelated sibling (chain $\text{Suc } (\text{Suc } 0)$) stays blinded. Selective disclosure is alive at the path level.

definition *demo-src* :: *chain-id rose-tree* **where**

$\text{demo-src} = \text{Tree } (0, [\text{Tree } (\text{Suc } 0, []), \text{Tree } (\text{Suc } (\text{Suc } 0), [])])$

lemma *reg-view-inclusion-nonvacuous-endpoint*:

$\text{reg-chains } (\text{tree-of-zipper}_m (\text{embed-reg-path } [], \text{embed-source-tree rv-embed } \text{demo-src}))$
 $\quad = \{0, \text{Suc } 0, \text{Suc } (\text{Suc } 0)\}$

by (*auto simp: demo-src-def embed-path-def*)

lemma *reg-view-inclusion-nonvacuous-selective:*

$\exists z \in \text{set } (\text{zippers-reg-view } ([], \text{demo-src})). \text{reg-chains } (\text{tree-of-zipper}_m z) = \{0, \text{Suc } 0\}$

by (*force simp: demo-src-def zippers-rose-tree.simps zipper-children.simps
blind-path-def embed-path-def*)

Load-bearing regression witness. A forged zipper that fabricates a different chain is neither in the legitimate enumeration nor matched by the same-hash guarantee, so the membership premise of *reg-view-inclusion-same-hash* is load-bearing; dropping it makes the theorem false.

lemma *reg-view-inclusion-forgery-excluded:*

$([], \text{embed-source-tree } \text{rv-embed } (\text{Tree } (\text{Suc } 0, [])))$
 $\notin \text{set } (\text{zippers-reg-view } ([], \text{Tree } (0::\text{chain-id}, [])))$
 $\wedge \text{rv-hash } (\text{tree-of-zipper}_m ([], \text{embed-source-tree } \text{rv-embed } (\text{Tree } (\text{Suc } 0, []))))$
 $\neq \text{rv-hash } (\text{tree-of-zipper}_m (\text{embed-reg-path } [], \text{embed-source-tree } \text{rv-embed } (\text{Tree } (0::\text{chain-id}, []))))$
by (*simp add: zippers-rose-tree.simps zipper-children.simps embed-path-def
hash-rt-F_m-alt-def*)

end

AI-assisted Review Disclosure

Large language models (Anthropic Claude and OpenAI Codex) were used extensively in the development of this entry: drafting and revising the Isabelle/HOL theories, proofs, and documentation under the author's direction, and auditing the entry for proof-scope, documentation, and release-packaging issues. The author reviewed and accepted the retained changes and remains responsible for the formal statements, proofs, source files, and manuscript. These tools were not part of the trusted proof-checking path; the formal sources were checked by Isabelle/HOL.

64 Conclusion

The entry establishes kernel-checked model-level laws for preservation morphisms, authenticated extractable views, degree-indexed forgetting, finite-set priority selection, aggregate fair-discharge progress, and bounded existential recovery. Its explicit boundaries are equally part of the result: no model-to-code refinement, probabilistic leader election, BFT or network execution, request-identity fairness, or concurrent lock protocol is claimed.

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